True or false?

A temperature that is 7°C at t = 0, 23°C at t = 12 h, and repeats every 24 hours is modeled by $15 - 8\cos(\pi t/12)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q3/5

A blood pressure that starts at 100 mm Hg and rises, varies between 80 and 120, and repeats every 0.8 s is modeled by

A. $100 + 20\sin(0.8t)$

B. $20 + 100\sin(2.5\pi t)$

C. $120\sin(2.5\pi t)$

D. $100 + 20\sin(2.5\pi t)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q4/5

A piston's position starts at 0 cm and increases, varies between −3 and 3 cm, and repeats every 1 s. Write it in the form p = A·sin(Bt) with A > 0 and B > 0. Give B exactly, in terms of π.

A = _____

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q5/5

SET 44 • HSF-IF.B.5 • INTERPRETING FUNCTIONS

State domains of radical, rational, logarithmic, and polynomial models

MAIN	Algebra II main question	**HSF-IF.B.5**
BACKWARD 1	Domains of linear, quadratic, and exponential models	**HSF-IF.B.5**
BACKWARD 2	Division by zero	**7.NS.A.2.b**
BACKWARD 3	Solve linear inequalities	**HSA-REI.B.3**
FORWARD 1	Domains defined by nonlinear conditions	**HSF-IF.B.5**
FORWARD 2	Domains and ranges of inverse functions	**HSF-BF.B.4.a**

SET 44 • HSF-IF.B.5

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5

A car's speed is $s(d) = \sqrt{(22.5d)}$ mph for a skid of d feet. Which gives the domain of the function and a domain that makes sense in context?

A. Function: $d > 22.5$; context: $d > 22.5$, up to the longest skid that can occur

B. Function: $d \geq 0$; context: $d \geq 0$, up to the longest skid that can actually occur

C. Function: $d \geq 0$; context: all real numbers, since any skid length is possible

D. Function: all real numbers; context: $d \geq 0$, up to the longest skid that can occur

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q1/5

The average cost of x shirts is $A(x) = (500 + 4x)/x$ dollars. Which gives the domain of the function and the domain in context?

A. Function: $x > 0$; context: all real numbers

B. Function: $x \neq 500$; context: $x \geq 500$

C. Function: all real numbers; context: whole numbers $x \geq 0$

D. Function: all real $x \neq 0$; context: whole numbers $x \geq 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q2/5

$pH = -\log[H^+]$, where $[H^+]$ is the hydrogen ion concentration. What is the domain?

A. $[H^+] > 0$

B. $[H^+] \geq 0$

C. All real numbers

D. $0 < [H^+] < 14$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q3/5

A box made from a 12-in square sheet has volume $V(x) = x(12 - 2x)^2$. Which gives the domain of the polynomial and the domain in context?

A. Polynomial: all real numbers; context: $0 < x < 12$

B. Polynomial: all real numbers; context: $0 < x < 6$

C. Polynomial: $x \geq 0$; context: $x \geq 6$

D. Polynomial: $0 < x < 6$; context: all real numbers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q4/5

Which is the domain of $f(x) = \sqrt{(x - 2)/(x - 5)}$?

A. $x > 5$

B. $x \geq 2$

C. $x \geq 2$ and $x \neq 5$

D. All real numbers except 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q5/5

SET 44 • HSF-IF.B.5

BACKWARD 1

Domains of linear, quadratic, and exponential models • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

$h(t) = -16t^2 + 64t$ models height until the ball lands. Give the domain in context.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q1/5

$C(n) = 2n + 5$ is the cost in dollars of n tickets. Which domain makes sense in context?

A. Negative numbers

B. All real numbers

C. All real numbers $n \geq 5$

D. Whole numbers $n \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q2/5

True or false?

The domain of $f(x) = 2^x$ is $x > 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q3/5

The input n of a model counts people. Which domain fits?

A. Fractions between 0 and 1

B. All real numbers

C. Nonnegative whole numbers

D. Negative numbers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q4/5

Why might the domain that makes sense for a model be smaller than the domain of its formula?

A. The context restricts the inputs, for example to nonnegative numbers or whole numbers.

B. The formula is always wrong outside small inputs, so only small inputs make sense.

C. The domain of a model must always be all real numbers, as for its formula.

D. Formulas never have restricted domains, so only the context can restrict them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • HSF-IF.B.5

BACKWARD 2

Division by zero • Grade 7 • The Number System • 7.NS.A.2.b

For which x is $1/(x - 5)$ undefined?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q1/5

For which x is (500 + 4x)/x undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q2/5

True or false?

$3/0 = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q3/5

$(x + 1)/(x^2 - 9)$ is undefined at

A. $x = 0$

B. $x = 9$

C. $x = -1$

D. $x = \pm 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q4/5

Give the value of 0/5.

0/5 = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q5/5

SET 44 • HSF-IF.B.5

BACKWARD 3

Solve linear inequalities • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $x - 2 \geq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q1/5

Solve $12 - 2x > 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q2/5

True or false?

$22.5d \geq 0$ means $d \geq 0$.

A. True

B. False

$3 - x \geq 0$ means

A. $x \geq 3$

B. $x \leq 3$

C. $x \geq -3$

D. $x \leq -3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q4/5

Solve $2x + 8 > 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q5/5

SET 44 • HSF-IF.B.5

FORWARD 1

Domains defined by nonlinear conditions • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5

Give the domain of $f(x) = \sqrt{(9 - x^2)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q1/5

Which is the domain of $g(x) = \log(x^2 - 4)$?

A. $x \geq 2$

B. $-2 < x < 2$

C. $x > 2$ only

D. $x < -2$ or $x > 2$

$h(x) = 1/(x^2 - x - 6)$. List the real numbers excluded from the domain.

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q3/5

True or false?

The domain of $\sqrt{(x^2)}$ is $x \geq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q4/5

The domain of $\log(x - 1) + \sqrt{(5 - x)}$ is

A. $1 < x \leq 5$

B. $x \leq 5$

C. $1 \leq x \leq 5$

D. $x > 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q5/5

SET 44 • HSF-IF.B.5

FORWARD 2

Domains and ranges of inverse functions • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

$f(x) = \sqrt{(x - 2)}$ has domain $x \geq 2$ and range $y \geq 0$. Write $f^{-1}(x)$ and give its domain.

$f^{-1}(x) =$ _____

Domain of f^{-1}: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q1/5

$f(x) = 2^x$ has range $y > 0$. What is the domain of $f^{-1}(x) = \log_2 x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q2/5

True or false?

The domain of an inverse function is the range of the original function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q3/5

$f(x) = 1/x + 3$ $(x \neq 0)$ has an inverse with domain

A. $x \neq 0$

B. $x > 3$

C. all real x

D. $x \neq 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q4/5

$s(d) = \sqrt{(22.5d)}$ gives a car's speed s from its skid length d. Why does its inverse $d = s^2/22.5$ need the restriction $s \geq 0$?

A. Because $s^2/22.5$ is undefined for $s < 0$, so the formula cannot be used for negative speeds.

B. Because d must be a whole number of feet, and $s \geq 0$ makes $s^2/22.5$ a whole number.

C. The range of $s(d)$ is $s \geq 0$, so $d = s^2/22.5$ undoes s only for $s \geq 0$; without the restriction it is not one-to-one.

D. Because 22.5 is positive, so s must also be positive for the quotient $s^2/22.5$ to make sense.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q5/5

SET 45 • HSF-IF.B.6 • INTERPRETING FUNCTIONS

Calculate and interpret average rates of change of nonlinear models

MAIN	Algebra II main question	**HSF-IF.B.6**
BACKWARD 1	Average rate of change of simpler functions	**HSF-IF.B.6**
BACKWARD 2	Slope between two points	**8.F.B.4**
BACKWARD 3	Evaluate nonlinear functions	**HSF-IF.A.2**
FORWARD 1	Compare average rates on different intervals	**HSF-IF.B.6**
FORWARD 2	Compare rates of change across representations	**HSF-IF.C.9**

SET 45 • HSF-IF.B.6

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6

A culture has $N(t) = 200 \cdot 2^t$ bacteria after t hours. Find the average rate of change from $t = 1$ to $t = 3$.

Average rate (bacteria per hour): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q1/5

A stone's height is $h(t) = 64 - 16t^2$ ft after t seconds. Find the average rate of change from t = 0.5 to t = 1.5 s.

Average rate (ft per second): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q2/5

The table shows a car's value. Find the average rate of change from t = 2 to t = 4 years, to the nearest cent.

t (years)	0	1	2	3	4	5
V ($)	20000	17000	14450	12282.50	10440.06	8874.11

Average rate (dollars per year): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q3/5

Speed from skid length is $s(d) = \sqrt{(24d)}$ mph. Find the average rate of change from $d = 24$ to $d = 96$ ft.

Average rate (mph per foot): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q4/5

Temperature is $T(t) = 15 - 8\cos(\pi t/12)$ °C after t hours. Find the average rate of change from $t = 0$ to $t = 6$.

Average rate (°C per hour): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q5/5

SET 45 • HSF-IF.B.6

BACKWARD 1

Average rate of change of simpler functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = x^2$. Find the average rate of change from $x = 1$ to $x = 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q1/5

A table gives f(2) = 10 and f(5) = 1. Find the average rate of change.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q2/5

True or false?

The average rate of change from a to b is $(f(b) - f(a))/(b - a)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q3/5

The average rate of change of f(x) = 3x + 1 on any interval is

A. it varies

B. 1

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q4/5

f is in dollars and x is in years. What are the units of the average rate of change of f?

A. Years per dollar

B. Dollars per year

C. Dollar-years

D. Dollars

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • HSF-IF.B.6

BACKWARD 2

Slope between two points • Grade 8 • Functions • 8.F.B.4

Find the slope between (1, 400) and (3, 1600).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q1/5

Find the slope between (0.5, 60) and (1.5, 28).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q2/5

True or false?

A negative slope means the output increases as the input increases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q3/5

The slope between (24, 24) and (96, 48) is

A. 72/24

B. 1/3

C. 2

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q4/5

Find the slope between (0, 7) and (6, 15).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • HSF-IF.B.6

BACKWARD 3

Evaluate nonlinear functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Find $N(3)$ for $N(t) = 200 \cdot 2^t$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q1/5

Find $h(1.5)$ for $h(t) = 64 - 16t^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q2/5

True or false?

$\sqrt{(24 \cdot 96)} = 2304$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q3/5

$15 - 8\cos(\pi/2)$ equals

A. 15

B. 7

C. 23

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q4/5

Find $T(0)$ for $T(t) = 15 - 8\cos(\pi t/12)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q5/5

SET 45 • HSF-IF.B.6

FORWARD 1

Compare average rates on different intervals • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6

For $N(t) = 200 \cdot 2^t$ bacteria after t hours, find the average rates of change on [0, 1] and on [3, 4].

Rate on [0, 1] (bacteria per hour): _____

Rate on [3, 4] (bacteria per hour): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q1/5

For $s(d) = \sqrt{(24d)}$, find the average rates of change on $[0, 24]$ and on $[96, 150]$.

Rate on $[0, 24]$: _____

Rate on $[96, 150]$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q2/5

True or false?

For a function with period 24, the average rate of change over [0, 24] is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q3/5

For $f(x) = x^3$, the average rate on $[-1, 1]$ is

A. 3

B. 0

C. 2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q4/5

$T(t) = 15 - 8\cos(\pi t/12)$. On which interval of length 6 is the average rate of change $-4/3$?

A. [6, 12]

B. [3, 9]

C. [12, 18]

D. [0, 6]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q5/5

SET 45 • HSF-IF.B.6

FORWARD 2

Compare rates of change across representations • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9

$f(t) = 200 \cdot 2^t$, and g is given by $g(1) = 500$ and $g(3) = 1500$. Find the average rate of change of each on $[1, 3]$.

Rate of f: _____

Rate of g: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q1/5

A graph of a cooling liquid falls from 90°C at t = 0 to 50°C at t = 20 min. Find the average rate of change of the graph and of $T(t) = 90 - 1.5t$ on [0, 20].

Graph (°C per min): _____

Formula (°C per min): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q2/5

True or false?

A function with a greater average rate on one interval must be greater everywhere.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q3/5

A table shows $h(0) = 0$, $h(2) = 40$; the formula $k(t) = 5t^2$ gives

A. no rate for k

B. a greater rate for k

C. a greater rate for h

D. equal average rates on [0, 2]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q4/5

One function is given by a graph and another by a formula. How can their average rates of change on the same interval be compared?

A. Compare the two functions at the right endpoint only, since the larger final value means the larger average rate of change.

B. Read the two endpoint values from the graph, evaluate the formula at the same inputs, and compute both quotients (change in output)/(change in input).

C. Compare the steepest part of the graph with the leading coefficient of the formula, since both measure the rate of change on the interval.

D. Compare the y-intercepts of the graph and the formula, since the larger starting value has the larger average rate of change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q5/5

SET 46 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph square root and cube root functions with transformations

MAIN	Algebra II main question	**HSF-IF.C.7.b**
BACKWARD 1	Graph $y = \sqrt{x}$ and $y = \sqrt[3]{x}$ from a table	**HSF-IF.C.7.b**
BACKWARD 2	Translations and reflections of graphs	**HSF-BF.B.3**
BACKWARD 3	Square and cube roots of numbers	**8.EE.A.2**
FORWARD 1	Graph radical functions with horizontal stretches and reflections	**HSF-IF.C.7.b**
FORWARD 2	Graph a radical function and its inverse	**HSF-BF.B.4.a**

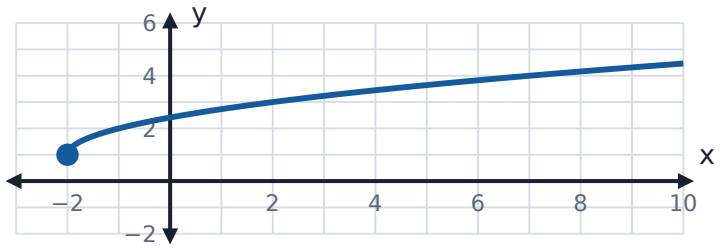
SET 46 • HSF-IF.C.7.b

MAIN

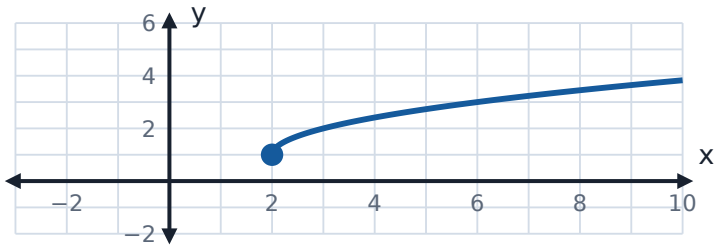
Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

Which graph shows $y = \sqrt{(x - 2)} + 1$?

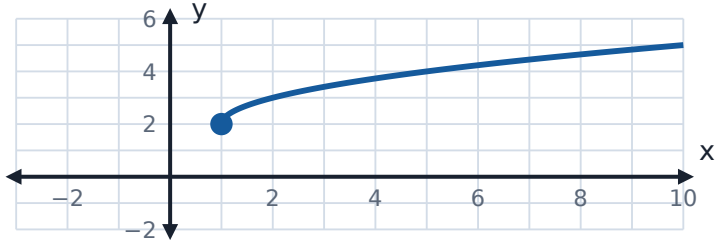
A



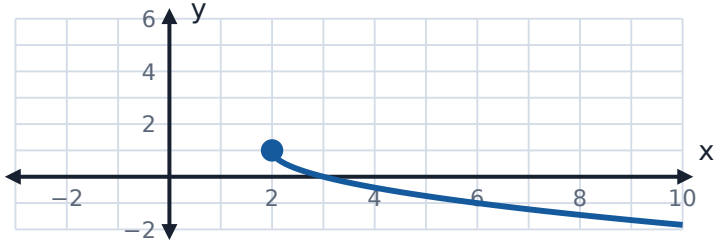
B



C



D



A. Graph A

B. Graph B

C. Graph C

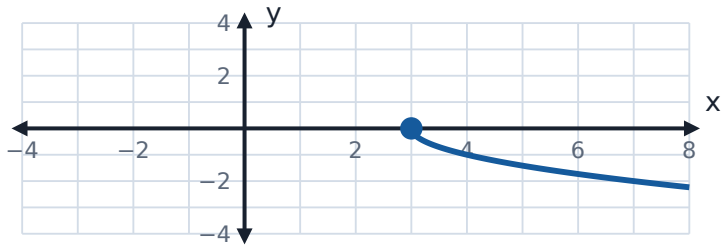
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-M-Q1**

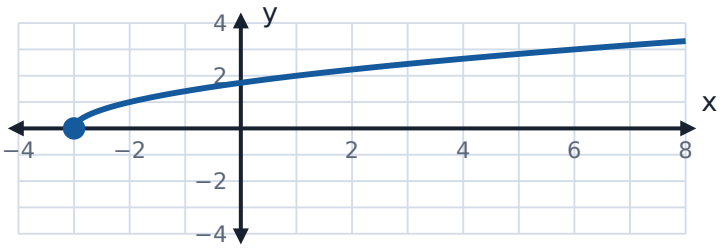
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q1/5

Which graph shows $y = -\sqrt{(x + 3)}$?

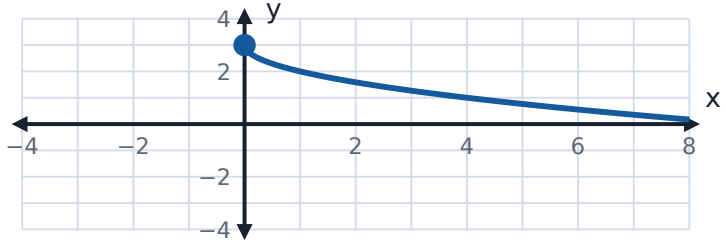
A



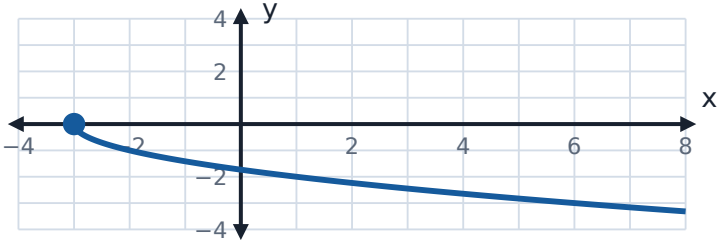
B



C



D



A. Graph A

B. Graph B

C. Graph C

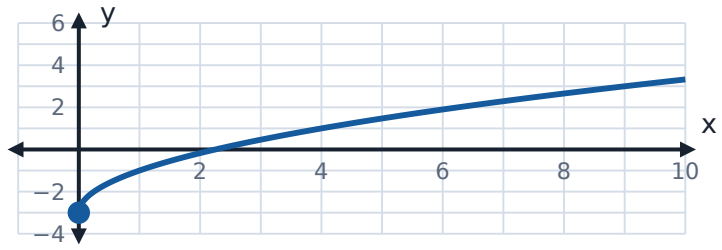
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-M-Q2**

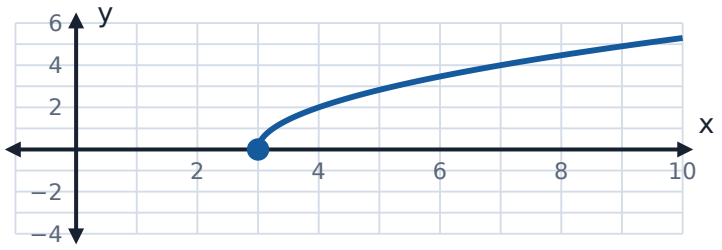
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q2/5

Which graph shows $y = 2\sqrt{x} - 3$?

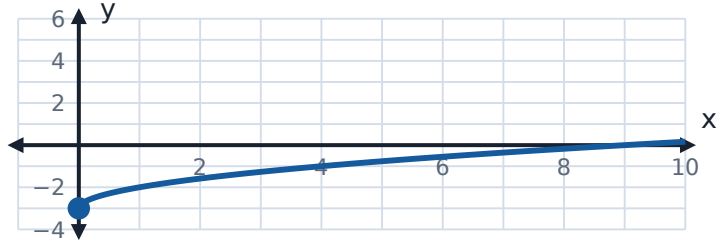
A



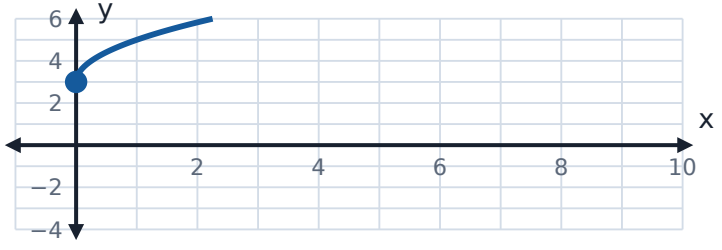
B



C



D



A. Graph A

B. Graph B

C. Graph C

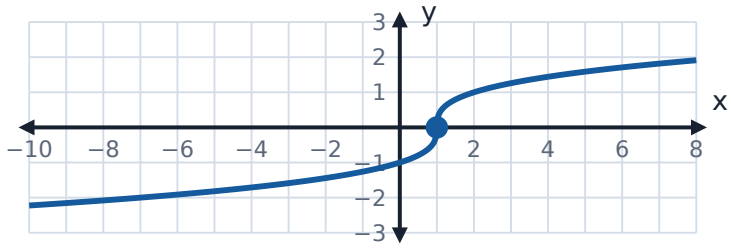
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-M-Q3**

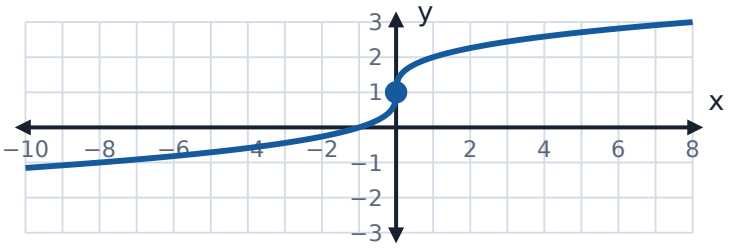
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q3/5

Which graph shows $y = \sqrt[3]{(x + 1)}$?

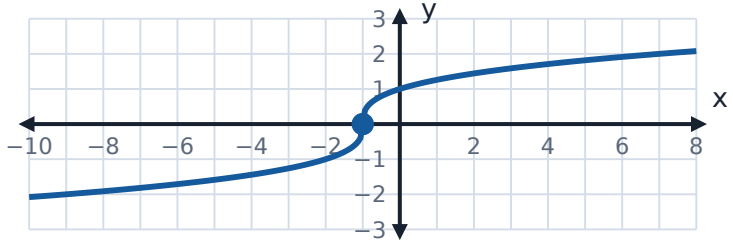
A



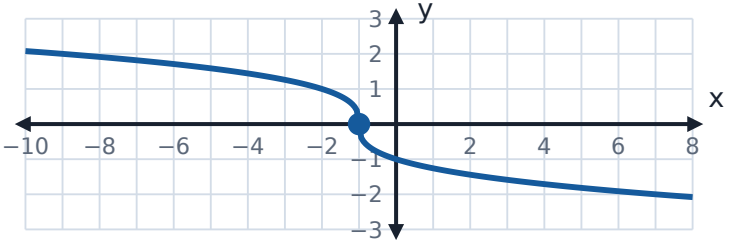
B



C



D



A. Graph A

B. Graph B

C. Graph C

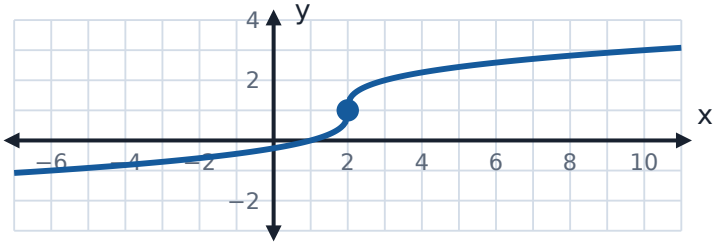
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-M-Q4**

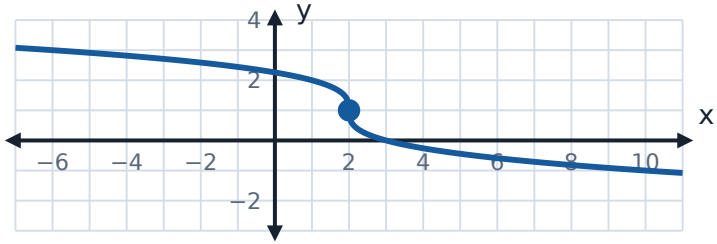
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q4/5

Which graph shows $y = -\sqrt[3]{(x - 2)} + 1$?

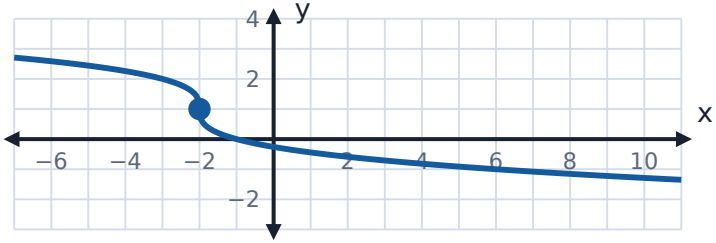
A



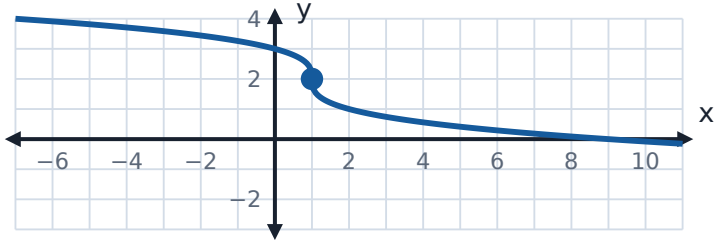
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q5/5

SET 46 • HSF-IF.C.7.b

BACKWARD 1

Graph $y = \sqrt{x}$ and $y = \sqrt[3]{x}$ from a table • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

Find $y = \sqrt{x}$ at $x = 0$, 1, 4, and 9, in that order.

y-values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q1/5

Find $y = \sqrt[3]{x}$ at $x = -8, -1, 0, 1,$ and 8, in that order.

y-values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q2/5

True or false?

The graph of $y = \sqrt{x}$ starts at the origin and decreases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q3/5

The domain of $y = \sqrt[3]{x}$ is

A. $x > 0$

B. all real numbers

C. $x \leq 0$

D. $x \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q4/5

Give the range of $y = \sqrt{x}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q5/5

SET 46 • HSF-IF.C.7.b

BACKWARD 2

Translations and reflections of graphs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

How is the graph of $y = \sqrt{(x - 2)} + 1$ obtained from the graph of $y = \sqrt{x}$?

A. Shift right 2 and up 1

B. Shift up 2 and right 1

C. Shift left 2 and up 1

D. Shift right 2 and down 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q1/5

How is the graph of $y = -\sqrt{(x + 3)}$ obtained from the graph of $y = \sqrt{x}$?

A. Shift right 3, then reflect over the x-axis

B. Shift down 3, then reflect over the x-axis

C. Shift left 3, then reflect over the y-axis

D. Shift left 3, then reflect over the x-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q2/5

True or false?

$y = 2\sqrt{x}$ is a vertical stretch of $y = \sqrt{x}$ by 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q3/5

The point (0, 0) of $y = \sqrt{x}$ moves to which point for $y = \sqrt{(x - 2)} + 1$?

A. (2, 1)

B. (1, 2)

C. (−2, 1)

D. (2, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q4/5

Where does (1, 1) of $y = \sqrt[3]{x}$ go under $y = \sqrt[3]{(x + 1)}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • HSF-IF.C.7.b

BACKWARD 3

Square and cube roots of numbers • Grade 8 • Expressions & Equations • 8.EE.A.2

Find $\sqrt{9}$ and $\sqrt[3]{(-8)}$.

$\sqrt{9} =$ _____

$\sqrt[3]{(-8)} =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q1/5

Find $\sqrt[3]{27}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q2/5

True or false?

$\sqrt{(-4)}$ is a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q3/5

$\sqrt[3]{(-1)}$ equals

A. i

B. undefined

C. -1

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q4/5

Find $\sqrt{16} - 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q5/5

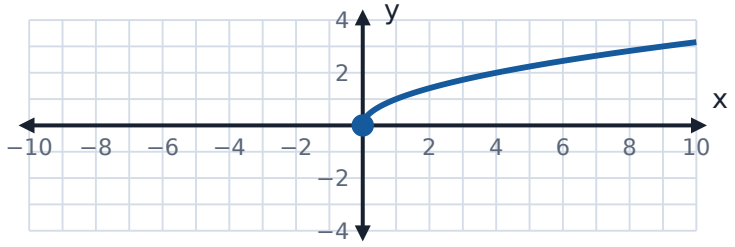
SET 46 • HSF-IF.C.7.b

FORWARD 1

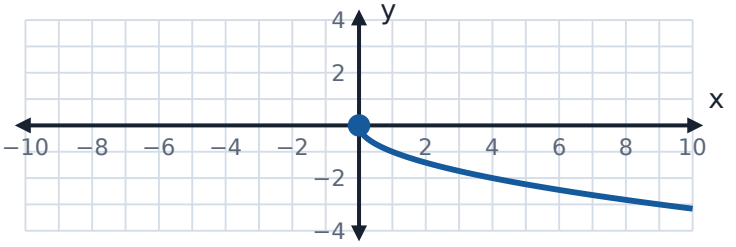
Graph radical functions with horizontal stretches and reflections • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

Which graph shows $y = \sqrt{(-x)}$?

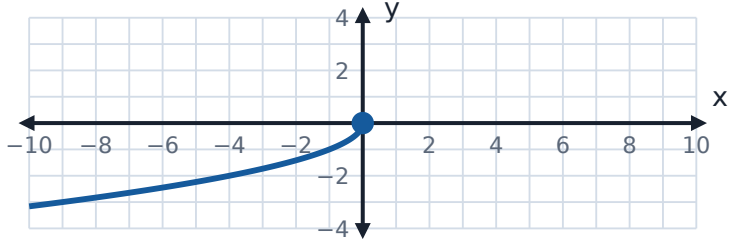
A



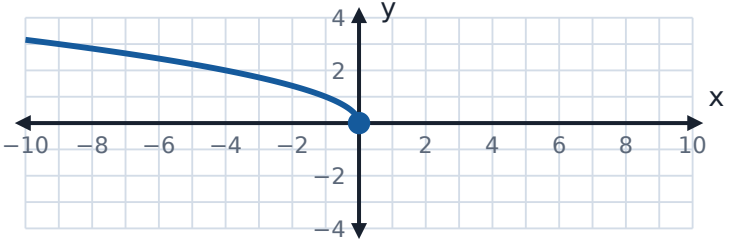
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q1/5

For $x \geq 0$, $\sqrt{(4x)} = 2\sqrt{x}$. What does this say about the graphs of $y = \sqrt{(4x)}$ and $y = 2\sqrt{x}$?

A. They are the same graph: a horizontal compression of $y = \sqrt{x}$ by a factor of 1/4 gives the same result as a vertical stretch by 2.

B. They are reflections of each other over the y-axis, because one changes the input and the other the output.

C. They are different graphs that meet only at the origin, because a horizontal change never equals a vertical one.

D. They are different graphs: $y = \sqrt{(4x)}$ is $y = \sqrt{x}$ shifted right 4, and $y = 2\sqrt{x}$ is $y = \sqrt{x}$ stretched up by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q2/5

True or false?

$y = \sqrt[3]{(-x)}$ is the same as $y = \sqrt[3]{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q3/5

The domain of $y = \sqrt{(3 - x)}$ is

A. all reals

B. $x \geq 3$

C. $x \leq 3$

D. $x \geq -3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q4/5

Which square root function has a graph that starts at (4, −2) and goes up to the left?

A. $y = \sqrt{(4 - x)} + 2$

B. $y = -\sqrt{(4 - x)} - 2$

C. $y = \sqrt{(x - 4)} - 2$

D. $y = \sqrt{(4 - x)} - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q5/5

SET 46 • HSF-IF.C.7.b

FORWARD 2

Graph a radical function and its inverse • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

$f(x) = \sqrt{(x - 2)} + 1$. Write $f^{-1}(x)$ and give its domain.

$f^{-1}(x) =$ _____

Domain of f^{-1}: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q1/5

Find the inverse of $g(x) = \sqrt[3]{(x + 1)}$.

$g^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q2/5

True or false?

The graph of an inverse is the reflection of the graph over y = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q3/5

The point (6, 3) on $f(x) = \sqrt{(x - 2)} + 1$ gives which point on f^{-1}?

A. (6, 3)

B. (3, 6)

C. (−6, −3)

D. (3, −6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q4/5

Why does the inverse of $f(x) = \sqrt{(x - 2)} + 1$ need the domain $x \geq 1$?

A. Because $f(x) = \sqrt{(x - 2)} + 1$ is undefined for $x < 1$, so its inverse is undefined there too.

B. Because every inverse of a square root function has domain $x \geq 1$, whatever the shifts are.

C. The range of f is $y \geq 1$, so f^{-1} has domain $x \geq 1$; otherwise $(x - 1)^2 + 2$ is not one-to-one.

D. Because the inverse of f is $\sqrt{(x - 1)} + 2$, and $\sqrt{(x - 1)}$ is undefined for $x < 1$, so the inverse needs $x \geq 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q5/5

SET 47 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph piecewise-defined and absolute value functions

MAIN	Algebra II main question	**HSF-IF.C.7.b**
BACKWARD 1	Evaluate and graph simple piecewise functions	**HSF-IF.C.7.b**
BACKWARD 2	Absolute value as distance	**6.NS.C.7**
BACKWARD 3	Open and closed endpoints	**6.EE.B.8**
FORWARD 1	Write piecewise functions from graphs and contexts	**HSF-IF.C.7.b**
FORWARD 2	Describe absolute value graphs as transformations, and identify…	**HSF-BF.B.3**

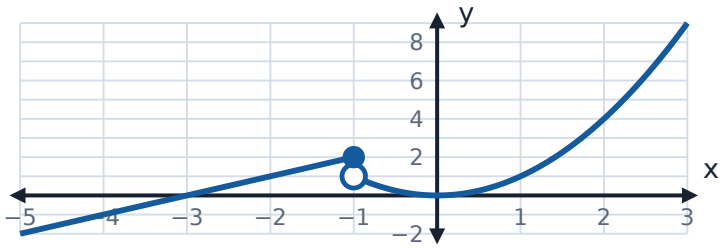
SET 47 • HSF-IF.C.7.b

MAIN

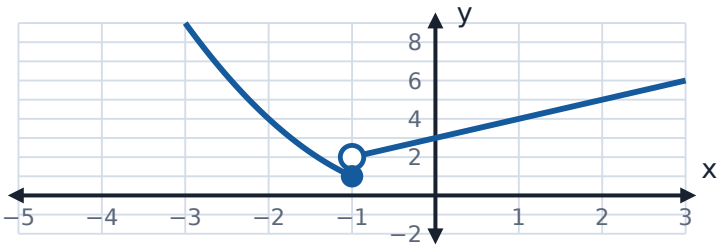
Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

Which graph shows $f(x) = x + 3$ for $x < -1$, and $f(x) = x^2$ for $x \geq -1$?

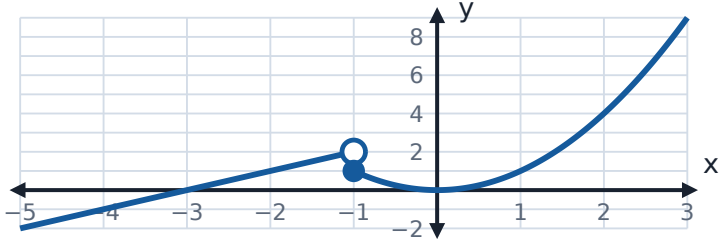
A



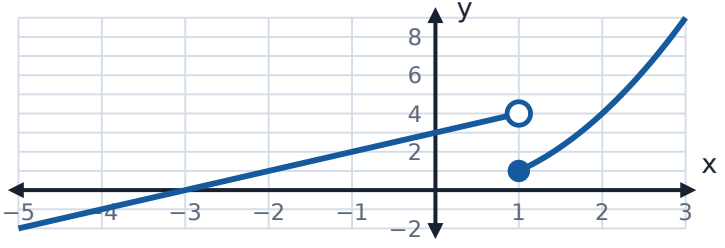
B



C



D



A. Graph A

B. Graph B

C. Graph C

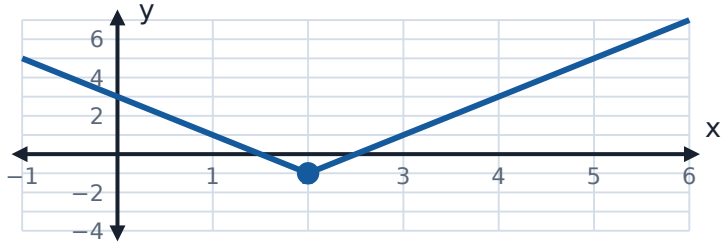
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-M-Q1**

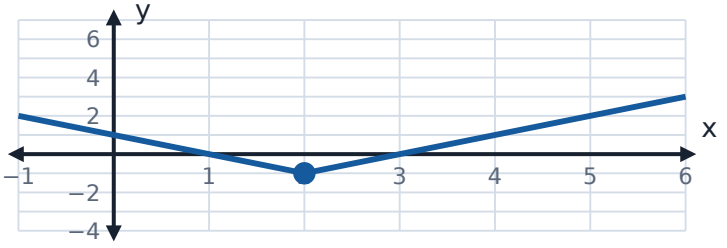
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q1/5

Which graph shows $y = |2x - 4| - 1$?

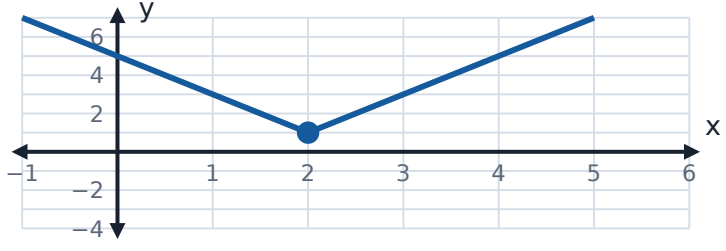
A



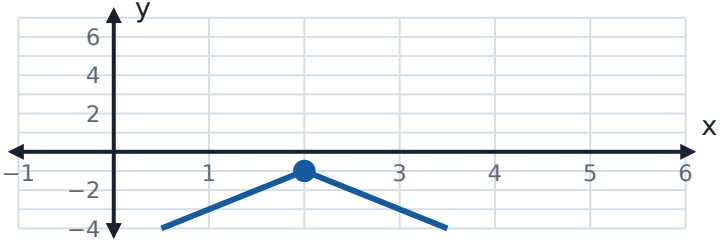
B



C



D



A. Graph A

B. Graph B

C. Graph C

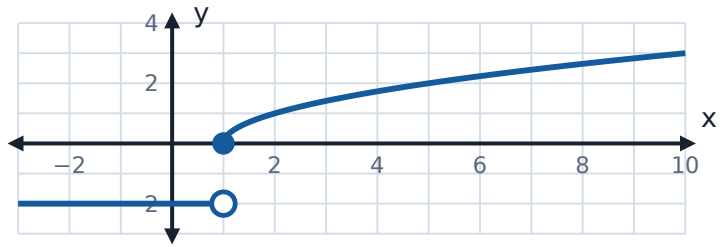
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-M-Q2**

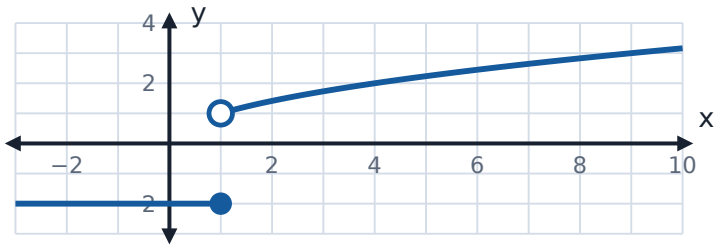
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q2/5

Which graph shows f(x) = −2 for x ≤ 1, and f(x) = √(x − 1) for x > 1?

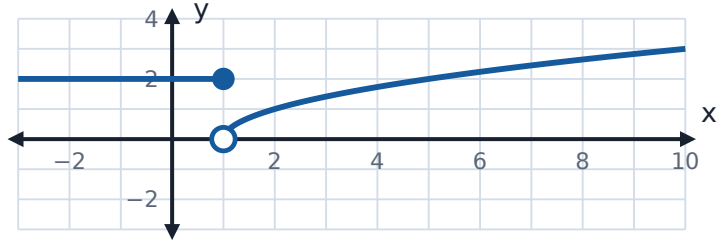
A



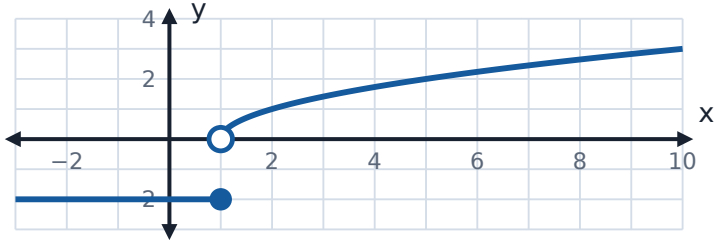
B



C



D



A. Graph A

B. Graph B

C. Graph C

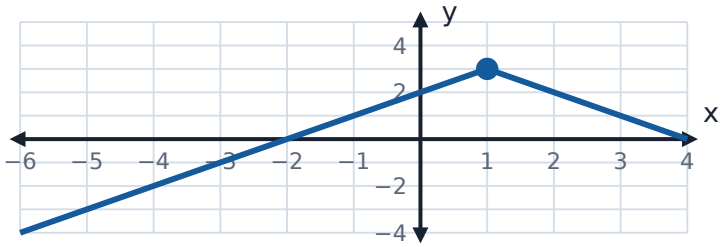
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-M-Q3**

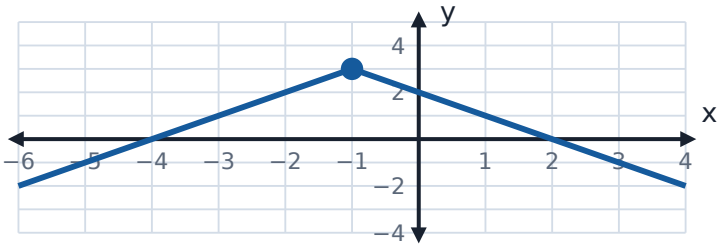
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q3/5

Which graph shows $y = -|x + 1| + 3$?

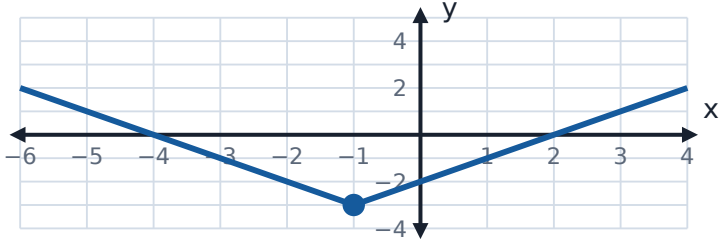
A



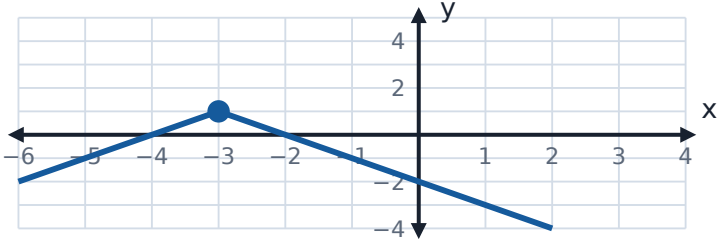
B



C



D



A. Graph A

B. Graph B

C. Graph C

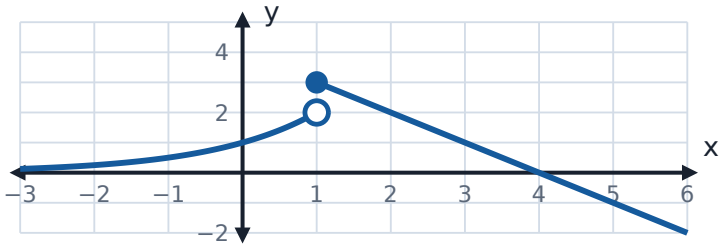
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-M-Q4**

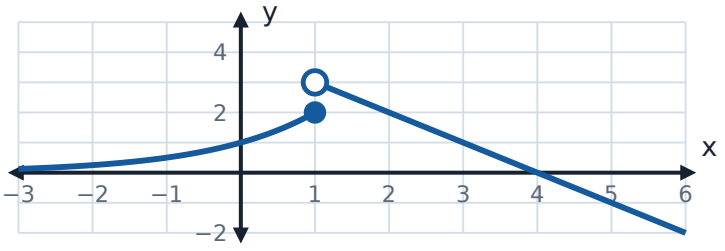
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q4/5

Which graph shows $f(x) = 2^x$ for $x < 1$, and $f(x) = 4 - x$ for $x \geq 1$?

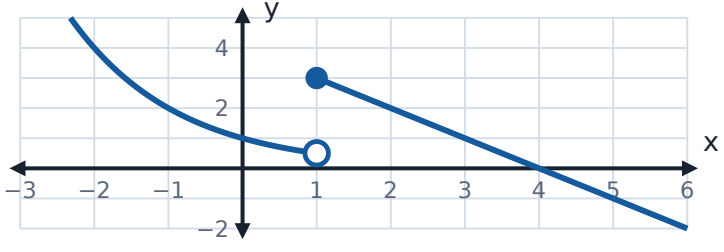
A



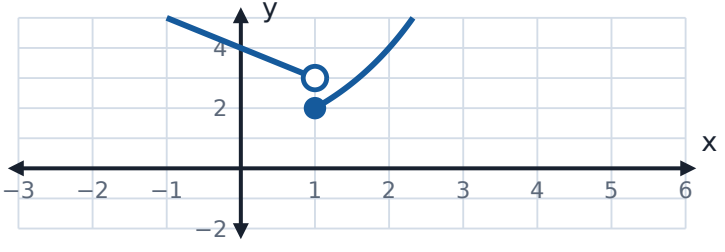
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q5/5

SET 47 • HSF-IF.C.7.b

BACKWARD 1

Evaluate and graph simple piecewise functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

$f(x) = 2$ for $x < 0$ and $f(x) = x$ for $x \geq 0$. Find $f(-3)$ and $f(5)$.

$f(-3) =$ _____

$f(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q1/5

$f(x) = 2$ for $x < 0$ and $f(x) = x$ for $x \geq 0$. Find $f(0)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q2/5

True or false?

A piecewise function uses different rules on different parts of its domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q3/5

For $f(x) = x + 3$ $(x < -1)$, x^2 $(x \geq -1)$, $f(-1)$ is

A. 3

B. 2

C. -1

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q4/5

$f(x) = x + 3$ for $x < -1$, and $f(x) = x^2$ for $x \geq -1$. Find $f(-2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q5/5

SET 47 • HSF-IF.C.7.b

BACKWARD 2

Absolute value as distance • Grade 6 • The Number System • 6.NS.C.7

Find $|-5|$ and $|3|$.

$|-5| =$ _____

$|3| =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q1

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q1/5

Find $|2(1) - 4| - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q2

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q2/5

True or false?

$|x|$ can be negative for negative x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q3/5

$|x + 1| = 0$ when

A. $x = 0$

B. $x = -1$

C. $x = 1$

D. never

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q4/5

Find $-|0 + 1| + 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • HSF-IF.C.7.b

BACKWARD 3

Open and closed endpoints • Grade 6 • Expressions & Equations • 6.EE.B.8

On a number line, is $x < -1$ drawn with an open or a closed circle at -1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q1/5

Is $x \geq 1$ drawn with an open or a closed circle at 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q2/5

True or false?

An open circle means the endpoint is included.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q3/5

On a number line, $x \leq 1$ is drawn with

A. an open circle and shading right

B. a closed circle at 1 and shading right

C. an open circle at 1 and shading left

D. a closed circle at 1 and shading left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q4/5

Write the inequality for "all numbers greater than 1".

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q5/5

SET 47 • HSF-IF.C.7.b

FORWARD 1

Write piecewise functions from graphs and contexts • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

A shipping company charges $5 for packages up to 2 lb and $5 plus $1.50 per pound over 2 lb. Which gives the cost C(w) for a package of w pounds?

A. $C(w) = 5$ for $0 < w \leq 2$, and $C(w) = 1.5(w - 2)$ for $w > 2$

B. $C(w) = 5$ for $0 < w \leq 2$, and $C(w) = 5 + 1.5(w - 2)$ for $w > 2$

C. $C(w) = 5$ for $0 < w < 2$, and $C(w) = 5 + 1.5w$ for $w \geq 2$

D. $C(w) = 5 + 1.5w$ for every $w > 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q1/5

Which writes $|x - 3|$ as a piecewise function without absolute value bars?

A. $3 - x$ for $x < 3$, and $x - 3$ for $x \geq 3$

B. $x - 3$ for every x

C. $3 - x$ for $x < 0$, and $x - 3$ for $x \geq 0$

D. $x - 3$ for $x < 3$, and $3 - x$ for $x \geq 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q2/5

True or false?

A piecewise function can have a jump and still be a function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q3/5

Which piecewise function is continuous at $x = 1$?

A. 2^x for $x < 1$, $4 - x$ for $x \geq 1$

B. x^2 for $x < 1$, $2x - 1$ for $x \geq 1$

C. 2 for $x < 1$, 3 for $x \geq 1$

D. x for $x < 1$, $x + 1$ for $x \geq 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q4/5

Find k so that $f(x) = kx$ for $x < 2$ and $f(x) = x^2$ for $x \geq 2$ has no jump at 2.

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q5/5

SET 47 • HSF-IF.C.7.b

FORWARD 2

Describe absolute value graphs as transformations, and identify symmetry • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

Which describes $y = |2x - 4| - 1$ as a transformation of $y = |x|$?

A. $y = 2|x - 4| - 1$: shift right 4, stretch vertically by 2, shift down 1

B. $y = 2|x - 2| - 1$: shift right 2, stretch vertically by 2, shift down 1

C. $y = 2|x + 2| - 1$: shift left 2, stretch vertically by 2, shift down 1

D. $y = |x - 2| - 1$: shift right 2, shift down 1, no vertical stretch

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q1/5

An absolute value function $y = a|x + 1| + 3$ has vertex $(-1, 3)$, opens down, and passes through $(2, 0)$. Find a.

$a =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q2/5

True or false?

$y = |x|$ is an odd function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q3/5

$y = |x - 2|$ is symmetric about

A. the origin

B. $y = 2$

C. $x = 0$

D. $x = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q4/5

$f(x) = -|x + 1| + 3$. Find $f(1)$ and $f(-1)$, and decide whether f is even.

$f(1) =$ _____

$f(-1) =$ _____

Is f even? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q5/5

SET 85 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph step functions

MAIN	Algebra II main question	**HSF-IF.C.7.b**
BACKWARD 1	Evaluate step functions	**HSF-IF.C.7.b**
BACKWARD 2	Included and excluded endpoints of intervals	**6.EE.B.8**
BACKWARD 3	Integers just below a number	**6.NS.C.7**
FORWARD 1	Write and interpret step-function models	**HSF-IF.C.7.b**
FORWARD 2	Transform step functions	**HSF-BF.B.3**

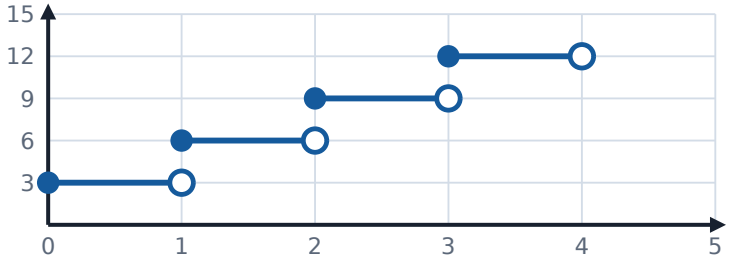
SET 85 • HSF-IF.C.7.b

MAIN

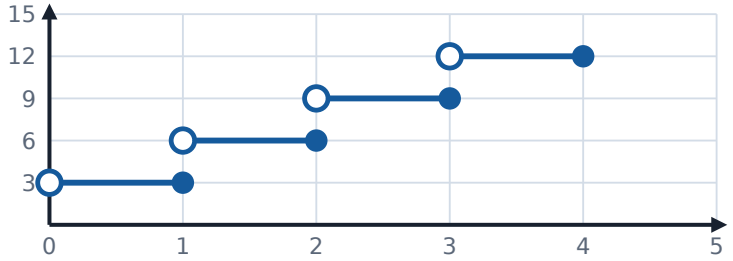
Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

A parking garage charges $3 for each hour or part of an hour, so C(t) = 3[t] dollars for 0 < t ≤ 4 hours, where [t] is the least integer greater than or equal to t. Which graph shows C?

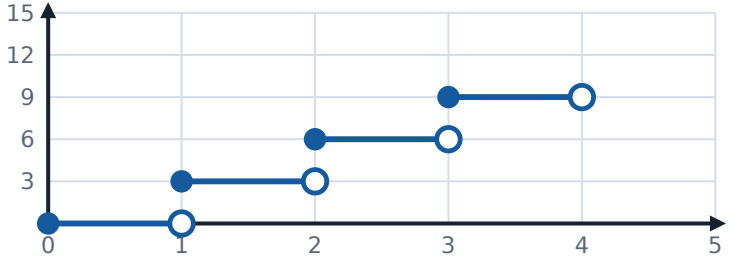
A



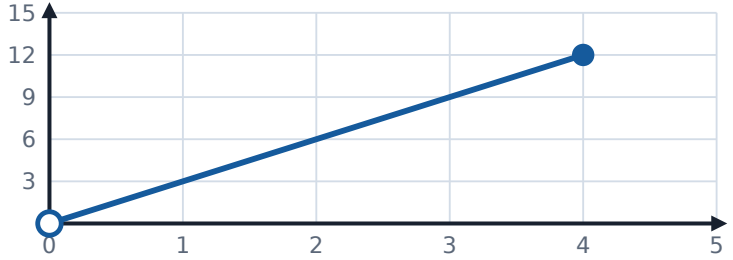
B



C



D



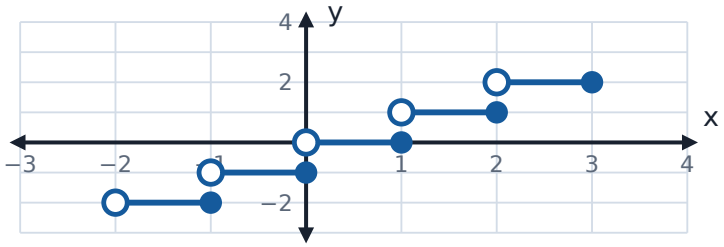
A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S85-M-Q1**

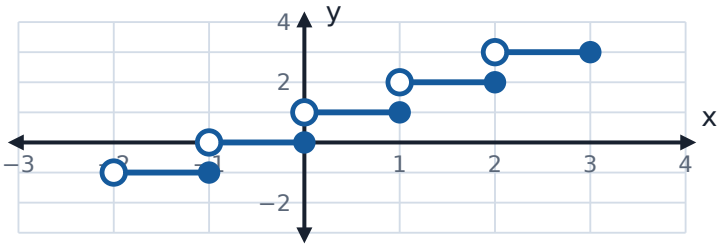
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • MAIN • Q1/5

$f(x) = \lfloor x \rfloor$, the greatest integer less than or equal to x, for $-2 \le x < 3$. Which graph shows f?

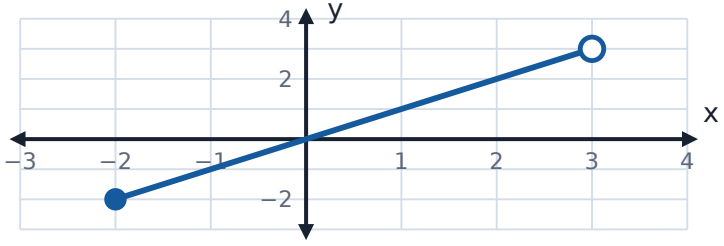
A



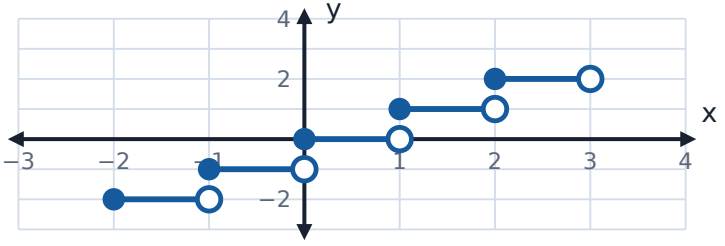
B



C



D



A. Graph A

B. Graph B

C. Graph C

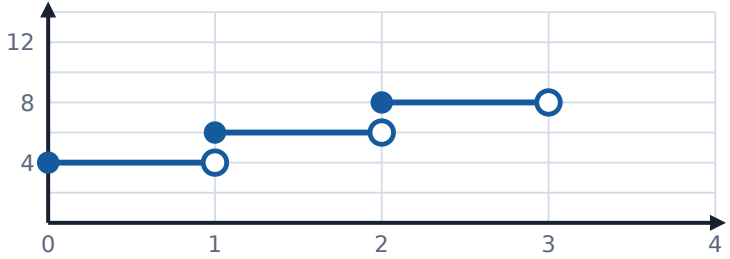
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S85-M-Q2**

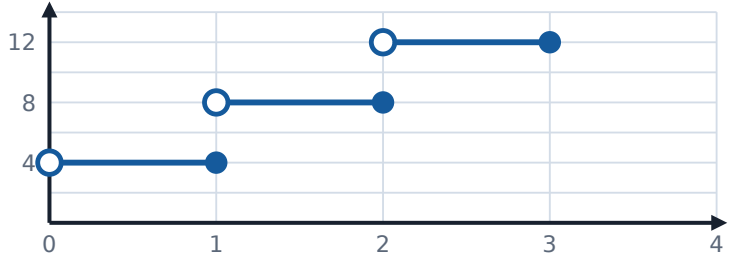
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • MAIN • Q2/5

A courier charges $4 for a package up to 1 lb, $6 for more than 1 lb up to 2 lb, and $8 for more than 2 lb up to 3 lb. Which graph shows the cost for weights $0 < w \leq 3$?

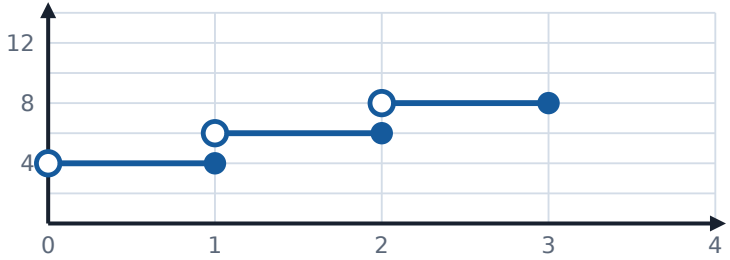
A



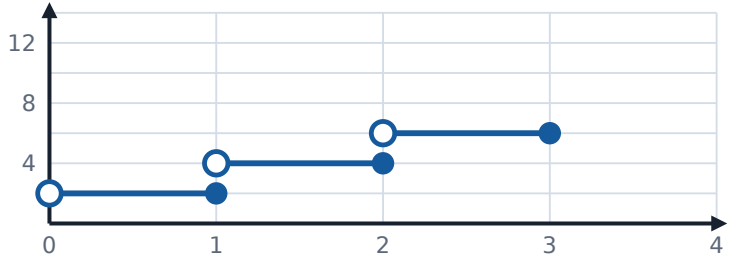
B



C



D



A. Graph A

B. Graph B

C. Graph C

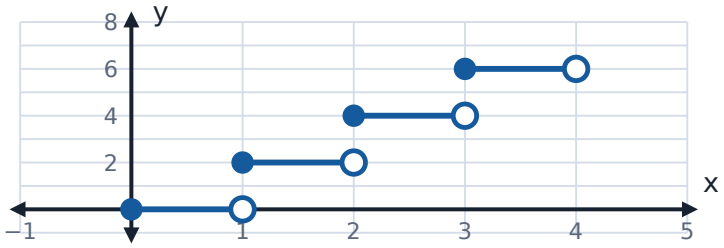
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S85-M-Q3**

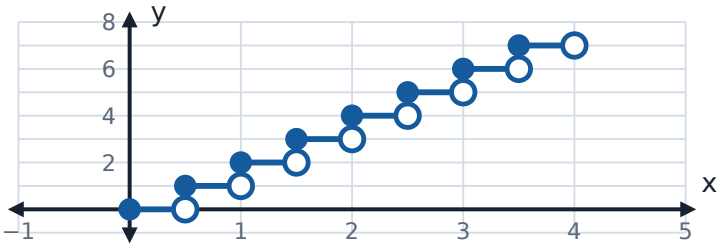
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • MAIN • Q3/5

$g(x) = 2\lfloor x \rfloor$ for $0 \le x < 4$, where $\lfloor x \rfloor$ is the greatest integer less than or equal to x. Which graph shows g?

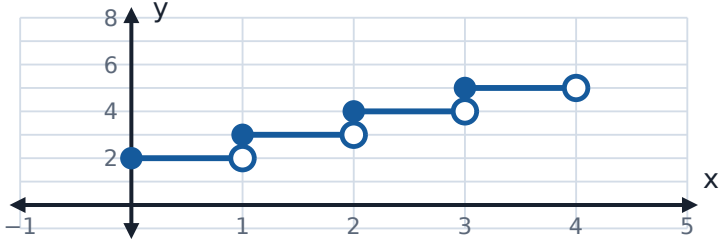
A



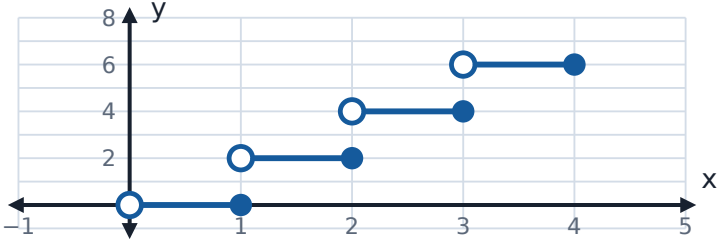
B



C



D



A. Graph A

B. Graph B

C. Graph C

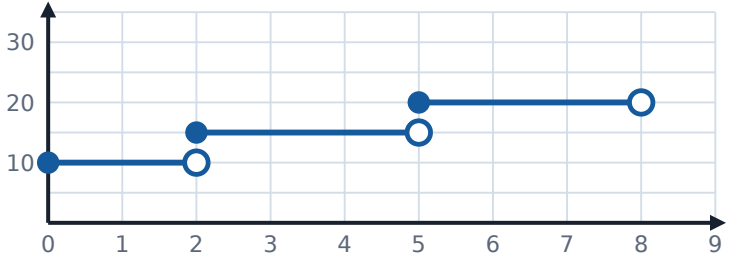
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S85-M-Q4**

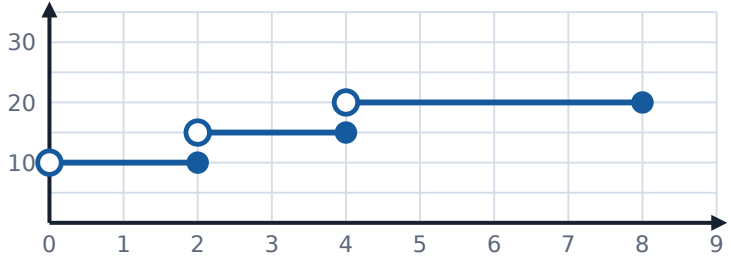
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • MAIN • Q4/5

A phone plan costs $10 for up to 2 GB of data, $15 for more than 2 GB up to 5 GB, and $20 for more than 5 GB up to 8 GB. Which graph shows the monthly cost for data use $0 < d \leq 8$?

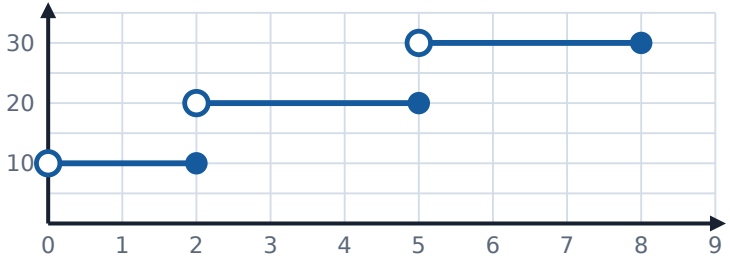
A



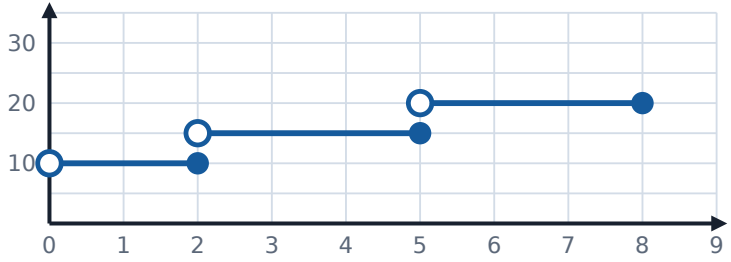
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S85-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • MAIN • Q5/5

SET 85 • HSF-IF.C.7.b

BACKWARD 1

Evaluate step functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

$\lfloor x \rfloor$ is the greatest integer less than or equal to x. Find $\lfloor 2.7 \rfloor$ and $\lfloor -1.3 \rfloor$.

$\lfloor 2.7 \rfloor =$ _____

$\lfloor -1.3 \rfloor =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • BACKWARD 1 • Q1/5

A parking garage charges $3 for each hour or part of an hour. How many dollars does it charge for 2.5 hours?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • BACKWARD 1 • Q2/5

True or false?

$\lfloor x \rfloor$, the greatest integer less than or equal to x, equals x exactly when x is an integer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • BACKWARD 1 • Q3/5

$\lfloor x \rfloor$ is the greatest integer less than or equal to x. What is $\lfloor -0.5 \rfloor$?

A. 1

B. 0

C. -1

D. -0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • BACKWARD 1 • Q4/5

A parking garage charges $3 for each hour or part of an hour. How many dollars does it charge for exactly 4 hours?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • BACKWARD 1 • Q5/5

SET 85 • HSF-IF.C.7.b

BACKWARD 2

Included and excluded endpoints of intervals • Grade 6 • Expressions & Equations • 6.EE.B.8

Is 2 included in the interval $1 < t \leq 2$? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 85 • BACKWARD 2 • Q1/5

Is 1 included in the interval $1 < t \leq 2$? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 85 • BACKWARD 2 • Q2/5

True or false?

On a graph, a closed dot at the end of a segment means that endpoint is included.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 85 • BACKWARD 2 • Q3/5

Which inequality describes all t greater than 2 and at most 3?

A. $2 < t < 3$

B. $2 \leq t < 3$

C. $2 \leq t \leq 3$

D. $2 < t \leq 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 85 • BACKWARD 2 • Q4/5

A horizontal segment from (1, 6) to (2, 6) has an open circle at (1, 6) and a closed dot at (2, 6). Which inputs does it represent?

A. $1 < x \leq 2$

B. $1 \leq x < 2$

C. $1 < x < 2$

D. $1 \leq x \leq 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 85 • BACKWARD 2 • Q5/5

SET 85 • HSF-IF.C.7.b

BACKWARD 3

Integers just below a number • Grade 6 • The Number System • 6.NS.C.7

What is the greatest integer that is less than or equal to 3.2?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B3-Q1

Grade 6 • The Number System • 6.NS.C.7 • Set 85 • BACKWARD 3 • Q1/5

What is the greatest integer that is less than or equal to -3.2?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B3-Q2

Grade 6 • The Number System • 6.NS.C.7 • Set 85 • BACKWARD 3 • Q2/5

True or false?

On a number line, -4 is to the left of -3.2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B3-Q3

Grade 6 • The Number System • 6.NS.C.7 • Set 85 • BACKWARD 3 • Q3/5

Which integer is the greatest integer less than or equal to -0.1?

A. -0.1

B. 1

C. -1

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B3-Q4

Grade 6 • The Number System • 6.NS.C.7 • Set 85 • BACKWARD 3 • Q4/5

What is the least integer that is greater than or equal to 2.1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B3-Q5

Grade 6 • The Number System • 6.NS.C.7 • Set 85 • BACKWARD 3 • Q5/5

SET 85 • HSF-IF.C.7.b

FORWARD 1

Write and interpret step-function models • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

A parking garage charges $3 for each hour or part of an hour, up to a maximum of $15 per day. $\lceil t \rceil$ is the least integer greater than or equal to t. Which rule gives the cost C(t) for $0 < t \leq 24$ hours?

A. $C(t) = 3\lceil t \rceil$ for $0 < t \leq 5$, and $C(t) = 15$ for $5 < t \leq 24$

B. $C(t) = 3\lceil t \rceil$ for every t with $0 < t \leq 24$

C. $C(t) = 3t$ for $0 < t \leq 5$, and $C(t) = 15$ for $5 < t \leq 24$

D. $C(t) = 3\lfloor t \rfloor$ for $0 < t \leq 5$, and $C(t) = 15$ for $5 < t \leq 24$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S85-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • FORWARD 1 • Q1/5

Which function has a jump of 2 at every integer and steps of width 1?

A. $y = \lfloor 2x \rfloor$

B. $y = \lfloor x \rfloor + 2$

C. $y = \lfloor x/2 \rfloor$

D. $y = 2\lfloor x \rfloor$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • FORWARD 1 • Q2/5

True or false?

For $f(x) = \lfloor x \rfloor$, the average rate of change over any interval inside one step, such as $1.2 \leq x \leq 1.8$, is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • FORWARD 1 • Q3/5

For $f(x) = \lfloor x \rfloor$, how many solutions does $f(x) = 3$ have?

A. none

B. infinitely many: every x with $3 \leq x < 4$

C. exactly one: $x = 3$

D. exactly two: $x = 3$ and $x = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S85-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • FORWARD 1 • Q4/5

Postage is $0.70 for a letter up to 1 oz plus $0.25 for each additional ounce or part of an ounce. What is the postage, in dollars, for a 2.3-oz letter? Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • FORWARD 1 • Q5/5

SET 85 • HSF-IF.C.7.b

FORWARD 2

Transform step functions • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

How is the graph of $y = 3\lfloor x \rfloor$ related to the graph of $y = \lfloor x \rfloor$?

A. shifted right 3: each step moves 3 units to the right

B. compressed horizontally by 3: each step is 1/3 as wide

C. stretched vertically by 3: each step is 3 times as high

D. shifted up 3: each step is 3 units higher than before

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S85-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 85 • FORWARD 2 • Q1/5

How is the graph of $y = \lfloor x/2 \rfloor$ related to the graph of $y = \lfloor x \rfloor$?

A. stretched horizontally by 2: each step is 2 units wide

B. shifted right 2: each step moves 2 units to the right

C. stretched vertically by 2: each step is 2 times as high

D. compressed horizontally by 2: each step is 1/2 unit wide

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S85-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 85 • FORWARD 2 • Q2/5

True or false?

The graphs of $y = \lfloor x \rfloor + 1$ and $y = \lfloor x + 1 \rfloor$ are the same.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 85 • FORWARD 2 • Q3/5

Which function has steps of width 1 at heights ..., −2, 0, 2, 4, ..., with a closed dot at the left end of each step?

A. $y = 2\lceil x \rceil$

B. $y = \lfloor 2x \rfloor$

C. $y = \lfloor x \rfloor + 2$

D. $y = 2\lfloor x \rfloor$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 85 • FORWARD 2 • Q4/5

$g(x) = 2\lfloor x - 1 \rfloor + 3$, where $\lfloor x \rfloor$ is the greatest integer less than or equal to x. Find g(2.5).

g(2.5) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 85 • FORWARD 2 • Q5/5

SET 48 • HSF-IF.C.7.c • INTERPRETING FUNCTIONS

Graph polynomial functions showing zeros and end behavior

MAIN	Algebra II main question	**HSF-IF.C.7.c**
BACKWARD 1	Factor out common factors and special products	**HSA-SSE.A.2**
BACKWARD 2	Crossing and touching from multiplicity	**HSA-APR.B.3**
BACKWARD 3	Evaluate polynomials at test points	**HSF-IF.A.2**
FORWARD 1	Locate relative extrema of polynomial graphs	**HSF-IF.C.7.c**
FORWARD 2	Interpret polynomial graphs in context	**HSF-IF.B.4**

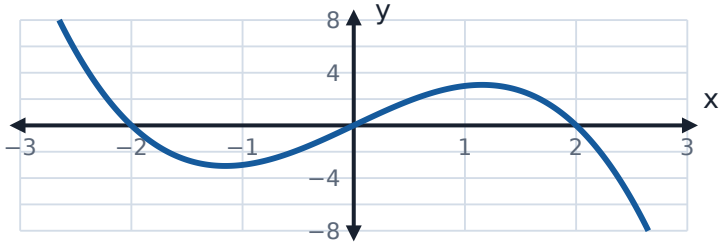
SET 48 • HSF-IF.C.7.c

MAIN

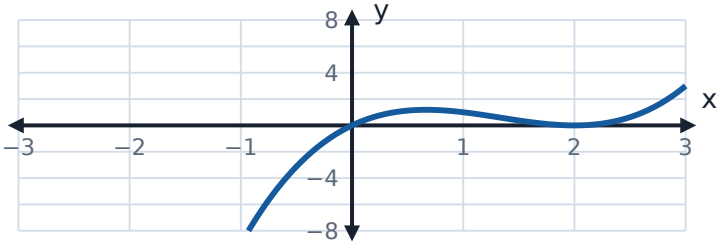
Algebra II main question • High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c

Which graph shows $y = x^3 - 4x$?

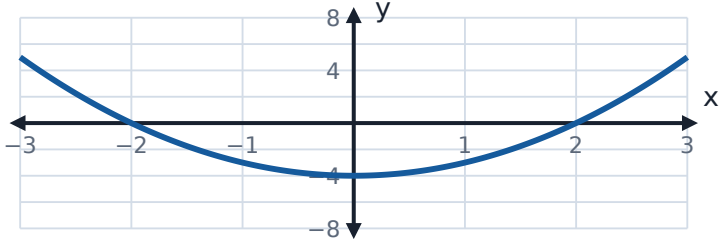
A



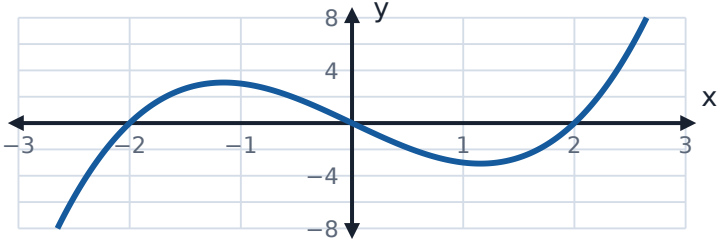
B



C



D



A. Graph A

B. Graph B

C. Graph C

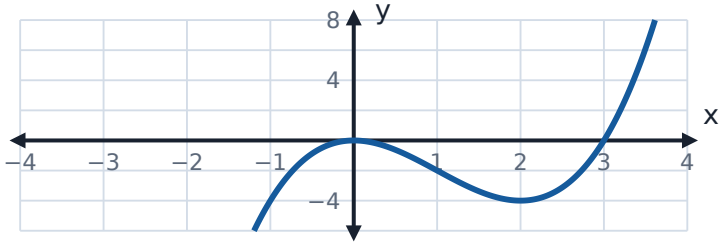
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-M-Q1**

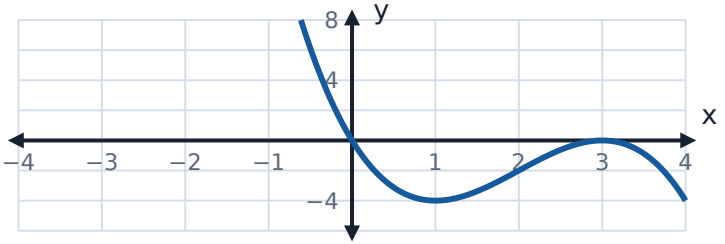
High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q1/5

Which graph shows $y = -x^3 + 3x^2$?

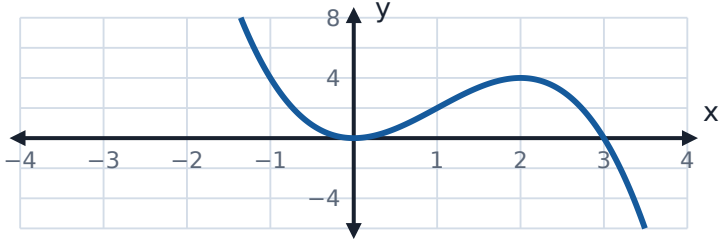
A



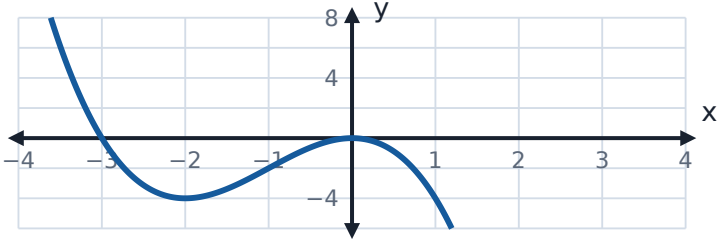
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

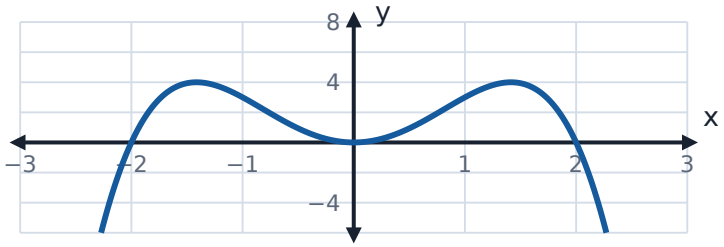
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

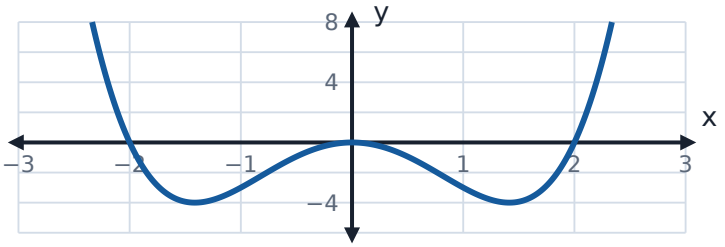
High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q2/5

Which graph shows $y = x^4 - 4x^2$?

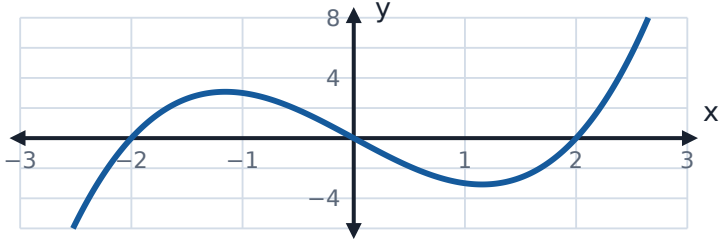
A



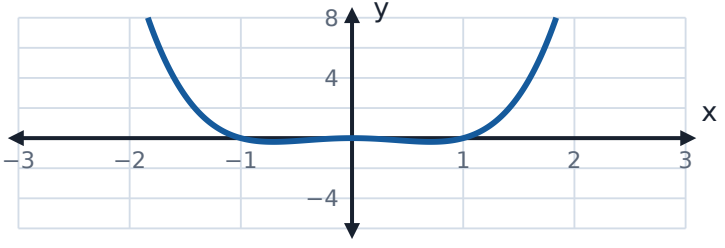
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

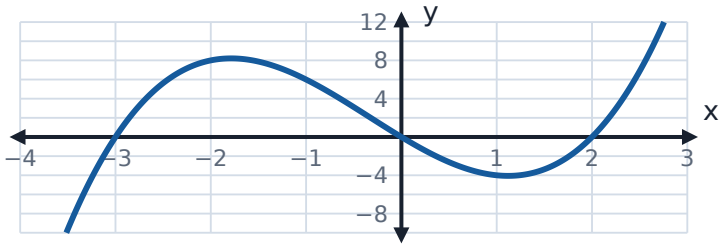
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

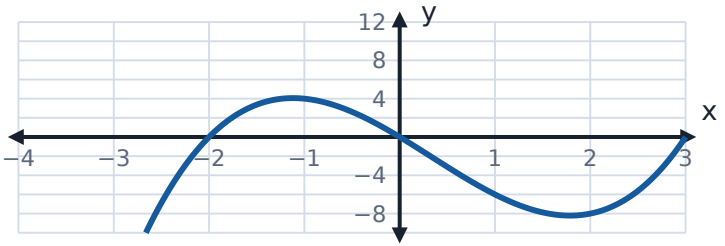
High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q3/5

Which graph shows $y = x^3 + x^2 - 6x$?

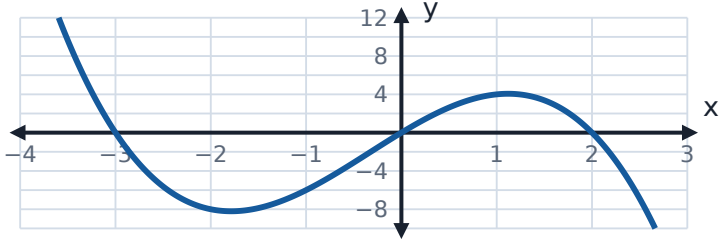
A



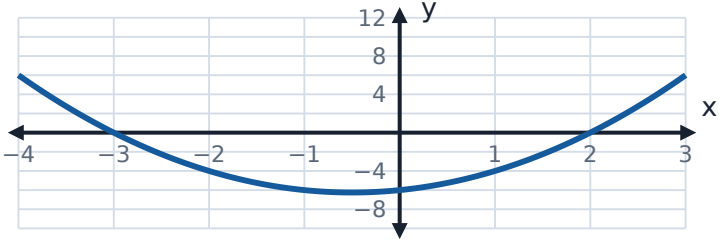
B



C



D



A. Graph A

B. Graph B

C. Graph C

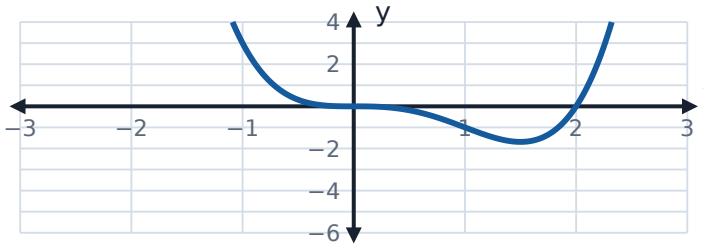
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-M-Q4**

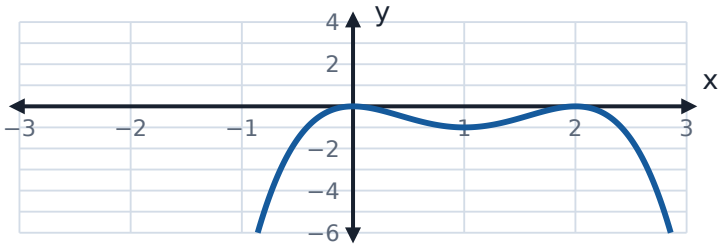
High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q4/5

Which graph shows $y = -x^4 + 2x^3$?

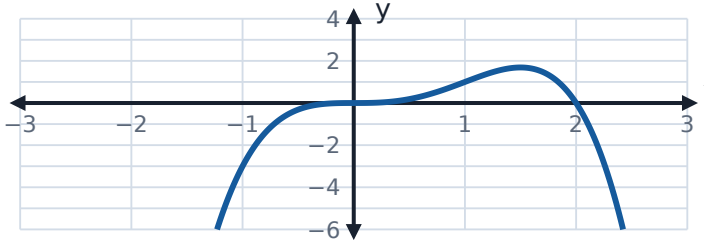
A



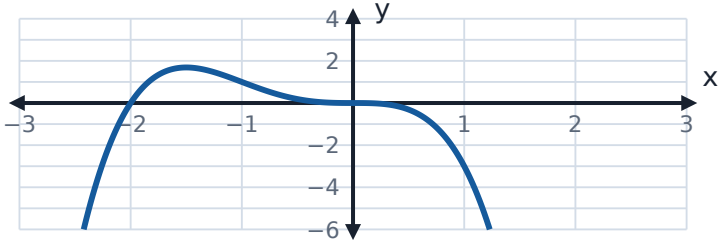
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q5/5

SET 48 • HSF-IF.C.7.c

BACKWARD 1

Factor out common factors and special products • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^3 - 4x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q1/5

Factor $-x^3 + 3x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q2/5

True or false?

$$x^4 - 4x^2 = x^2(x^2 - 4) = x^2(x - 2)(x + 2).$$

A. True

B. False

$x^3 + x^2 - 6x$ equals

A. $(x + 3)(x - 2)$

B. $x(x + 3)(x - 2)$

C. $x(x - 3)(x + 2)$

D. $x(x + 6)(x - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q4/5

Factor $-x^4 + 2x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • HSF-IF.C.7.c

BACKWARD 2

Crossing and touching from multiplicity • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Does $y = x^2(x - 3)$ cross or touch the x-axis at 0?
Answer Crosses or Touches.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q1/5

Does the graph of $y = x^3(x - 2)$ cross the x-axis at $x = 0$?

A. Yes, as a straight line (multiplicity 1).

B. Yes, and it flattens there (odd multiplicity 3).

C. No, $x = 0$ is not a zero.

D. No, it only touches there (even multiplicity).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q2/5

True or false?

A zero of odd multiplicity is a crossing point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q3/5

The graph of $y = (x + 2)^2(x - 1)$ at $x = -2$

A. crosses

B. touches

C. has an asymptote

D. is undefined

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q4/5

Which describes the end behavior of $y = -x^4 + 2x^3$?

A. Rises at both ends

B. Rises to the left and falls to the right

C. Falls to the left and rises to the right

D. Falls at both ends

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • HSF-IF.C.7.c

BACKWARD 3

Evaluate polynomials at test points • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Evaluate $x^3 - 4x$ at $x = -1$ and at $x = 1$.

At $x = -1$: _____

At $x = 1$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q1/5

Evaluate $-x^3 + 3x^2$ at $x = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q2/5

True or false?

$x^4 - 4x^2$ at $x = 1$ equals 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q3/5

$x^3 + x^2 - 6x$ at $x = -2$ equals

A. 8

B. 0

C. -8

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q4/5

Evaluate $-x^4 + 2x^3$ at $x = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q5/5

SET 48 • HSF-IF.C.7.c

FORWARD 1

Locate relative extrema of polynomial graphs • High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c

Use a table with steps of 0.1 (or technology) to estimate the relative maximum of $y = x^3 - 4x$. Give the x-coordinate to the nearest tenth and the y-value to the nearest hundredth. A calculator may be used.

$x \approx$ _____

$y \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q1

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q1/5

Use technology or a table to estimate the relative minimum of $y = x^3 + x^2 - 6x$ on $0 < x < 2$. Give the x-coordinate to the nearest tenth and the y-value to the nearest hundredth. A calculator may be used.

$x \approx$ _____

$y \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q2

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q2/5

True or false?

A polynomial of degree 4 always has exactly 3 turning points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q3/5

How many turning points can $y = x^4 - 4x^2$ have?

A. 1

B. 4

C. 3

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q4/5

Why does the graph of $y = x^3 - 4x$ have a turning point between $x = 0$ and $x = 2$?

A. Because y is positive between 0 and 2 (for example $y(1) = 3$), so the graph must come back down to 0 at $x = 2$.

B. Because the graph crosses the x-axis at $x = 1$, halfway between 0 and 2, and a graph turns at every crossing.

C. y is 0 at $x = 0$ and at $x = 2$ and negative between them (for example $y(1) = -3$), so the graph must turn back up somewhere between them.

D. Because every cubic with a positive leading coefficient turns at $x = 1$, which lies between 0 and 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q5/5

SET 48 • HSF-IF.C.7.c

FORWARD 2

Interpret polynomial graphs in context • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

A box's volume is $V(x) = x(12 - 2x)^2$ in³, where x is the side (in) of the squares cut from a 12-in square sheet. Which interval of x is meaningful, and what is the maximum volume there (it occurs at x = 2)?

Meaningful interval: _____

Maximum volume (in³): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q1/5

A company's profit (thousands of dollars) is $P(x) = -x^3 + 9x^2 - 15x$ for x hundred items, $0 \le x \le 8$. For which x is the profit positive? Give the endpoints to the nearest hundredth.

Profit positive for: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q2/5

True or false?

A zero of a profit model is a break-even point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q3/5

A population model $P(t) = -t^3 + 6t^2 + 15t + 100$ rises then falls. The relative maximum means

A. the largest population before decline

B. the starting population

C. the end behavior

D. the time the population is 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q4/5

Why does the end behavior of $V(x) = x(12 - 2x)^2$ not matter for the box made from a 12-in sheet?

A. The context uses only $0 < x < 6$, so the behavior for large $|x|$ is never used.

B. Because $V(x) = x(12 - 2x)^2$ has no end behavior at all.

C. Because the volume of the box is always 128 cubic inches.

D. Because x must be negative for the box, so large positive x is never used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q5/5

SET 49 • HSF-IF.C.7.e • INTERPRETING FUNCTIONS

Graph exponential functions with intercepts, asymptote, and end behavior

MAIN	Algebra II main question	**HSF-IF.C.7.e**
BACKWARD 1	Graph basic exponential functions	**HSF-IF.C.7.e**
BACKWARD 2	Zero and negative exponents	**8.EE.A.1**
BACKWARD 3	Vertical and horizontal shifts and reflections	**HSF-BF.B.3**
FORWARD 1	Compare and describe exponential graphs	**HSF-IF.C.7.e**
FORWARD 2	Graph the inverse of an exponential function	**HSF-BF.B.4.a**

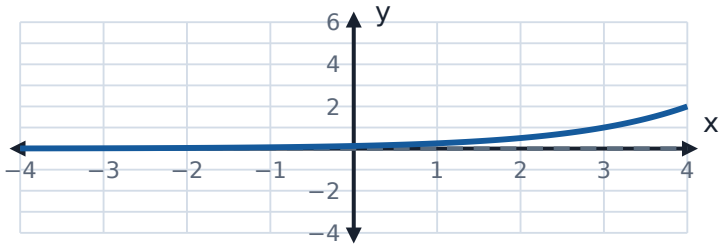
SET 49 • HSF-IF.C.7.e

MAIN

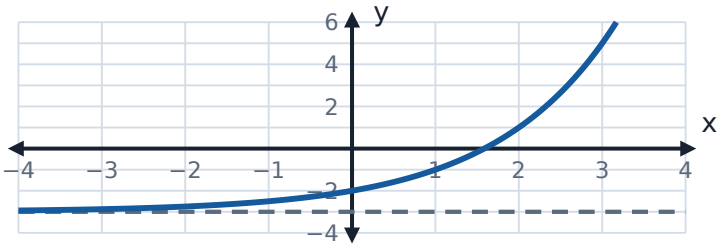
Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

Which graph shows $y = 2^x - 3$?

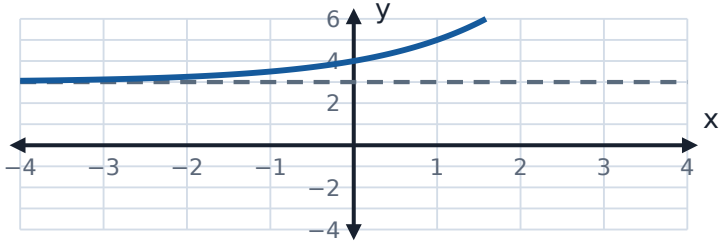
A



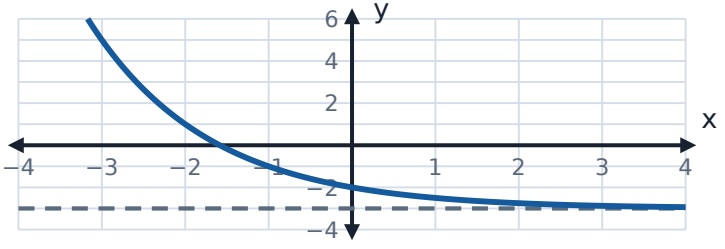
B



C



D



A. Graph A

B. Graph B

C. Graph C

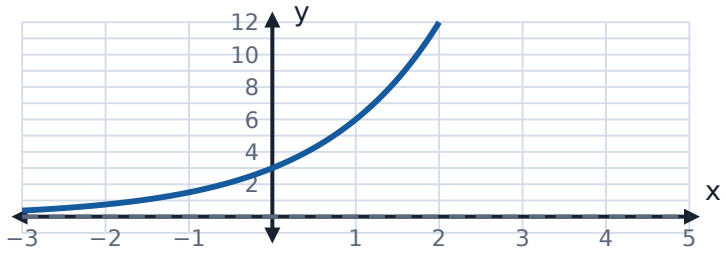
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-M-Q1**

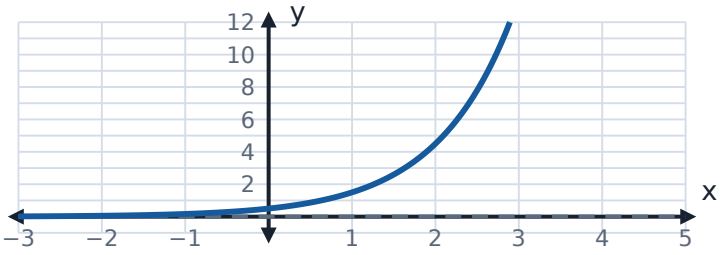
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q1/5

Which graph shows $y = 3(0.5)^x$?

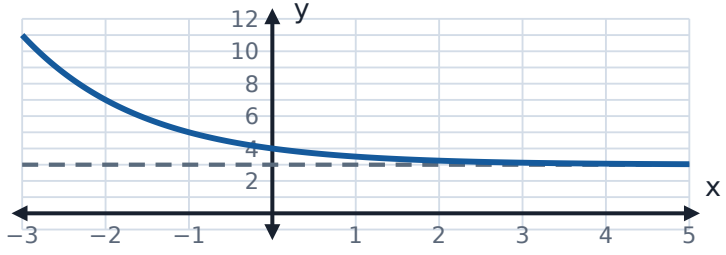
A



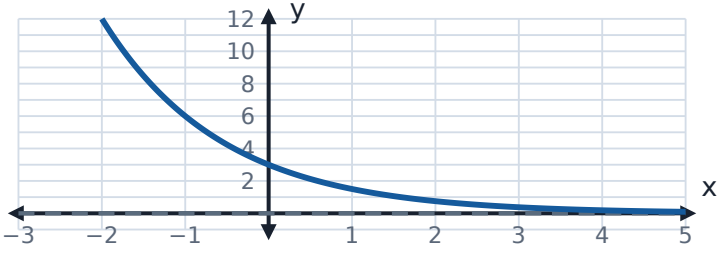
B



C



D



A. Graph A

B. Graph B

C. Graph C

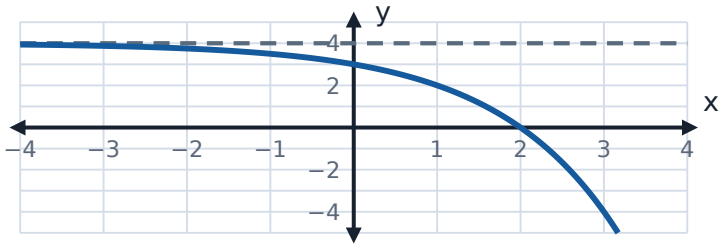
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-M-Q2**

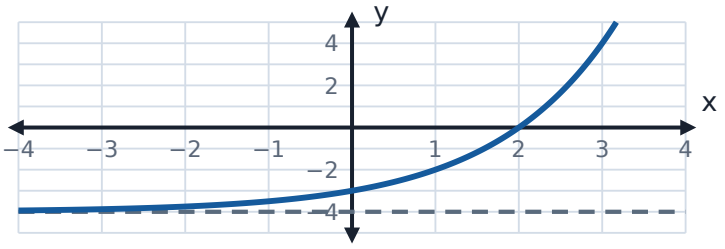
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q2/5

Which graph shows $y = -2^x + 4$?

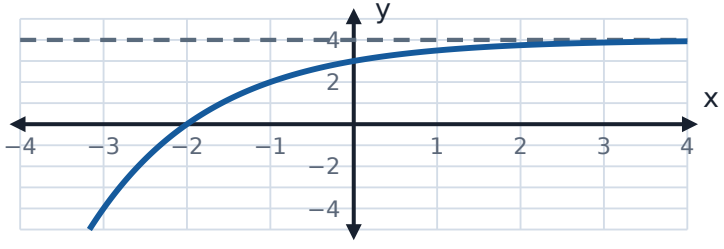
A



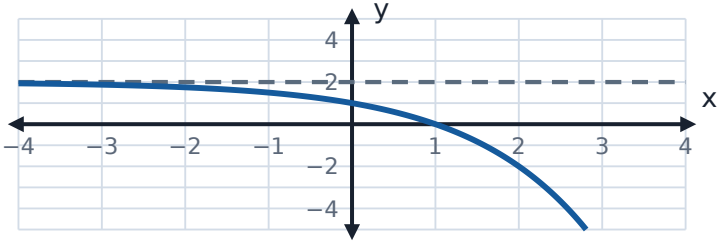
B



C



D



A. Graph A

B. Graph B

C. Graph C

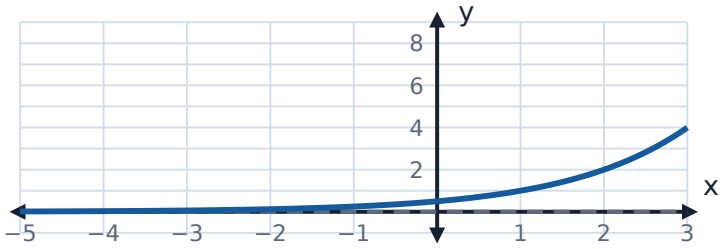
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-M-Q3**

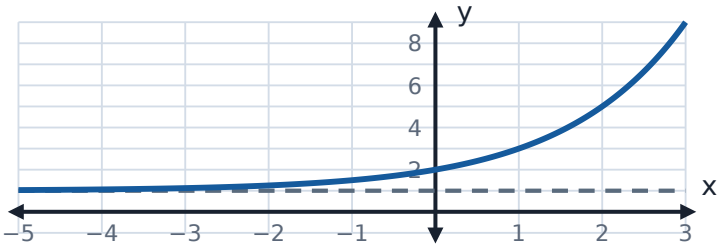
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q3/5

Which graph shows $y = 2^{x+1}$?

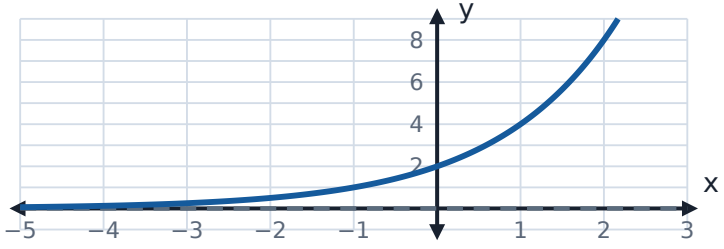
A



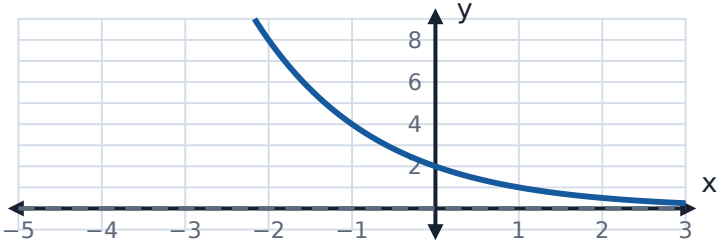
B



C



D



A. Graph A

B. Graph B

C. Graph C

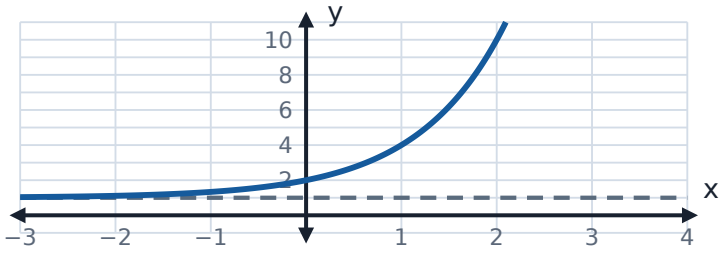
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-M-Q4**

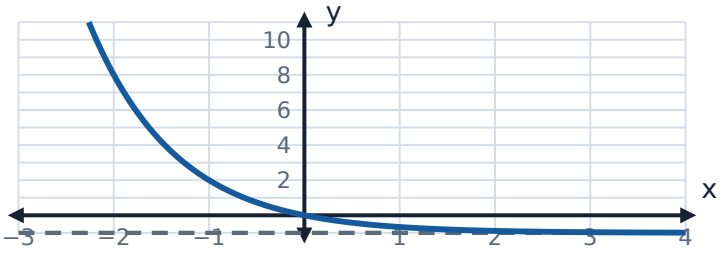
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q4/5

Which graph shows $y = (1/3)^x + 1$?

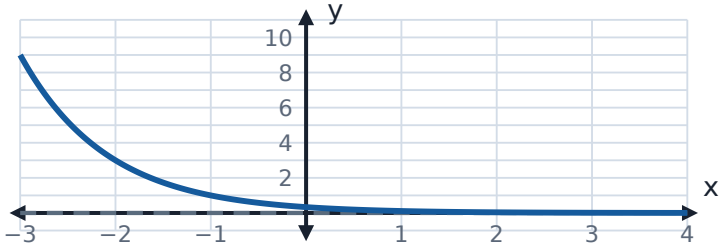
A



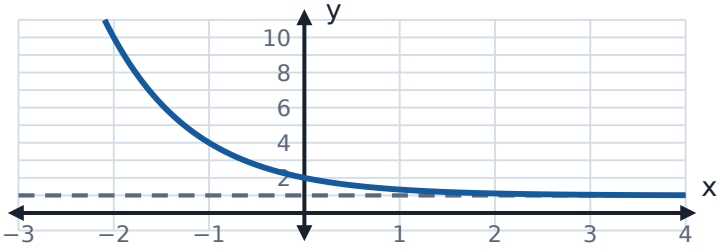
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q5/5

SET 49 • HSF-IF.C.7.e

BACKWARD 1

Graph basic exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

Find $y = 2^x$ at $x = -2, -1, 0, 1$, and 2, in that order.

y-values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q1/5

What is the y-intercept of $y = 3(0.5)^x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q2/5

True or false?

$y = 2^x$ has the horizontal asymptote $y = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q3/5

$y = (0.5)^x$ is

A. linear

B. increasing

C. constant

D. decreasing

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q4/5

Does the graph of $y = 2^x$ have an x-intercept?

A. Yes: at $x = 0$

B. No: $2^x > 0$ for every x, so y is never 0

C. Yes: at $x = -1$

D. No: the graph does not cross the y-axis either

SET 49 • HSF-IF.C.7.e

BACKWARD 2

Zero and negative exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Evaluate 2^{-2}.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q1/5

Evaluate $(1/3)^{-2}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q2/5

True or false?

$b^0 = 0$ for $b \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q3/5

$3(0.5)^{-1}$ equals

A. 6

B. 3

C. -1.5

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q4/5

Evaluate $2^{-1} - 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • HSF-IF.C.7.e

BACKWARD 3

Vertical and horizontal shifts and reflections • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

How is the graph of $y = 2^x - 3$ related to the graph of $y = 2^x$?

A. Shifted left 3

B. Shifted up 3

C. Shifted right 3

D. Shifted down 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q1/5

How is the graph of $y = 2^{x+1}$ related to the graph of $y = 2^{x}$?

A. Shifted left 1

B. Shifted up 1

C. Shifted down 1

D. Shifted right 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q2/5

True or false?

A vertical shift moves the horizontal asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q3/5

$y = -2^x + 4$ has asymptote

A. $y = 4$

B. $y = 0$

C. $y = -4$

D. $x = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q4/5

Give the asymptote of $y = (1/3)^x + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q5/5

SET 49 • HSF-IF.C.7.e

FORWARD 1

Compare and describe exponential graphs • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

Why are the graphs of $y = 2^x$ and $y = (1/2)^x$ reflections of each other over the y-axis?

A. Because 1/2 is less than 1, so $y = (1/2)^x$ decreases while $y = 2^x$ increases.

B. $(1/2)^x = 2^{-x}$, so $y = (1/2)^x$ is $y = 2^x$ with x replaced by $-x$.

C. $(1/2)^x = -2^x$, so $y = (1/2)^x$ is $y = 2^x$ with every output made negative.

D. Because both graphs pass through (0, 1), which lies on the y-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q1/5

$2^{x+1} = 2 \cdot 2^x$. What does this say about two transformations of $y = 2^x$?

A. A shift right by 1 gives the same graph as a vertical stretch by 2.

B. A shift left by 1 gives the same graph as a shift up by 2.

C. A shift left by 1 gives the same graph as a vertical stretch by 2.

D. They are never the same graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q2/5

True or false?

$y = 5 \cdot 3^x$ and $y = 3^x$ have different horizontal asymptotes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q3/5

Which exponential graph passes through (0, 4) and (1, 2)?

A. $y = 2^x + 3$

B. $y = 4 \cdot 2^x$

C. $y = 0.5 \cdot 4^x$

D. $y = 4(0.5)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q4/5

Which exponential function is increasing, has asymptote $y = -1$, has y-intercept 2, and passes through (1, 5)?

A. $y = 3(1/2)^x - 1$

B. $y = 3 \cdot 3^x - 1$

C. $y = 3 \cdot 2^x - 1$

D. $y = 2 \cdot 2^x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q5/5

SET 49 • HSF-IF.C.7.e

FORWARD 2

Graph the inverse of an exponential function • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

Which is the inverse of $f(x) = 2^x$?

A. $f^{-1}(x) = \log_2 x$

B. $f^{-1}(x) = x^2$

C. $f^{-1}(x) = (1/2)^x$

D. $f^{-1}(x) = x/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q1/5

The point (3, 8) is on $y = 2^x$. Which point is on its inverse?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q2/5

True or false?

The inverse of $y = 2^x$ has the vertical asymptote $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q3/5

The inverse of $y = 2^x - 3$ is

A. $y = \log_2(x - 3)$

B. $y = \log_2 x - 3$

C. $y = 2^{x+3}$

D. $y = \log_2(x + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q4/5

Which gives the domain and range of $y = \log_2 x$?

A. Domain: all real numbers; range: $y > 0$

B. Domain: $x \geq 0$; range: $y \geq 0$

C. Domain: $x > 1$; range: $y > 0$

D. Domain: $x > 0$; range: all real numbers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q5/5

SET 50 • HSF-IF.C.7.e • INTERPRETING FUNCTIONS

Graph logarithmic functions with intercept, asymptote, and end behavior

MAIN	Algebra II main question	**HSF-IF.C.7.e**
BACKWARD 1	Graph exponential functions	**HSF-IF.C.7.e**
BACKWARD 2	Find inverses by swapping inputs and outputs	**HSF-BF.B.4.a**
BACKWARD 3	Shift and reflect graphs	**HSF-BF.B.3**
FORWARD 1	Graph logarithms with reflections that change the domain	**HSF-IF.C.7.e**
FORWARD 2	Solve equations with logarithms	**HSF-LE.A.4**

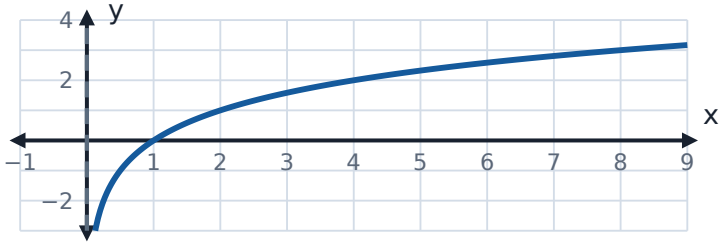
SET 50 • HSF-IF.C.7.e

MAIN

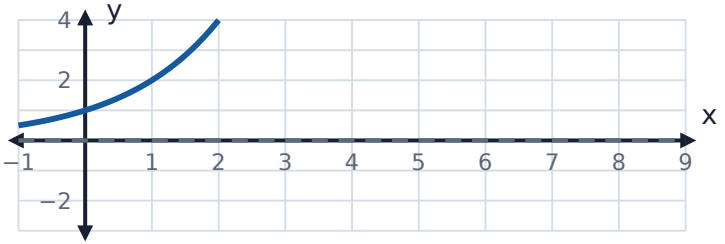
Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

Which graph shows $y = \log_2 x$?

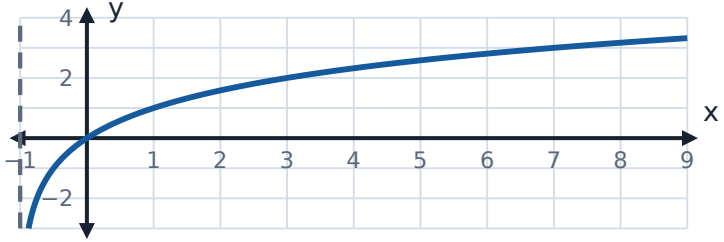
A



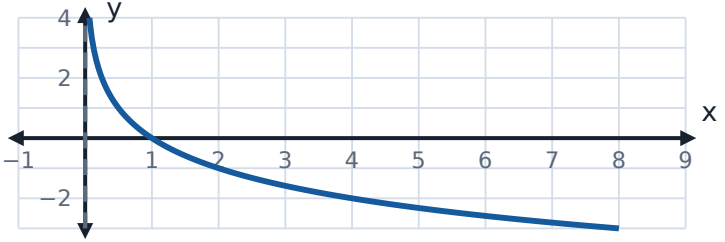
B



C



D



A. Graph A

B. Graph B

C. Graph C

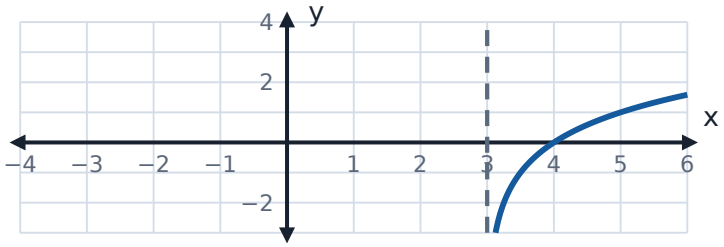
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-M-Q1**

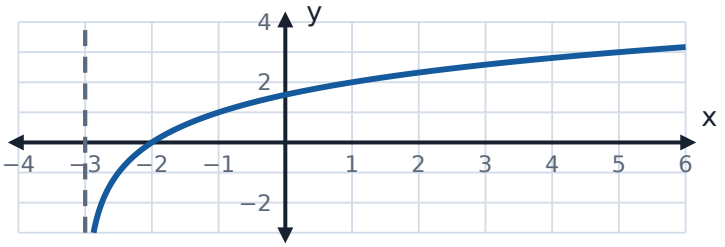
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q1/5

Which graph shows $y = \log_2(x + 3)$?

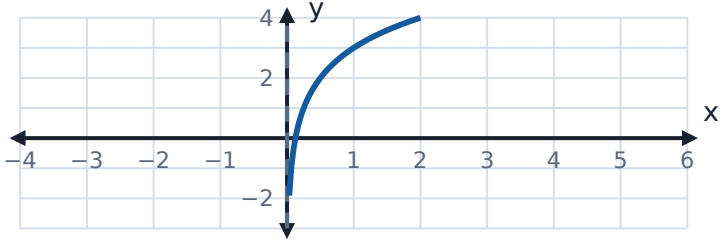
A



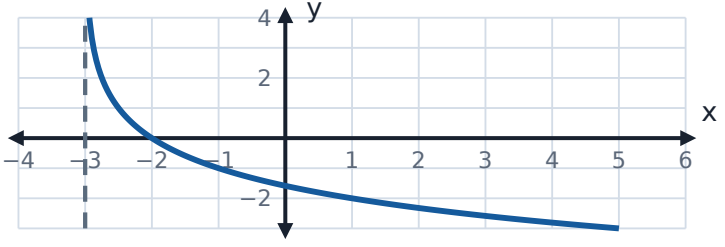
B



C



D



A. Graph A

B. Graph B

C. Graph C

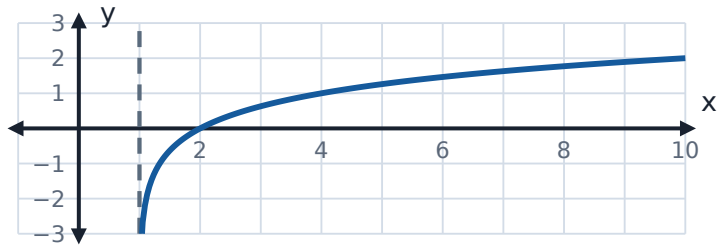
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-M-Q2**

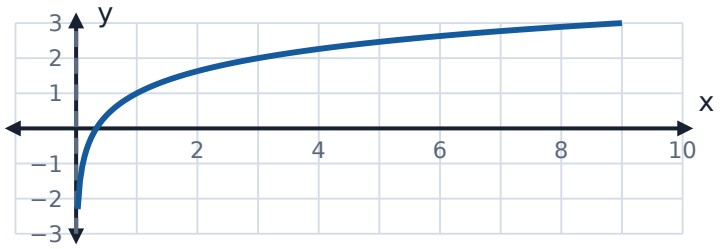
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q2/5

Which graph shows $y = \log_3 x - 1$?

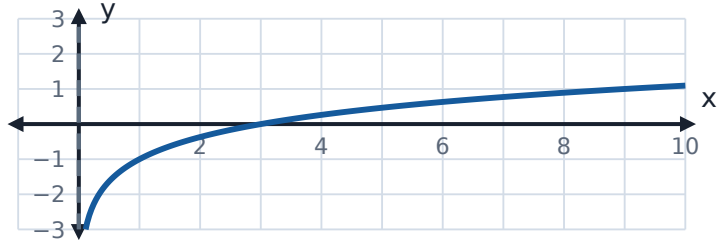
A



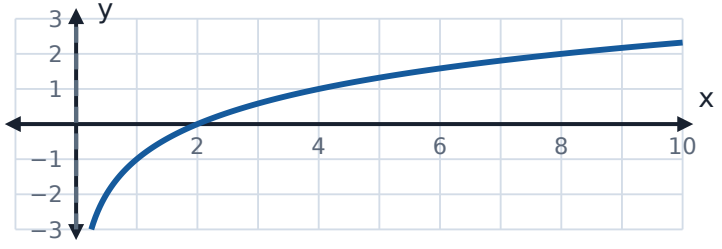
B



C



D



A. Graph A

B. Graph B

C. Graph C

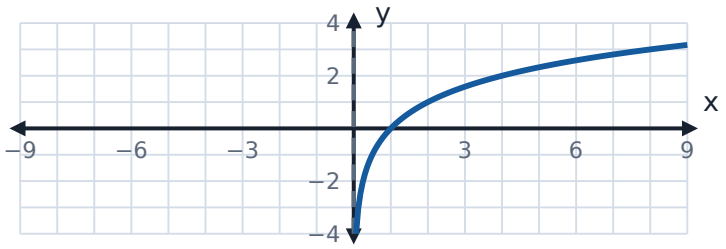
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-M-Q3**

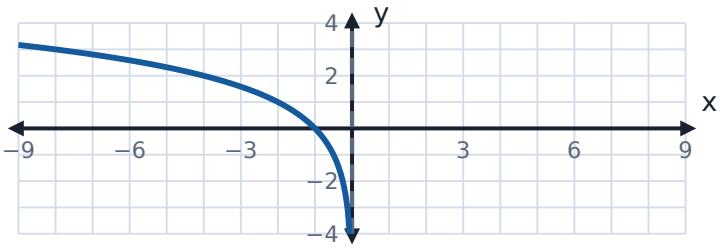
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q3/5

Which graph shows $y = -\log_2 x$?

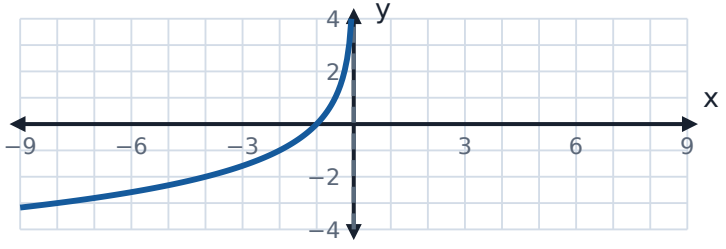
A



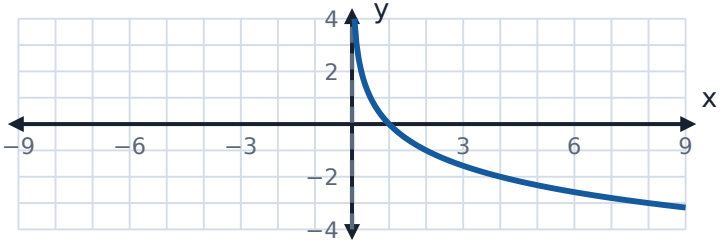
B



C



D



A. Graph A

B. Graph B

C. Graph C

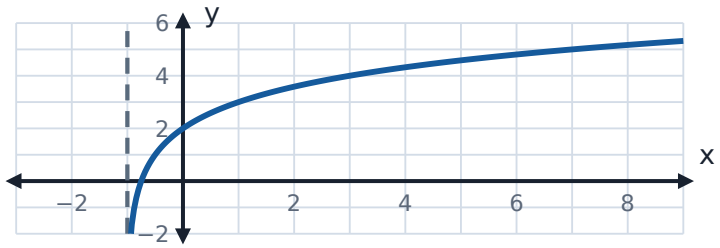
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-M-Q4**

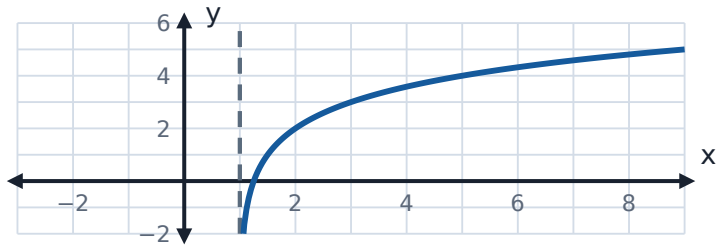
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q4/5

Which graph shows $y = \log_2(x - 1) + 2$?

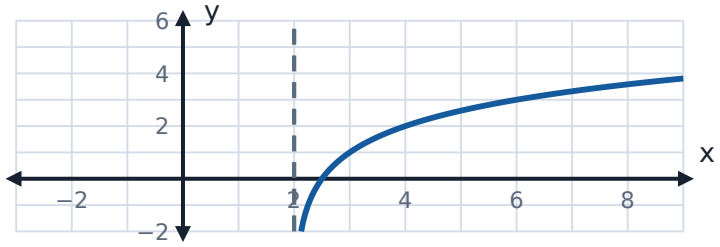
A



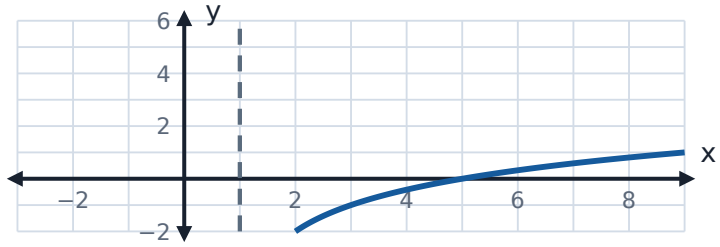
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q5/5

SET 50 • HSF-IF.C.7.e

BACKWARD 1

Graph exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

Find the y-values of $y = 2^x$ at $x = 0$, 1, and 2, in that order, to get three points on the graph.

y-values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q1/5

Give the asymptote of $y = 2^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q2/5

True or false?

$y = 3^x$ passes through (3, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q3/5

$y = 2^x$ passes through

A. (2, 1)

B. (−1, −2)

C. (1/2, −1)

D. (−1, 1/2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q4/5

Give the range of $y = 3^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q5/5

SET 50 • HSF-IF.C.7.e

BACKWARD 2

Find inverses by swapping inputs and outputs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

(2, 4) is on $y = 2^x$. Which point is on its inverse?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q1/5

Which is the inverse of $y = 3^x$?

A. $y = x^3$

B. $y = \log_3 x$

C. $y = x/3$

D. $y = 3^{-x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q2/5

True or false?

Swapping x and y in $y = 2^x$ gives $x = 2^y$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q3/5

Which point is on $y = \log_2 x$?

A. (3, 8)

B. (2, 4)

C. (8, 3)

D. (0, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q4/5

What is $\log_2 16$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q5/5

SET 50 • HSF-IF.C.7.e

BACKWARD 3

Shift and reflect graphs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

How is the graph of $y = \log_2(x + 3)$ related to the graph of $y = \log_2 x$?

A. Shifted right 3

B. Shifted down 3

C. Shifted left 3

D. Shifted up 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q1/5

How is the graph of $y = -\log_2 x$ related to the graph of $y = \log_2 x$?

A. Reflected over the line $y = x$

B. Reflected over the y-axis

C. Reflected over the x-axis

D. Shifted down 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q2/5

True or false?

A horizontal shift moves the vertical asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q3/5

$y = \log_2(x - 1) + 2$ has asymptote

A. $x = -1$

B. $y = 2$

C. $x = 1$

D. $x = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q4/5

Where does (1, 0) of $y = \log_3 x$ move under $y = \log_3 x - 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q5/5

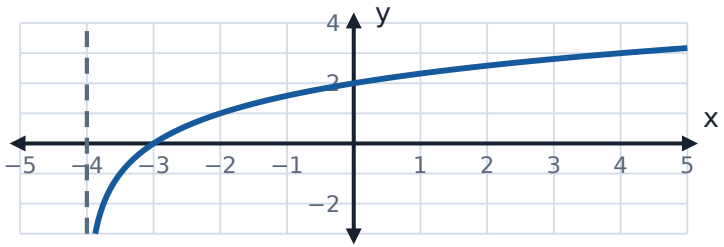
SET 50 • HSF-IF.C.7.e

FORWARD 1

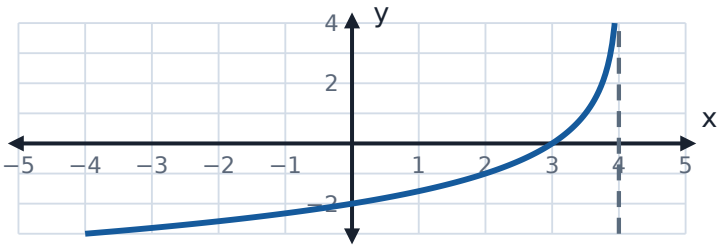
Graph logarithms with reflections that change the domain • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

Which graph shows $y = \log_2(4 - x)$?

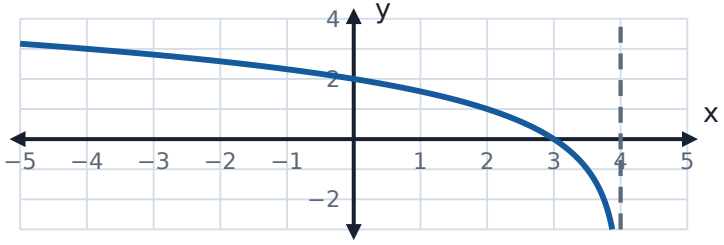
A



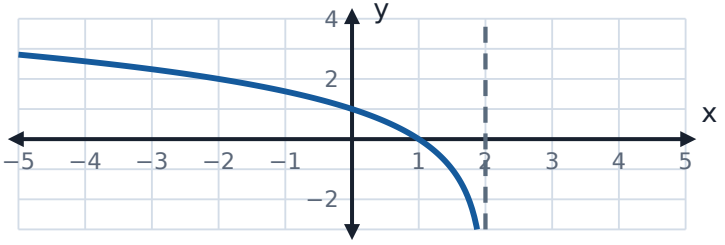
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q1/5

Why does the graph of $y = \log(x^2 - 1)$ have two separate branches?

A. Every logarithm graph has two branches, one on each side of its asymptote.

B. Its domain is $x < -1$ or $x > 1$: $x^2 - 1 > 0$ on two separate intervals.

C. $\log(x^2 - 1)$ is undefined at $x = 0$ only, which splits the graph in two.

D. Its domain is $-1 < x < 1$, an interval split in two at $x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q2/5

True or false?

$y = \log_2(2x)$ is $y = \log_2 x$ shifted up 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q3/5

The x-intercept of $y = \log_2(x + 3)$ is

A. -2

B. 1

C. -3

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q4/5

Find the x-intercept of $y = \log_2(x - 1) + 2$ exactly.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q5/5

SET 50 • HSF-IF.C.7.e

FORWARD 2

Solve equations with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $2^x = 12$. Give x to the nearest hundredth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q1/5

Solve $5 \cdot 10^{x} = 40$. Give x to the nearest thousandth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q2/5

True or false?

$\log_2 12$ is between 4 and 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q3/5

The solution of $3^x = 7$ is

A. log 7/log 3

B. log 3/log 7

C. 7/3

D. log 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q4/5

Use a graphing calculator (or the logarithm key) to estimate $\log_2 6$ to the nearest tenth. A calculator may be used.

$\log_2 6 \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q5/5

SET 51 • HSF-IF.C.7.e • INTERPRETING FUNCTIONS

Graph trigonometric functions showing period, midline, and amplitude

MAIN	Algebra II main question	**HSF-IF.C.7.e**
BACKWARD 1	Unit-circle values at quarter turns	**HSF-TF.A.2**
BACKWARD 2	Read amplitude, period, and midline from a formula	**HSF-TF.B.5**
BACKWARD 3	Vertical stretches, shifts, and reflections	**HSF-BF.B.3**
FORWARD 1	Graph shifted sinusoids and the tangent function	**HSF-IF.C.7.e**
FORWARD 2	Horizontal stretches f(kx) of trigonometric graphs	**HSF-BF.B.3**

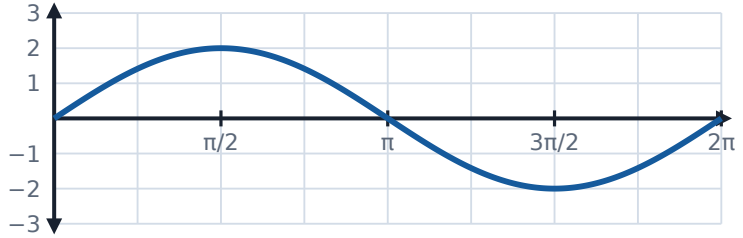
SET 51 • HSF-IF.C.7.e

MAIN

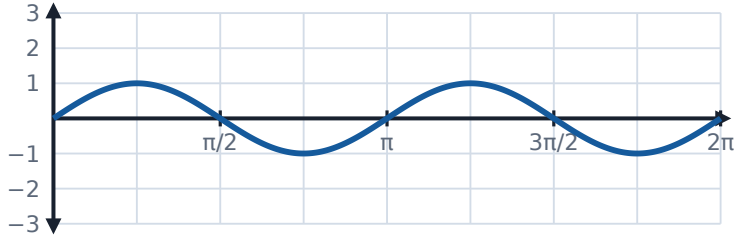
Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

Which graph shows $y = 2\sin x$ on $0 \le x \le 2\pi$?

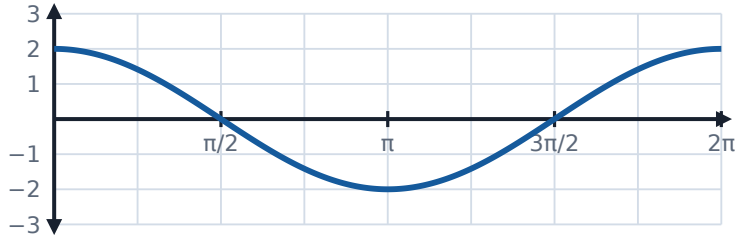
A



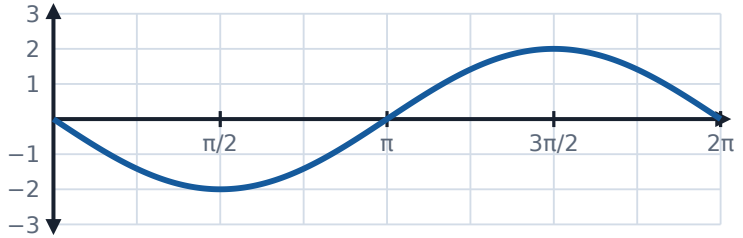
B



C



D



A. Graph A

B. Graph B

C. Graph C

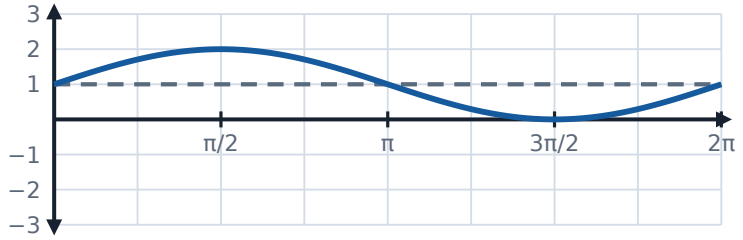
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-M-Q1**

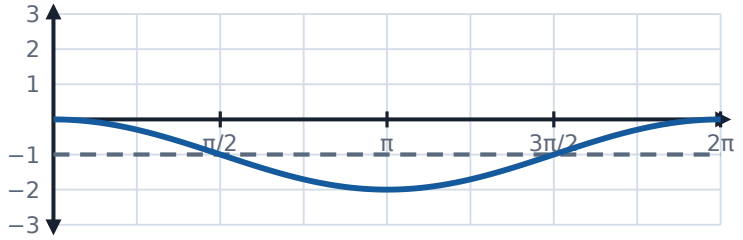
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q1/5

Which graph shows $y = \cos x + 1$ on $0 \le x \le 2\pi$?

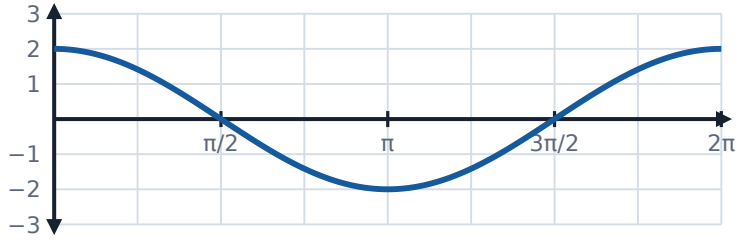
A



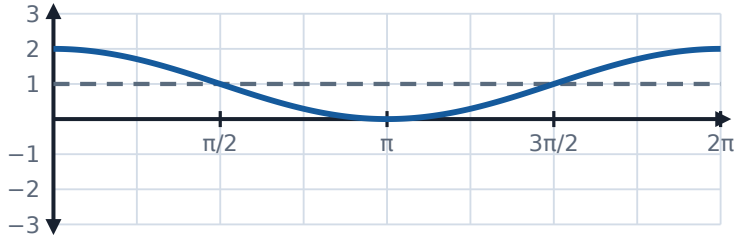
B



C



D



A. Graph A

B. Graph B

C. Graph C

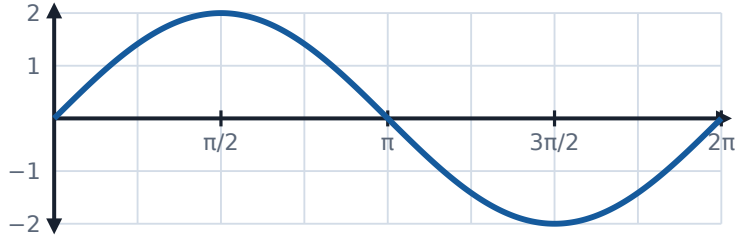
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-M-Q2**

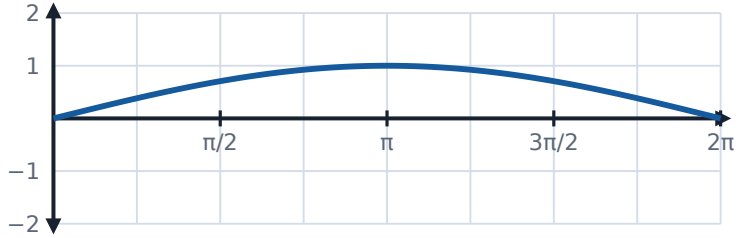
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q2/5

Which graph shows $y = \sin(2x)$ on $0 \le x \le 2\pi$?

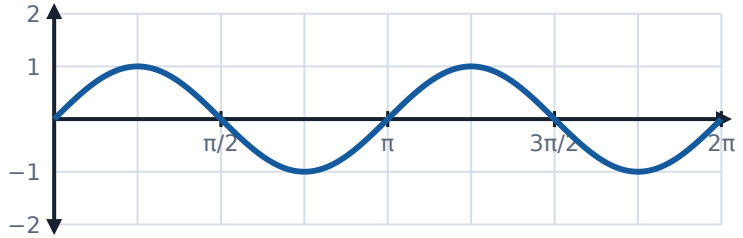
A



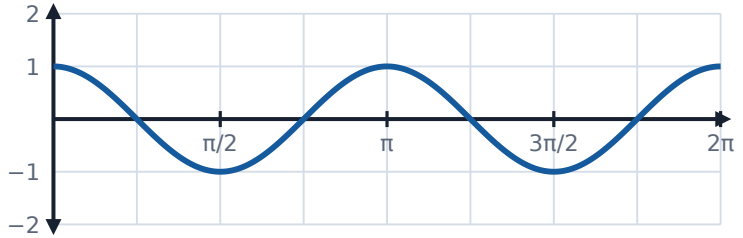
B



C



D



A. Graph A

B. Graph B

C. Graph C

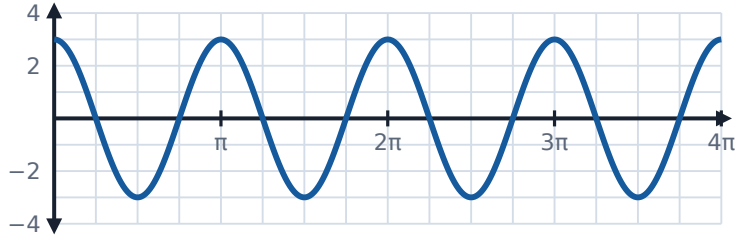
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-M-Q3**

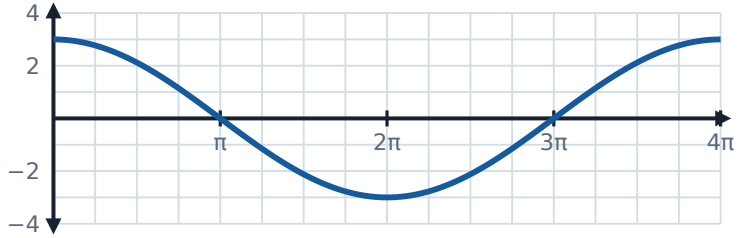
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q3/5

Which graph shows $y = 3\cos(x/2)$ on $0 \le x \le 4\pi$?

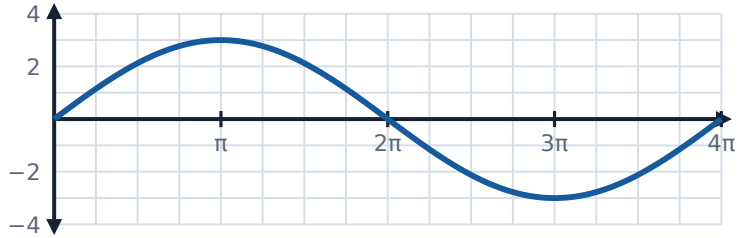
A



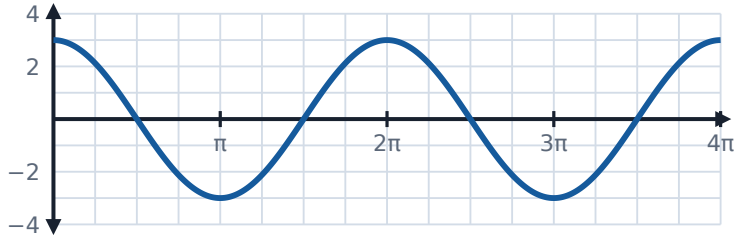
B



C



D



A. Graph A

B. Graph B

C. Graph C

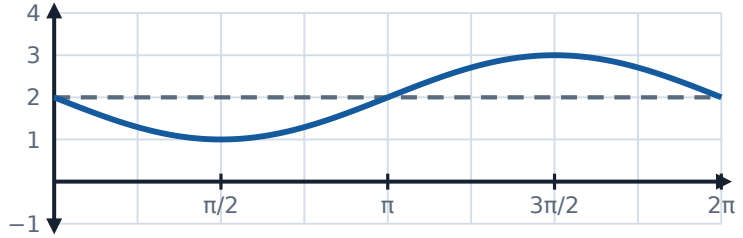
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-M-Q4**

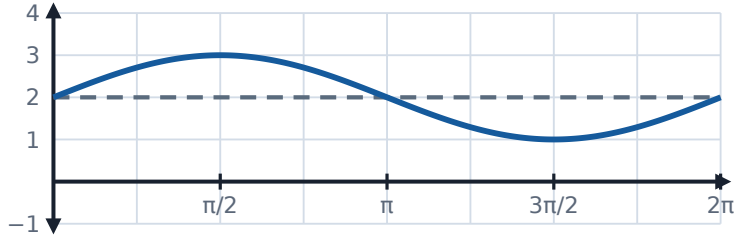
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q4/5

Which graph shows $y = -\sin x + 2$ on $0 \le x \le 2\pi$?

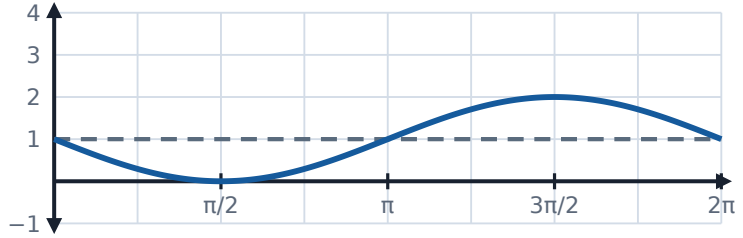
A



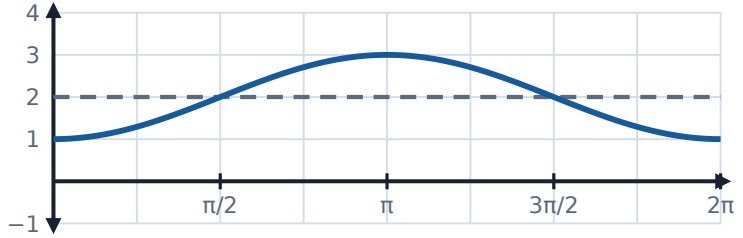
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q5/5

SET 51 • HSF-IF.C.7.e

BACKWARD 1

Unit-circle values at quarter turns • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Give $\sin x$ at $x = 0, \pi/2, \pi, 3\pi/2, 2\pi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q1/5

Give cos x at x = 0, π/2, π, 3π/2, 2π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q2/5

True or false?

$\sin(3\pi/2) = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q3/5

cos(π) equals

A. π

B. 1

C. −1

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q4/5

Give $\sin(\pi/4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q5/5

SET 51 • HSF-IF.C.7.e

BACKWARD 2

Read amplitude, period, and midline from a formula • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

Give the amplitude and the period of $y = 3\cos(x/2)$.

Amplitude: _____

Period: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q1/5

Give the midline of $y = -\sin x + 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q2/5

True or false?

$y = \sin(2x)$ has period 4π.

A. True

B. False

$y = 2\sin x$ has amplitude

A. 1

B. 2

C. π

D. 2π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q4/5

Give the period of $y = \cos(3x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q5/5

SET 51 • HSF-IF.C.7.e

BACKWARD 3

Vertical stretches, shifts, and reflections • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

How does the graph of $y = \cos x + 1$ compare with the graph of $y = \cos x$?

A. Shifted left 1

B. Stretched vertically by 1

C. Shifted down 1

D. Shifted up 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q1/5

How does the graph of $y = -\sin x$ compare with the graph of $y = \sin x$?

A. Reflected over the x-axis

B. Stretched vertically by 2

C. Shifted down 1

D. Unchanged

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q2/5

True or false?

$y = 2\sin x$ has the same zeros as $y = \sin x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q3/5

The maximum of $y = -\sin x + 2$ is

A. 1

B. 2

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q4/5

What is the minimum of $y = \cos x + 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q5/5

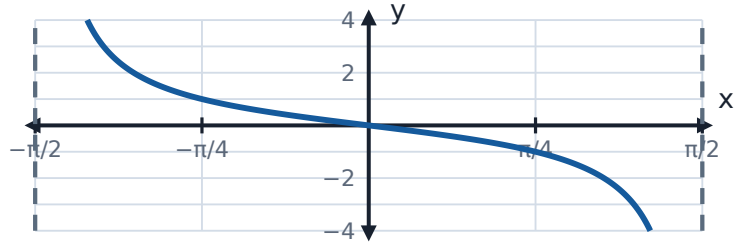
SET 51 • HSF-IF.C.7.e

FORWARD 1

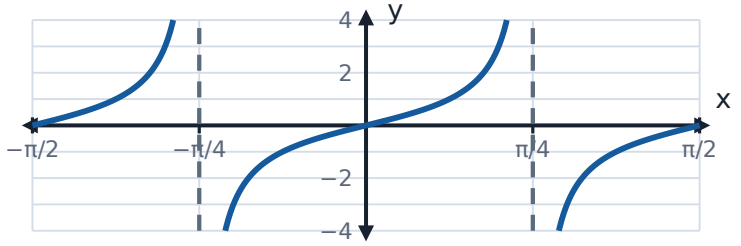
Graph shifted sinusoids and the tangent function • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

Which graph shows $y = \tan x$ for $-\pi/2 < x < \pi/2$?

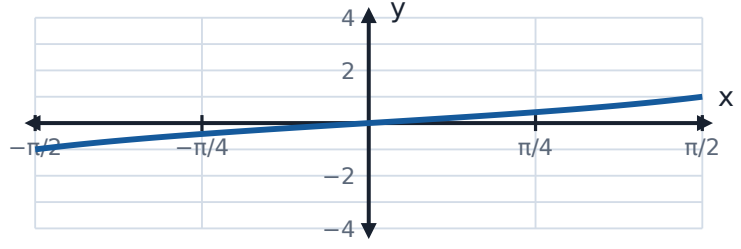
A



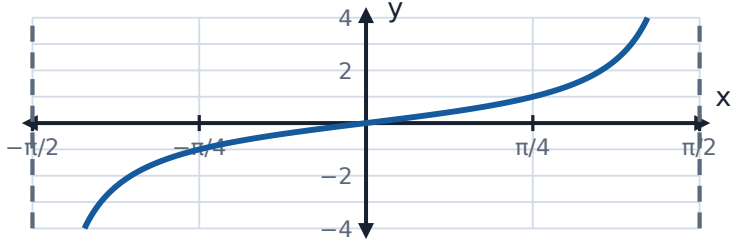
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q1/5

How does $y = \sin(x - \pi/2)$ compare with $y = \sin x$, and which simpler formula describes it?

A. It is shifted left $\pi/2$, and it equals $\cos x$.

B. It is shifted right $\pi/2$, and it equals $-\cos x$.

C. It is shifted down $\pi/2$, and it equals $\sin x - \pi/2$.

D. It is shifted right $\pi/2$, and it equals $\cos x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q2/5

True or false?

The period of $y = \tan x$ is 2π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q3/5

The first maximum of $y = \cos(x - \pi/3)$ for $x \geq 0$ is at

A. $x = 2\pi/3$

B. $x = \pi/3$

C. $x = 0$

D. $x = -\pi/3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q4/5

Give the x-values of the maximum points of $y = \sin(x + \pi/4)$ on $0 \leq x < 2\pi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q5/5

SET 51 • HSF-IF.C.7.e

FORWARD 2

Horizontal stretches f(kx) of trigonometric graphs • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

How is the graph of $y = \sin(2x)$ obtained from the graph of $y = \sin x$?

A. Horizontal compression by a factor of 1/2

B. Horizontal stretch by a factor of 2

C. Shift left 2

D. Vertical stretch by a factor of 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q1/5

How is the graph of $y = \cos(x/2)$ obtained from the graph of $y = \cos x$?

A. Shift right 2

B. Vertical compression by a factor of 1/2

C. Horizontal stretch by a factor of 2

D. Horizontal compression by a factor of 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q2/5

True or false?

Replacing x by kx (k > 1) shortens the period by the factor k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q3/5

$y = \sin(-x)$ is the same as

A. $\cos x$

B. $\sin x$

C. $-\cos x$

D. $-\sin x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q4/5

For which $k > 0$ does $y = \cos(kx)$ have period 6π?

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q5/5

SET 52 • HSF-IF.C.8.b • INTERPRETING FUNCTIONS

Rewrite exponential functions to reveal rates over other time units

MAIN	Algebra II main question	**HSF-IF.C.8.b**
BACKWARD 1	Read growth rates from exponential functions	**HSF-IF.C.8.b**
BACKWARD 2	Powers of powers and rational exponents	**HSN-RN.A.1**
BACKWARD 3	Percent change and growth factors	**7.RP.A.3**
FORWARD 1	Compare growth offers by rewriting to the same time unit	**HSF-IF.C.8.b**
FORWARD 2	Find doubling times and half-lives with logarithms	**HSF-LE.A.4**

SET 52 • HSF-IF.C.8.b

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b

$y = 100(1.08)^t$ with t in years. Rewrite it as $y = 100(b)^{12t}$ with a monthly growth factor b. Give b to 6 decimal places and the monthly percent rate to the nearest hundredth of a percent. A calculator may be used.

b = _____

Monthly rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q1/5

$y = 500(0.97)^t$ with t in years. Rewrite it as $y = 500(b)^{t/10}$ with a growth factor b per decade. Give b to 4 decimal places and the percent decrease per decade to the nearest hundredth of a percent. A calculator may be used.

b = _____

Percent decrease per decade: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q2/5

$Q = 50(2)^{t/3}$ with t in hours. Rewrite it as $Q = 50(b)^{t}$ with an hourly growth factor b. Give b to 4 decimal places and the hourly percent rate to the nearest hundredth of a percent. A calculator may be used.

b = _____

Hourly rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q3/5

$A = 1000(1.005)^{12t}$ with t in years. Rewrite it as $A = 1000(b)^{t}$ with an annual growth factor b. Give b to 6 decimal places and the annual percent rate to the nearest hundredth of a percent. A calculator may be used.

b = _____

Annual rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q4/5

$N = 800(0.5)^{t/10}$ with t in years. Rewrite it as $N = 800(b)^t$ with a yearly factor b. Give b to 6 decimal places and the yearly percent decrease to the nearest hundredth of a percent. A calculator may be used.

b = _____

Yearly decrease: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q5/5

SET 52 • HSF-IF.C.8.b

BACKWARD 1

Read growth rates from exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b

Is $y = 100(1.08)^t$ growth or decay, and by what percent per year? Answer Growth or Decay, then the percent.

Growth or Decay: _____

Percent per year: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q1/5

Is $y = 500(0.97)^t$ growth or decay, and by what percent per year? Answer Growth or Decay, then the percent.

Growth or Decay: _____

Percent per year: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q2/5

True or false?

$y = 2(1.25)^t$ grows 125% per unit of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q3/5

$y = 40(0.9)^t$ decreases by

A. 9% per unit

B. 10% per unit

C. 90% per unit

D. 0.9% per unit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q4/5

What is the growth factor for 6.17% growth?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q5/5

SET 52 • HSF-IF.C.8.b

BACKWARD 2

Powers of powers and rational exponents • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1

Rewrite 1.08^{t} as a power of $1.08^{1/12}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q1/5

Rewrite $2^{t/3}$ as $(2^{1/3})^{t}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q2/5

True or false?

$(b^m)^n = b^{mn}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q3/5

1.005^{12t} equals

A. $(1.005^{12})^{t}$

B. $(1.005^{t})/12$

C. $12 \cdot 1.005^{t}$

D. $1.005^{12} + t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q4/5

Write $8^{1/3}$ as a whole number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • HSF-IF.C.8.b

BACKWARD 3

Percent change and growth factors • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

Write the growth factor for a 25.99% increase.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q1/5

A factor 0.7374 means what percent decrease?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q2/5

True or false?

A factor 1.006434 means about 0.64% growth.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q3/5

A 3% decrease corresponds to the factor

A. 0.03

B. 0.97

C. 0.7

D. 1.03

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q4/5

An amount is multiplied by 0.933. By what percent does it decrease?

Percent decrease: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q5/5

SET 52 • HSF-IF.C.8.b

FORWARD 1

Compare growth offers by rewriting to the same time unit • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b

Account A grows 0.5% per month. Account B grows 6.1% per year. Find A's annual growth factor (to 4 decimal places), and say which account grows faster (A or B). A calculator may be used.

A's annual factor: _____

Faster account (A or B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q1/5

Culture 1 doubles every 3 hours; culture 2 triples every 5 hours. Find each hourly growth factor to 3 decimal places, and say which grows faster (1 or 2). A calculator may be used.

Hourly factor of culture 1: _____

Hourly factor of culture 2: _____

Faster culture (1 or 2): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q2/5

True or false?

Growing 1% per month is the same as growing 12% per year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q3/5

Which is the greatest annual growth?

A. 4% per half year

B. 0.6% per month

C. 8% per year

D. 2% per quarter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q4/5

A substance with a half-life of 10 years and a substance that decays 7% per year are compared. Which decays faster? A calculator may be used.

A. 7% per year: its yearly factor 0.93 is less than $0.5^{1/10} \approx 0.933$.

B. 7% per year: 7 is less than 10.

C. The half-life substance: its yearly factor 0.5 is less than 0.93.

D. They decay at the same rate, because $0.5^{1/10} \approx 0.93$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q5/5

SET 52 • HSF-IF.C.8.b

FORWARD 2

Find doubling times and half-lives with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Find the doubling time of $y = 100(1.08)^t$, in years to the nearest hundredth. A calculator may be used.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q1/5

Find the half-life of $y = 500(0.97)^t$, in years to the nearest hundredth. A calculator may be used.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q2/5

True or false?

The doubling time of $y = 50(2)^{t/3}$ is 6 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q3/5

The doubling time of a quantity growing 6.17% per year is about

A. 16.2 years

B. 11.6 years

C. 6.17 years

D. 3.1 years

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q4/5

$y = 100(1.08)^t$ is rewritten as $y = 100 \cdot 2^{t/9.01}$. Why does the rewritten form show the doubling time?

A. Because 9.01 is the growth rate in percent per year, and doubling depends only on the rate.

B. Because 2 is the starting value, and the exponent t/9.01 counts how many times it is used.

C. Because $y = 200$ when $t = 1$, so y doubles from 100 to 200 in the first year and in each 9.01-year period.

D. The exponent t/9.01 increases by 1 every 9.01 years, and each such increase multiplies y by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q5/5

SET 53 • HSF-IF.C.9 • INTERPRETING FUNCTIONS

Compare properties of Algebra II functions in different representations

MAIN	Algebra II main question	**HSF-IF.C.9**
BACKWARD 1	Compare linear, quadratic, and exponential functions	**HSF-IF.C.9**
BACKWARD 2	Recognize a pattern in a table	**8.F.A.3**
BACKWARD 3	Domains, asymptotes, and periods of Algebra II families	**HSF-IF.B.5**
FORWARD 1	Compare rates of change and long-run behavior across represent…	**HSF-IF.C.9**
FORWARD 2	Build a difference function to compare	**HSF-BF.A.1.b**

SET 53 • HSF-IF.C.9

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9

$f(x) = 3(1.1)^x$. The table shows the linear function g. Give the greater y-intercept, f(20) to the nearest hundredth, and g(20). A calculator may be used.

x	0	1	2	3	4
g(x)	10	15	20	25	30

Greater y-intercept: _____

f(20) = _____

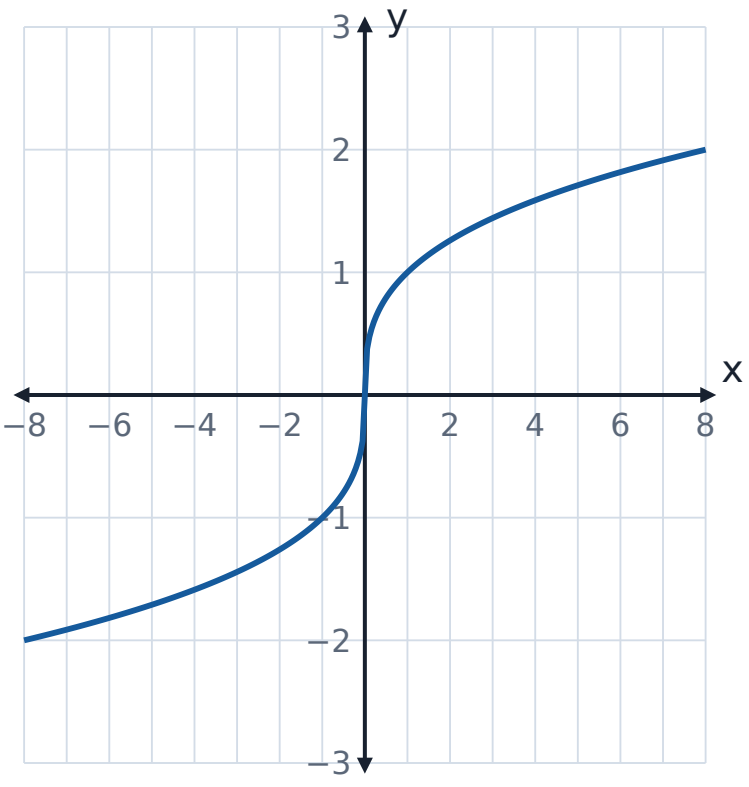
g(20) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q1/5

$f(x) = \sqrt{(x - 1)}$. The graph of g is shown. Compare their domains.



A. f: $x > 1$; g: all real numbers

B. f: $x \geq 1$; g: $x \geq 0$

C. f: $x \geq 1$; g: all real numbers

D. f: all real numbers; g: $x \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q2/5

$h(t) = 12 + 10\sin(\pi t/15)$ gives one Ferris wheel's height (m, t in s). A second wheel has radius 8 m, its center 10 m high, and turns once every 40 s. Give the maximum height and the period of each wheel.

First wheel maximum (m): _____

First wheel period (s): _____

Second wheel maximum (m): _____

Second wheel period (s): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q3/5

$f(x) = \log_2 x$. The table shows g. Find f(9) to the nearest hundredth and g(9). A calculator may be used.

x	1	3	9	27
g(x)	0	1	2	3

f(9) = _____

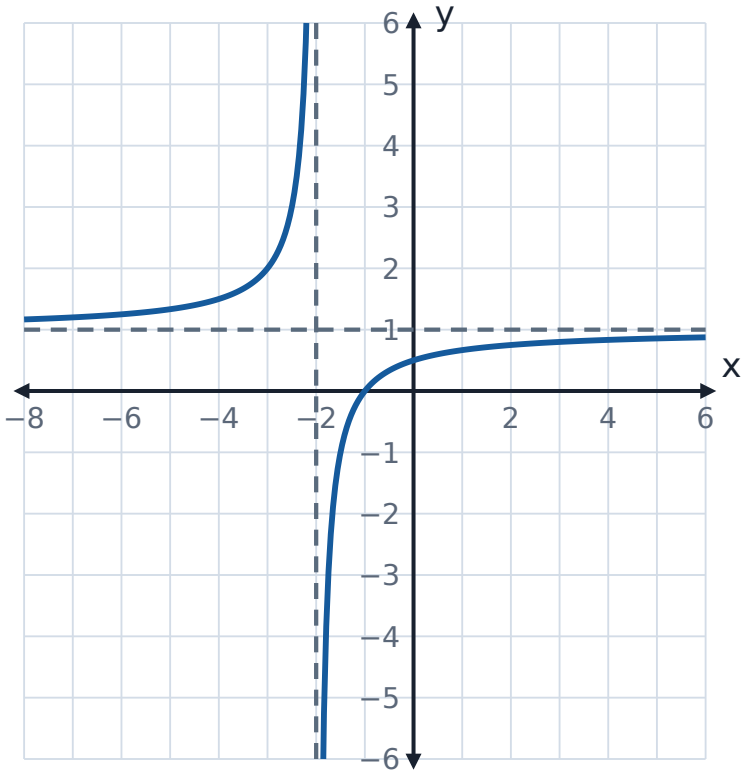
g(9) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q4/5

$f(x) = (2x + 1)/(x - 3)$. The graph of g shows its asymptotes. Give the horizontal and vertical asymptotes of f and of g as equations.



f, horizontal: _____

f, vertical: _____

g, horizontal: _____

g, vertical: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q5/5

SET 53 • HSF-IF.C.9

BACKWARD 1

Compare linear, quadratic, and exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9

$f(x) = 2x + 1$, and g has the table $g(0) = 3$, $g(1) = 4$. Which function has the greater y-intercept?

A. f, with y-intercept 2

B. f, with y-intercept 1

C. g, with y-intercept 3

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q1/5

$f(x) = x^2$ and $g(x) = 2^x$. Find f(10) and g(10).

f(10) = _____

g(10) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q2/5

True or false?

An increasing exponential eventually exceeds any linear function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q3/5

$f(x) = -(x - 1)^2 + 5$ and g with maximum 4 in a graph. The greater maximum is

A. f: 5

B. cannot tell

C. equal

D. g: 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q4/5

A table's outputs increase by 5 at each equal step of the input. What kind of function fits it?

A. Exponential

B. Linear

C. Quadratic

D. Logarithmic

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q5/5

SET 53 • HSF-IF.C.9

BACKWARD 2

Recognize a pattern in a table • Grade 8 • Functions • 8.F.A.3

Is the table x: 0, 1, 2 and y: 10, 15, 20 linear?

A. No: it starts at 10, not 0, and a linear table must start at 0

B. Yes: every table of numbers can be joined by straight segments, so every table is linear

C. No: the outputs 10, 15, 20 are not all multiples of 10, so they are not linear

D. Yes: the outputs have a constant difference of 5 for equal steps in x

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q1/5

Is the table x: 1, 3, 9, 27 and y: 0, 1, 2, 3 linear?

A. No: each output step of 1 needs the input multiplied by 3, so it is logarithmic ($y = \log_3 x$)

B. Yes: the inputs increase each time, and any table with increasing inputs is linear

C. No: it is exponential, because each input is 3 times the one before it ($y = 3^x$)

D. Yes: the outputs increase by 1 each time, so the table has a constant rate of change

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q2/5

True or false?

A table with constant ratios of outputs is exponential.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q3/5

Which table is exponential?

A. 1, 4, 9, 16

B. 1, 2, 4, 8

C. 2, 2, 2, 2

D. 1, 3, 5, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q4/5

What is the slope of a linear table with outputs 10, 15, 20 at x = 0, 1, 2?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • HSF-IF.C.9

BACKWARD 3

Domains, asymptotes, and periods of Algebra II families • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

Give the domain of $\sqrt{(x - 1)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q1/5

Give the vertical and horizontal asymptotes of $y = (2x + 1)/(x - 3)$ as equations.

Vertical asymptote: _____

Horizontal asymptote: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q2/5

True or false?

The period of $12 + 10\sin(\pi t/15)$ is 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q3/5

The domain of $\log_2 x$ is

A. all reals

B. $x \geq 0$

C. $x > 0$

D. $x \neq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q4/5

A wheel of radius 8 m with center 10 m high has what maximum height?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q5/5

SET 53 • HSF-IF.C.9

FORWARD 1

Compare rates of change and long-run behavior across representations • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9

$f(x) = 3(1.1)^x$ and $g(x) = 10 + 5x$. Estimate the first whole number x at which f exceeds g. A calculator may be used. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q1/5

$f(x) = \log_2 x$, and g has $g(1) = 0$, $g(3) = 1$, $g(9) = 2$. Find the average rates of change of f and of g on $1 \le x \le 9$, each to the nearest hundredth. A calculator may be used.

Rate of f: _____

Rate of g: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q2/5

True or false?

A function with a larger y-intercept is always larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q3/5

Which grows fastest for large x?

A. $\sqrt{(x - 1)}$

B. $3(1.1)^x$

C. $5x + 10$

D. $\log_2 x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q4/5

Why does $g(x) = \log_3 x$ stay below $f(x) = \log_2 x$ for $x > 1$?

A. Because $3 > 2$ makes every value of $\log_3 x$ larger, so g stays below f only for $0 < x < 1$.

B. Because both equal 0 at $x = 1$, and two graphs that share a point never cross after it.

C. Because $\log_3 x$ is negative for $x > 1$ while $\log_2 x$ is positive there, so g is below f.

D. $\log_3 x = \log_2 x / \log_2 3$, and $\log_2 3 > 1$, so for $x > 1$ (where $\log_2 x > 0$), g is a smaller positive multiple of f.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q5/5

SET 53 • HSF-IF.C.9

FORWARD 2

Build a difference function to compare • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b

Write $d(x) = f(x) - g(x)$ for $f(x) = 3(1.1)^x$ and $g(x) = 5x + 10$.

$d(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q1/5

$d(x) = f(x) - g(x)$ is the difference of two functions. What does the sign of $d(x)$ tell you?

A. $d(x) > 0$ means f is increasing and g is decreasing at x.

B. $d(x) > 0$ means $f(x) > g(x)$; $d(x) < 0$ means $g(x) > f(x)$; $d(x) = 0$ means they are equal.

C. $d(x) > 0$ means both functions are positive at x; $d(x) < 0$ means both are negative.

D. The sign of $d(x)$ gives the slope of f: positive means f rises and negative means f falls.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q2/5

True or false?

A zero of $d(x) = f(x) - g(x)$ is an x where $f(x) = g(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q3/5

For $h_1(t) = 12 + 10\sin(\pi t/15)$ and $h_2(t) = 10 + 8\sin(\pi t/20)$, $h_1(0) - h_2(0)$ is

A. 2

B. 22

C. -2

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q4/5

Two Ferris wheel heights are $h_1(t) = 12 + 10\sin(\pi t/15)$ and $h_2(t) = 10 + 8\sin(\pi t/20)$. Which is $h_1(t) - h_2(t)$?

A. $2 + 10\sin(\pi t/15) + 8\sin(\pi t/20)$

B. $2 + 2\sin(\pi t/60)$

C. $2 + 10\sin(\pi t/15) - 8\sin(\pi t/20)$

D. $22 + 10\sin(\pi t/15) + 8\sin(\pi t/20)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q5/5

Build models by combining Algebra II function types

MAIN	Algebra II main question	**HSF-BF.A.1.b**
BACKWARD 1	Combine linear and quadratic functions	**HSF-BF.A.1.b**
BACKWARD 2	Multiply and subtract polynomials	**HSA-APR.A.1**
BACKWARD 3	Evaluate exponential and trigonometric expressions	**HSF-IF.A.2**
FORWARD 1	Interpret the parts of a combined model	**HSF-BF.A.1.b**
FORWARD 2	Compose functions in context	**HSF-BF.A.1.c**

SET 54 • HSF-BF.A.1.b

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b

Revenue is $R(x) = x(50 - 0.5x)$ and cost is $C(x) = 200 + 10x$ dollars for x items. Write the profit $P(x) = R(x) - C(x)$ with its terms multiplied out, and find $P(30)$.

$P(x) =$ _____

$P(30) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q1/5

The cost of x posters is C(x) = 1200 + 3x dollars. Write the average cost per poster A(x) = C(x)/x and find A(100).

A(x) = _____

A(100) ($) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q2/5

A town has $P(t) = 5000(1.02)^t$ people and $F(t) = 6000 + 100t$ units of food after t years. Write the food per person $F(t)/P(t)$, and find it at $t = 10$ to the nearest hundredth. A calculator may be used.

Food per person = _____

At t = 10: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q3/5

A spring starts at displacement +10 cm at t = 0. Its amplitude envelope is 10 cm and shrinks by 10% each second, and it oscillates once per second, so $y(t) = A(t)\cos(2\pi t)$. Write the envelope $A(t)$ and find $y(1)$.

$A(t) =$ _____

$y(1) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-M-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q4/5

Coffee starts 70°C above a 20°C room, and the difference shrinks 5% each minute. Write its temperature T(t) after t minutes as a sum of a constant and an exponential, and find T(10) to the nearest hundredth. A calculator may be used.

T(t) = _____

T(10) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q5/5

SET 54 • HSF-BF.A.1.b

BACKWARD 1

Combine linear and quadratic functions • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b

$R(x) = 8x$ and $C(x) = 3x + 50$. Write $P = R - C$.

$P(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q1/5

$f(x) = x + 2$ and $g(x) = x - 2$. Write $f \cdot g$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q2/5

True or false?

Profit is revenue minus cost.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q3/5

Average cost is

A. total cost times the number of items

B. total cost divided by the number of items

C. total cost minus total revenue

D. total revenue divided by total cost

$f(x) = 2x$ and $g(x) = x^2 + 1$. Find $(f + g)(3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q5/5

SET 54 • HSF-BF.A.1.b

BACKWARD 2

Multiply and subtract polynomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Expand $x(50 - 0.5x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q1/5

Simplify $(50x - 0.5x^2) - (200 + 10x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q2/5

True or false?

$-(200 + 10x) = -200 + 10x.$

A. True

B. False

$(3x + 1) - (x - 4)$ equals

A. $2x + 5$

B. $4x - 3$

C. $2x - 3$

D. $2x + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-B2-Q4**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q4/5

Expand $2(x^2 - 3x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • HSF-BF.A.1.b

BACKWARD 3

Evaluate exponential and trigonometric expressions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Evaluate $70(0.95)^{10}$ to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q1/5

Evaluate $5000(1.02)^{10}$ to the nearest whole number. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q2/5

True or false?

$\cos(2\pi) = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q3/5

$10(0.9)^2$ equals

A. 9

B. 18

C. 81

D. 8.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q4/5

Evaluate $\cos(2\pi \cdot 0.5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q5/5

SET 54 • HSF-BF.A.1.b

FORWARD 1

Interpret the parts of a combined model • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b

Coffee in a 20°C room has temperature $T(t) = 20 + 70(0.95)^t$ °C after t minutes. What do 20 and $70(0.95)^t$ represent?

A. 20 is the room temperature; $70(0.95)^t$ is how far the coffee is above room temperature, which shrinks 5% each minute.

B. 20 is the starting temperature of the coffee; $70(0.95)^t$ is the room temperature, which falls 5% each minute.

C. 20 is the number of minutes the coffee cools; $70(0.95)^t$ is the temperature of the coffee at time t.

D. 20 is the room temperature; $70(0.95)^t$ is the drop in the coffee's temperature during each minute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q1/5

A spring's position is $y(t) = 10(0.9)^t \cos(2\pi t)$. Which factor controls the size of the swings and which controls their timing?

A. $10(0.9)^t$ controls the size (it shrinks 10% each second); $\cos(2\pi t)$ controls the timing (one swing per second).

B. $\cos(2\pi t)$ controls the size (it shrinks each second); $10(0.9)^t$ controls the timing (one swing per 0.9 seconds).

C. Both factors control only the size of the swings; the timing is set by the 10 in front.

D. 10 controls the timing (10 swings per second); 0.9 controls the size (it shrinks the swings).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q2/5

True or false?

In A(x) = (1200 + 3x)/x = 1200/x + 3, the term 1200/x is the fixed cost spread over x posters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q3/5

As t grows, $T(t) = 20 + 70(0.95)^t$ approaches

A. 20

B. 0

C. 90

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q4/5

A town has $P(t) = 5000(1.02)^t$ people and $F(t) = 6000 + 100t$ units of food after t years. Why does the food per person $F(t)/P(t)$ eventually decrease?

A. $F(t)/P(t)$ is constant because both F and P increase every year, so their ratio does not change as time passes.

B. $P(t) = 5000(1.02)^t$ decreases over time, so there are fewer people for the same food.

C. $F(t) = 6000 + 100t$ decreases over time, so there is less food for the same number of people.

D. P grows exponentially (factor 1.02 each year) while F grows linearly, so P eventually grows much faster.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q5/5

SET 54 • HSF-BF.A.1.b

FORWARD 2

Compose functions in context • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c

A balloon's radius is $r(t) = 2t$ cm after t seconds and its volume is $V(r) = (4/3)\pi r^3$. Write $V(r(t))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q1/5

A store takes 20% off with d(p) = 0.8p and then adds a $5 fee with f(p) = p + 5. Write f(d(p)) and find it for p = 50.

f(d(p)) = _____

f(d(50)) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q2/5

True or false?

f(g(x)) and g(f(x)) are always equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q3/5

If $f(x) = x^2$ and $g(x) = x + 1$, then $f(g(2)) =$

A. 4

B. 9

C. 3

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q4/5

How does composing two functions differ from multiplying them?

A. Composition f(g(x)) adds the two outputs f(x) + g(x); multiplication f(x) · g(x) multiplies them.

B. Composition f(g(x)) uses the output of g as the input of f; multiplication f(x) · g(x) multiplies the two outputs.

C. Composition f(g(x)) multiplies the inputs of f and g; multiplication f(x) · g(x) multiplies the outputs.

D. They are the same operation written two ways: f(g(x)) and f(x) · g(x) are always equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q5/5

Identify vertical shifts, vertical stretches, and horizontal shifts of Algebra II functions

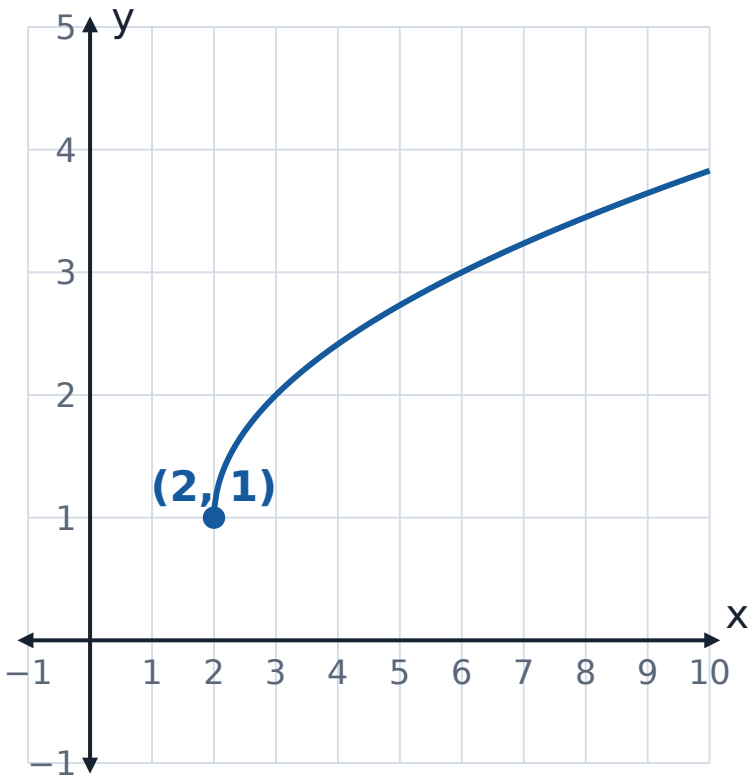
MAIN	Algebra II main question	**HSF-BF.B.3**
BACKWARD 1	Shifts and stretches of linear and quadratic graphs	**HSF-BF.B.3**
BACKWARD 2	Translate points in the plane	**8.G.A.3**
BACKWARD 3	Key points of parent functions	**HSF-IF.C.7.e**
FORWARD 1	Write transformed functions from descriptions and order of transfo..	**HSF-BF.B.3**
FORWARD 2	Read sinusoid parameters as transformations	**HSF-TF.B.5**

SET 55 • HSF-BF.B.3

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

$f(x) = \sqrt{x}$. The graph of $g(x) = f(x - h) + k$ is shown. Find h and k.



h = _____

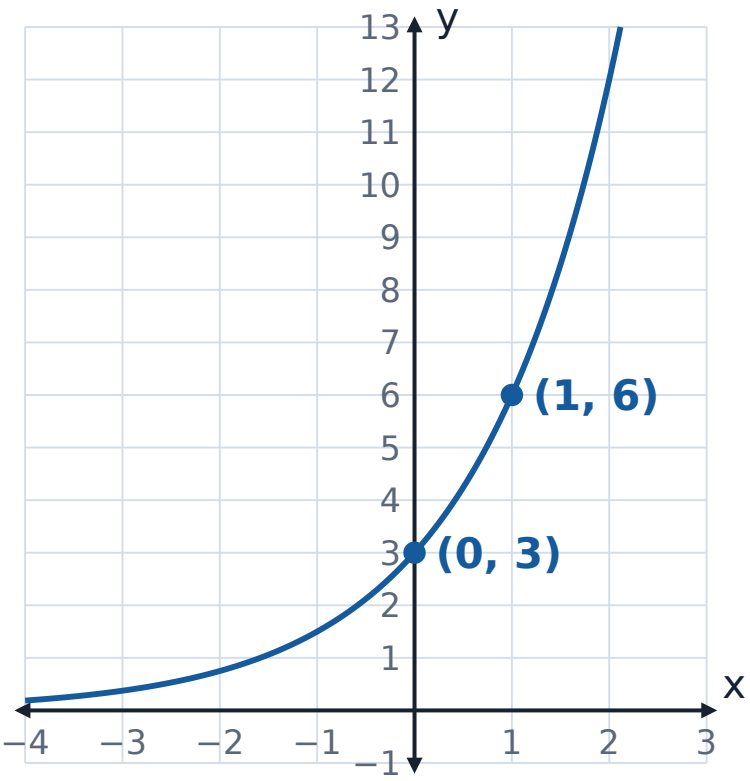
k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q1/5

$f(x) = 2^x$. The graph of $g(x) = a \cdot f(x)$ is shown. Find a.



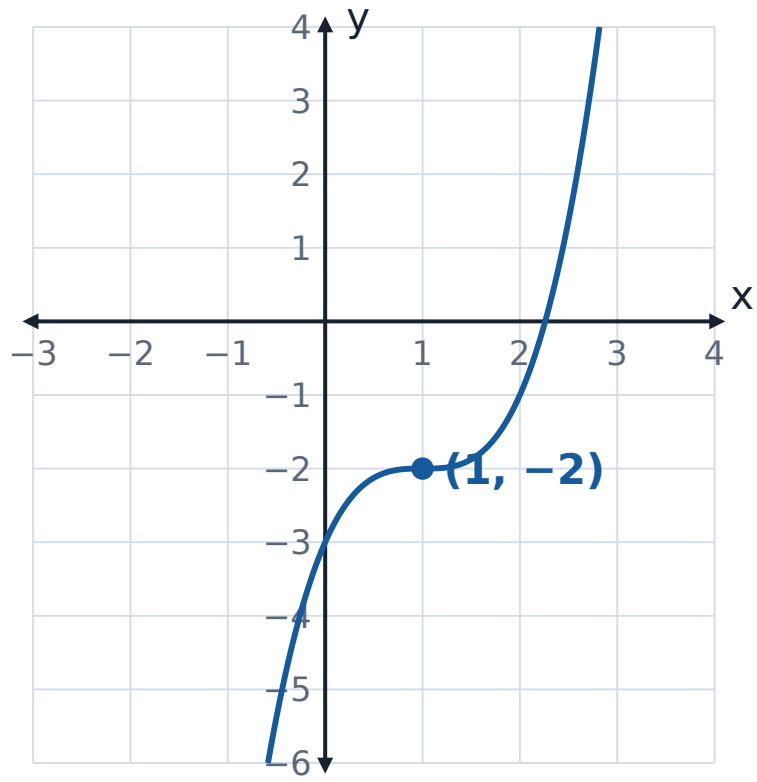
a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q2/5

$f(x) = x^3$. The graph of $g(x) = f(x - h) + k$ is shown. Find h and k.



h = _____

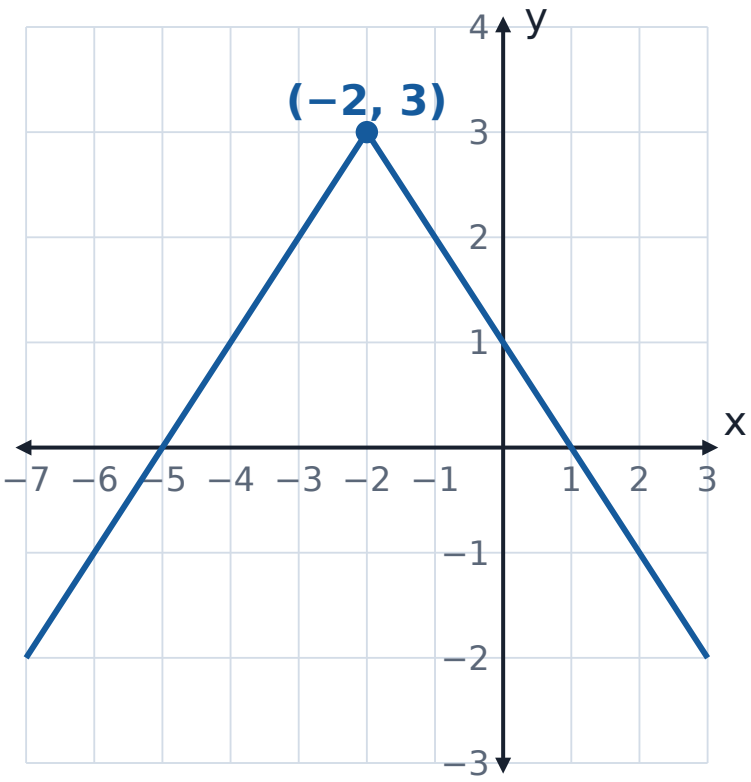
k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q3/5

$f(x) = |x|$. The graph of $g(x) = a \cdot f(x - h) + k$ is shown. Find a, h, and k.



a = _____

h = _____

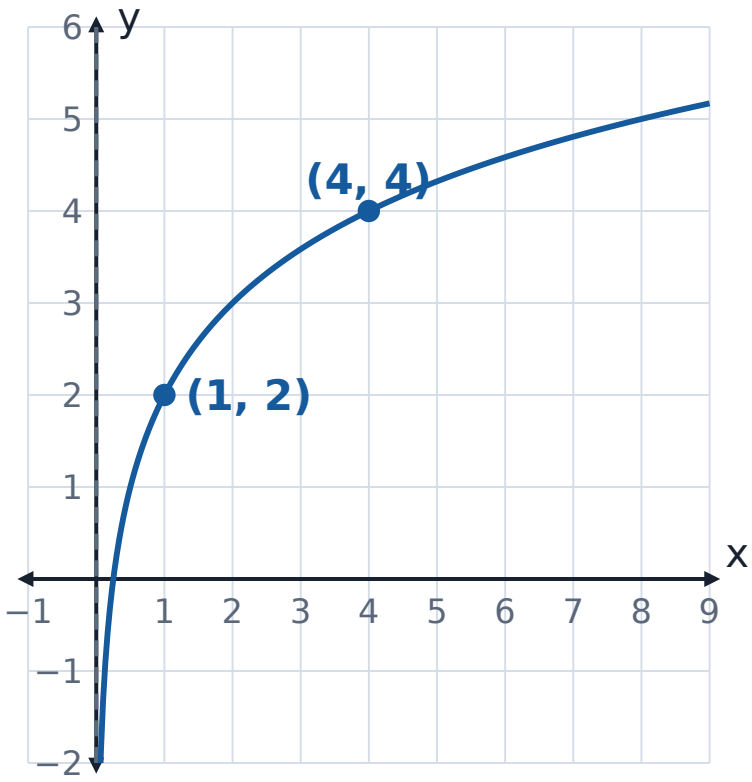
k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q4/5

$f(x) = \log_2 x$. The graph of $g(x) = f(x) + k$ is shown. Find k.



$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q5/5

SET 55 • HSF-BF.B.3

BACKWARD 1

Shifts and stretches of linear and quadratic graphs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

How is the graph of $y = (x - 3)^2 + 1$ obtained from the graph of $y = x^2$?

A. Shift left 3 and up 1

B. Shift up 3 and right 1

C. Shift right 3 and up 1

D. Shift right 3 and down 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q1/5

How is the graph of $y = 2x^2$ obtained from the graph of $y = x^2$?

A. Shift up 2

B. Vertical stretch by 2

C. Vertical compression by 1/2

D. Horizontal stretch by 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q2/5

True or false?

$y = f(x + 4)$ is the graph of f shifted right 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q3/5

The vertex of $y = (x + 2)^2 - 5$ is

A. $(2, 5)$

B. $(2, -5)$

C. $(-2, -5)$

D. $(-2, 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q4/5

Write $y = x^2$ shifted left 1 and down 4.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q5/5

SET 55 • HSF-BF.B.3

BACKWARD 2

Translate points in the plane • Grade 8 • Geometry • 8.G.A.3

Translate (0, 0) right 2 and up 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q1/5

Translate (1, 1) left 2 and up 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q2/5

True or false?

Translating (0, 0) down 2 gives (−2, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q3/5

$(x, y) \to (x - 2, y + 3)$ moves $(0, 0)$ to

A. $(-2, 3)$

B. $(-2, -3)$

C. $(2, 3)$

D. $(3, -2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q4/5

Which translation takes the point (0, 0) to (1, −2)?

A. Up 1 and left 2

B. Right 2 and down 1

C. Right 1 and down 2

D. Left 1 and up 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • HSF-BF.B.3

BACKWARD 3

Key points of parent functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

Give the point of $y = \sqrt{x}$ at $x = 0$ and at $x = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q1/5

Give the y-intercept of $y = 2^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q2/5

True or false?

The graph of $y = \log_2 x$ passes through (1, 0).

A. True

B. False

The vertex of $y = |x|$ is

A. (1, 0)

B. (1, 1)

C. (0, 1)

D. (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q4/5

Give the center (point of symmetry) of $y = x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q5/5

SET 55 • HSF-BF.B.3

FORWARD 1

Write transformed functions from descriptions and order of transformations • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

Write g, the graph of $f(x) = 2^x$ stretched vertically by 3, then shifted down 4.

$g(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q1/5

Which is g, the graph of $f(x) = \sqrt{x}$ shifted down 4 and then stretched vertically by 3?

A. $g(x) = 3\sqrt{(x - 4)}$

B. $g(x) = 3\sqrt{x} - 4$

C. $g(x) = \sqrt{(3x)} - 4$

D. $g(x) = 3\sqrt{x} - 12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q2/5

True or false?

A vertical stretch then a shift can give a different graph than the same shift then the stretch.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q3/5

$g(x) = 2f(x) + 1$ moves the point (3, 4) of f to

A. (6, 5)

B. (4, 9)

C. (3, 9)

D. (3, 10)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q4/5

$g(x) = f(x + 2) - 3$ and f has asymptote $x = 0$. Where is g's vertical asymptote?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q5/5

SET 55 • HSF-BF.B.3

FORWARD 2

Read sinusoid parameters as transformations • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

$y = 3\sin x + 2$ is a transformation of $y = \sin x$. Give the vertical stretch factor, the upward shift, and the maximum value.

Vertical stretch factor: _____

Upward shift: _____

Maximum value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q1/5

The graph of $y = \cos x$ is stretched vertically by 4 and then shifted down 1. Write the result in the form $y = A \cdot \cos x + D$.

A = _____

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q2/5

True or false?

In $y = A \sin x + D$, A is the midline.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q3/5

$y = -2\cos x + 5$ has amplitude and midline

A. 5 and $y = 2$

B. -2 and $y = 5$

C. 2 and $y = -5$

D. 2 and $y = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q4/5

A model is $y = A \sin x + D$ with maximum 9 and minimum 1. Find A ($A > 0$) and D.

$A =$ _____

$D =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q5/5

Horizontal stretches, compressions, and reflections f(kx)

MAIN	Algebra II main question	**HSF-BF.B.3**
BACKWARD 1	Reflections and dilations of points	**8.G.A.3**
BACKWARD 2	Solve kx = c for x	**6.EE.B.7**
BACKWARD 3	Period of y = sin(Bx)	**HSF-TF.B.5**
FORWARD 1	Combine horizontal stretches with shifts	**HSF-BF.B.3**
FORWARD 2	Graph exponential and trigonometric functions with horizontal cha…	**HSF-IF.C.7.e**

SET 56 • HSF-BF.B.3

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

$f(x) = \sqrt{x}$ passes through (4, 2). For $g(x) = \sqrt{(4x)}$, by what number are the x-coordinates of points of f multiplied, and which point of g has y = 2?

x-coordinates are multiplied by: _____

Point of g with y = 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q1/5

$f(x) = \sin x$ has period 2π. For $g(x) = \sin(3x)$, by what number are the x-coordinates of points of f multiplied, and what is the period of g?

x-coordinates are multiplied by: _____

Period of g: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q2/5

$f(x) = 2^x$ passes through (1, 2). For $g(x) = 2^{-x}$, by what number are the x-coordinates of points of f multiplied, and which point of g comes from (1, 2)?

x-coordinates are multiplied by: _____

Point of g: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q3/5

f passes through (−2, 1), (0, 3), and (4, −1). List the corresponding points of g(x) = f(2x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q4/5

f passes through (2, 5). For g(x) = f(−x/2), by what number are the x-coordinates of points of f multiplied, and which point must be on the graph of g?

x-coordinates are multiplied by: _____

Point of g: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q5/5

SET 56 • HSF-BF.B.3

BACKWARD 1

Reflections and dilations of points • Grade 8 • Geometry • 8.G.A.3

Reflect (1, 2) over the y-axis.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q1/5

Multiply the x-coordinate of $(4, -1)$ by $1/2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q2/5

True or false?

Reflecting over the y-axis changes the sign of the x-coordinate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q3/5

(x, y) → (x/3, y) applied to (6, 2) gives

A. (18, 2)

B. (6, 2/3)

C. (2, 2)

D. (3, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q4/5

Apply $(x, y) \to (-2x, y)$ to $(2, 5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • HSF-BF.B.3

BACKWARD 2

Solve kx = c for x • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve 2x = 4.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q1/5

Solve $-x/2 = 2$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q2/5

True or false?

If 4x = 4, then x = 16.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q3/5

$3x = 2\pi$ gives $x =$

A. $2\pi/3$

B. $2\pi - 3$

C. $\pi/3$

D. 6π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q4/5

Solve $-x = 1$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • HSF-BF.B.3

BACKWARD 3

Period of y = sin(Bx) • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

Give the period of $y = \sin(2x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q1/5

Give the period of $y = \cos(x/2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q2/5

True or false?

For y = sin(Bx) with B > 0, a larger B gives a longer period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q3/5

$y = \sin(4x)$ has period

A. 8π

B. 4π

C. $\pi/4$

D. $\pi/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q4/5

For $y = \sin(Bx)$ with $B > 0$, which B gives period 6? Give B exactly, in terms of π.

$B =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q5/5

SET 56 • HSF-BF.B.3

FORWARD 1

Combine horizontal stretches with shifts • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

Rewrite $g(x) = f(2x - 6)$ in the form $f(k(x - h))$. Give k and h.

$k =$ _____

$h =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q1/5

f passes through (4, 1). Which point is on $g(x) = f(2x - 6)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q2/5

True or false?

f(2x − 6) is f(2x) shifted right 6.

A. True

B. False

$y = \sqrt{(9x)}$ equals

A. $\sqrt{x} + 9$

B. $9\sqrt{x}$

C. $3\sqrt{x}$

D. $\sqrt{x}/3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q4/5

Write g, the graph of $f(x) = 2^x$ reflected over the y-axis and then shifted right 1.

$g(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q5/5

SET 56 • HSF-BF.B.3

FORWARD 2

Graph exponential and trigonometric functions with horizontal changes • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

For $y = 2^{-x} + 1$, give the y-intercept, the horizontal asymptote, and the y-values at $x = -1$ and at $x = 1$.

y-intercept: _____

Asymptote: $y =$ _____

y at $x = -1$: _____

y at $x = 1$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q1/5

On $0 \le x \le 2\pi$, at which x-values does $y = \cos(2x)$ reach its maximum? List them separated by commas, exactly in terms of π.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q2/5

True or false?

$y = 3^{2x} = 9^{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q3/5

$y = \sin(x/2)$ completes one cycle on

A. $0 \le x \le \pi$

B. $0 \le x \le 2\pi$

C. $0 \le x \le \pi/2$

D. $0 \le x \le 4\pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q4/5

Why is $y = 2^{2x}$ both a horizontal compression of $y = 2^x$ and an exponential function with base 4?

A. Because $2^{2x} = 2^x + 2^x$, which doubles each output, and doubling 2 gives the base 4.

B. Replacing x by 2x compresses the graph horizontally by a factor of 1/2, and $2^{2x} = (2^2)^x = 4^x$.

C. Because $2^{2x} = 2 \cdot 2^x$, a horizontal compression by 2, and $2 \cdot 2$ gives the base 4.

D. Replacing x by 2x stretches the graph vertically by a factor of 2, so the base 2 becomes $2 \cdot 2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q5/5

Recognize even and odd functions

MAIN	Algebra II main question	**HSF-BF.B.3**
BACKWARD 1	Signs of powers of negatives	**7.NS.A.2**
BACKWARD 2	Reflect points over the axes and the origin	**6.NS.C.6.b**
BACKWARD 3	Substitute -x into an expression	**HSF-IF.A.2**
FORWARD 1	Reason about sums and products of even and odd functions	**HSF-BF.B.3**
FORWARD 2	Explain trigonometric symmetry with the unit circle	**HSF-TF.A.4**

SET 57 • HSF-BF.B.3

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

Is $f(x) = x^4 - 3x^2$ even, odd, or neither?

A. Even: $f(0) = 0$, and a graph through the origin is even.

B. Even: $f(-x) = (-x)^4 - 3(-x)^2 = x^4 - 3x^2 = f(x)$.

C. Odd: $f(-x) = -x^4 + 3x^2 = -f(x)$, because $-x$ makes each term negative.

D. Neither: $f(1) = -2$ and $f(-1) = 2$, so $f(-1) \neq f(1)$ and f is not even.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-M-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q1/5

Is $f(x) = x^3 - x$ even, odd, or neither?

A. Odd: $f(0) = 0$, and every function through the origin is odd.

B. Even: $f(-x) = x^3 - x = f(x)$, because the signs cancel in each term.

C. Odd: $f(-x) = (-x)^3 - (-x) = -x^3 + x = -f(x)$.

D. Neither: $f(-x) = -x^3 - x$, which is neither $f(x)$ nor $-f(x)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q2/5

Is $f(x) = x^2 + x$ even, odd, or neither?

A. Even: $f(0) = 0$, and a graph through the origin is symmetric about the y-axis.

B. Even: x^2 is even, and the term of highest degree decides whether f is even.

C. Odd: x is odd, and adding an odd term to x^2 makes the whole function odd.

D. Neither: $f(1) = 2$ and $f(-1) = 0$, so $f(-1) \neq f(1)$ and $f(-1) \neq -f(1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q3/5

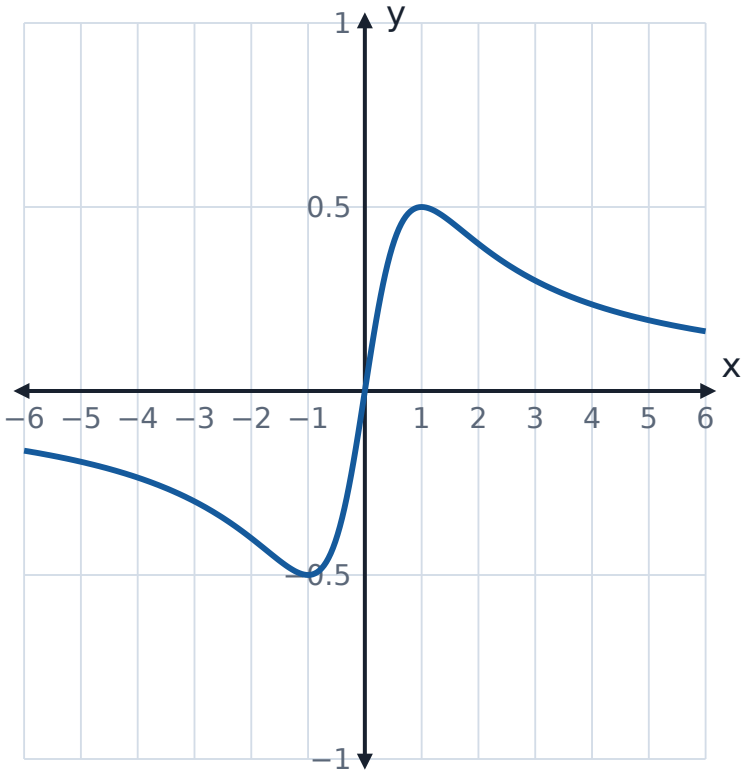
The graph of $f(x) = x/(x^2 + 1)$ is shown. Is f even, odd, or neither?

A. Even: the graph is symmetric about the y-axis (reflecting it over the y-axis gives the same graph).

B. Odd: the graph passes through the origin, and every graph through the origin is odd.

C. Odd: the graph is symmetric about the origin (rotating it 180° about the origin gives the same graph).

D. Neither: the graph has a peak on one side and a low point on the other, so it has no symmetry.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q4/5

Is $f(x) = \cos x + x^2$ even, odd, or neither?

A. Odd: cos is odd.

B. Even: $f(-x) = \cos(-x) + (-x)^2 = \cos x + x^2 = f(x)$.

C. Odd: $f(-x) = -\cos x + x^2$.

D. Neither: cos x and x^2 are different kinds of functions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q5/5

SET 57 • HSF-BF.B.3

BACKWARD 1

Signs of powers of negatives • Grade 7 • The Number System • 7.NS.A.2

Simplify $(-x)^4$ and $(-x)^3$.

$(-x)^4 =$ _____

$(-x)^3 =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q1/5

Simplify $(-x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q2/5

True or false?

$(-x)^5 = x^5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q3/5

$(-2)^2 - (-2)$ equals

A. -6

B. 2

C. 6

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q4/5

Simplify $-(-x)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q5/5

SET 57 • HSF-BF.B.3

BACKWARD 2

Reflect points over the axes and the origin • Grade 6 • The Number System • 6.NS.C.6.b

Reflect (2, 3) over the y-axis.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q1/5

Rotate (2, 3) 180° about the origin.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q2/5

True or false?

A graph symmetric about the y-axis contains $(-a, b)$ whenever it contains (a, b).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q3/5

Symmetry about the origin pairs (a, b) with

A. (−a, b)

B. (−a, −b)

C. (a, −b)

D. (b, a)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q4/5

A graph is symmetric about the origin and contains (1, 0.5). Name another point.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q5/5

SET 57 • HSF-BF.B.3

BACKWARD 3

Substitute −x into an expression • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

For $f(x) = x^2 + x$, find $f(-x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q1/5

For $f(x) = x^3 - x$, find $f(-2)$ and $f(2)$.

$f(-2) =$ _____

$f(2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q2/5

True or false?

For $f(x) = 3x$, $f(-x) = 3x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q3/5

For $f(x) = x^4 - 3x^2$, $f(-1)$ is

A. 2

B. 4

C. -2

D. -4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q4/5

For $f(x) = 1/x$, find $f(-x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q5/5

SET 57 • HSF-BF.B.3

FORWARD 1

Reason about sums and products of even and odd functions • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

Which argument proves that the sum of two odd functions f and g is odd?

A. x^3 and x^5 are odd and $x^3 + x^5$ is odd, so every sum of two odd functions is odd.

B. $(f + g)(-x) = f(-x) + g(-x) = -f(x) - g(x) = -(f(x) + g(x)) = -(f + g)(x)$.

C. $(f + g)(-x) = f(x) + g(x) = (f + g)(x)$, so the sum is odd.

D. $(f + g)(-x) = -f(-x) - g(-x) = -(f + g)(-x)$, so the sum is odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q1/5

Which argument proves that the product of two odd functions f and g is even?

A. $(fg)(-x) = f(-x)g(-x) = (-f(x))(-g(x)) = f(x)g(x) = (fg)(x)$.

B. $x \cdot x^3 = x^4$ is even, so every product of two odd functions is even.

C. Odd times odd is odd for whole numbers, so the product of two odd functions is odd.

D. $(fg)(-x) = f(-x)g(-x) = -f(x)g(x)$, so the product is odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q2/5

True or false?

The only function that is both even and odd is $f(x) = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q3/5

$f(x) = |x| + x^3$ is

A. even

B. both

C. neither

D. odd

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q4/5

Is $f(x) = x \sin x$ even, odd, or neither?

A. Odd: $f(-x) = (-x) \sin x = -x \sin x = -f(x)$, since only the first factor changes sign.

B. Odd: sin x is odd, and multiplying an odd function by x keeps it odd, so x sin x is odd.

C. Even: x and sin x are both odd, and a product of two odd functions is even.

D. Neither: x is not a trigonometric function, so the rules for even and odd functions do not apply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q5/5

SET 57 • HSF-BF.B.3

FORWARD 2

Explain trigonometric symmetry with the unit circle • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

Why is $\cos(-x) = \cos x$, using the unit circle?

A. The points at angles x and $-x$ are reflections of each other over the x-axis, which keeps the x-coordinate (the cosine).

B. The points at angles x and $-x$ are reflections of each other over the y-axis, which keeps the x-coordinate (the cosine).

C. The points at angles x and $-x$ are the same point on the unit circle, so they have the same x-coordinate.

D. Because cos is never negative, so $\cos(-x)$ and $\cos x$ always have the same value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q1/5

Is tan even, odd, or neither?

A. Odd: $\tan(-x) = \sin(-x)/\cos(-x) = -\sin x/\cos x = -\tan x$.

B. Even: cos is even, so $\tan(-x) = \sin x/\cos(-x) = \sin x/\cos x = \tan x$.

C. Even: $\tan(-x) = \sin x/\cos x = \tan x$.

D. Neither: tan has vertical asymptotes, so it cannot be even or odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q2/5

True or false?

sin is an even function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q3/5

Which is an even function?

A. cos x

B. tan x

C. sin x + cos x

D. sin x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q4/5

Use symmetry to find $\sin(-\pi/6)$ from $\sin(\pi/6) = 1/2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q5/5

Write inverses of radical, rational, cubic, exponential, and logarithmic functions

MAIN	Algebra II main question	**HSF-BF.B.4.a**
BACKWARD 1	Inverses of linear functions	**HSF-BF.B.4.a**
BACKWARD 2	Rearrange formulas with roots, powers, and fractions	**HSA-CED.A.4**
BACKWARD 3	Domain and range	**HSF-IF.B.5**
FORWARD 1	Inverses that are their own inverse, and checking by points	**HSF-BF.B.4.a**
FORWARD 2	Verify inverses by composition	**HSF-BF.B.4.b**

SET 58 • HSF-BF.B.4.a

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

Find the inverse of $f(x) = (x - 1)^3 + 2$ (all real x). It has the form $f^{-1}(x) = \sqrt[3]{(x - a)} + b$; give a and b.

$a =$ _____

$b =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q1/5

Find the inverse of $f(x) = 2x/(x - 3)$, $x \neq 3$, and the value excluded from its domain.

$f^{-1}(x) =$ _____

Excluded value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q2/5

Find the inverse of $f(x) = \sqrt{(x + 4)}$, $x \geq -4$, and its domain.

$f^{-1}(x) =$ _____

Domain of f^{-1}: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q3/5

Which gives the inverse of $f(x) = 5 \cdot 2^x$ and its domain?

A. $f^{-1}(x) = \log_5(x/2),\ x > 0$

B. $f^{-1}(x) = \log_2(x) - 5,\ x > 0$

C. $f^{-1}(x) = \log_2(x/5)$, all real x

D. $f^{-1}(x) = \log_2(x/5),\ x > 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q4/5

Which gives the inverse of $f(x) = \log_3(x - 1)$ and its domain?

A. $f^{-1}(x) = 3^x + 1,\ x > 1$

B. $f^{-1}(x) = 3^x + 1$, all real x

C. $f^{-1}(x) = 3^{x+1}$, all real x

D. $f^{-1}(x) = x^3 + 1$, all real x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q5/5

SET 58 • HSF-BF.B.4.a

BACKWARD 1

Inverses of linear functions • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

Find the inverse of $f(x) = 3x - 6$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q1/5

Solve 2x + 5 = y for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q2/5

True or false?

The inverse undoes the function: $f^{-1}(f(x)) = x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q3/5

The inverse of $f(x) = x/4 + 1$ is

A. $4(x - 1)$

B. $(x - 1)/4$

C. $4x - 1$

D. $x/4 - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q4/5

If $f(2) = 7$, what is $f^{-1}(7)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q5/5

SET 58 • HSF-BF.B.4.a

BACKWARD 2

Rearrange formulas with roots, powers, and fractions • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

Solve $y = \sqrt{(x + 4)}$ for x ($y \geq 0$).

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q1/5

Solve $y = 2x/(x - 3)$ for x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q2/5

True or false?

Solving $y = (x - 1)^3 + 2$ for x gives $x = \sqrt[3]{(y - 2)} + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q3/5

Solving $y = x^3$ for x gives

A. $y/3$

B. $\sqrt[3]{y}$

C. $\sqrt{y}$

D. y^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q4/5

Solve $y = x/(x + 1)$ for x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q5/5

SET 58 • HSF-BF.B.4.a

BACKWARD 3

Domain and range • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

Give the range of $f(x) = \sqrt{(x + 4)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q1/5

Give the range of $f(x) = 5 \cdot 2^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q2/5

True or false?

The range of $f(x) = 2x/(x - 3)$ excludes $y = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q3/5

The domain of $\log_3(x - 1)$ is

A. $x > 1$

B. all reals

C. $x > 0$

D. $x \geq 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q4/5

What is the range of $f(x) = \log_3(x - 1)$?

A. $y > 0$

B. $y > 1$

C. All real numbers

D. $y \geq 0$

SET 58 • HSF-BF.B.4.a

FORWARD 1

Inverses that are their own inverse, and checking by points • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

$f(x) = (x + 1)/(x - 1)$, $x \neq 1$. Find $f^{-1}(x)$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q1/5

$f(x) = 5 \cdot 2^x$ passes through (1, 10), and $f^{-1}(x) = \log_2(x/5)$. Check the inverse with this point: find $f^{-1}(10)$.

$f^{-1}(10) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q2/5

True or false?

Every function has an inverse function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q3/5

Which is its own inverse?

A. $f(x) = x^2$

B. $f(x) = x + 1$

C. $f(x) = 2x$

D. $f(x) = 1/x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q4/5

Use the horizontal line test on the graph of $f(x) = x^3 - x$. Does f have an inverse function?

A. No: the horizontal line y = 0 meets the graph three times (x = −1, 0, 1), so f is not one-to-one.

B. Yes: f is a polynomial, and every polynomial passes the horizontal line test, so f has an inverse function.

C. Yes: every cubic passes the horizontal line test, because its graph rises from left to right, so f is one-to-one.

D. No: the vertical line x = 0 meets the graph three times, so f fails the vertical line test and is not one-to-one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q5/5

SET 58 • HSF-BF.B.4.a

FORWARD 2

Verify inverses by composition • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

$f(x) = (x - 1)^3 + 2$ and $g(x) = \sqrt[3]{(x - 2)} + 1$. Which computation of $f(g(x))$ is correct?

A. $f(g(x)) = f(x) \cdot g(x) = ((x - 1)^3 + 2)(\sqrt[3]{(x - 2)} + 1) = x$

B. $f(g(x)) = ((\sqrt[3]{(x - 2)} + 1) - 1)^3 + 1 = (\sqrt[3]{(x - 2)})^3 + 1 = x - 1$

C. $f(g(x)) = ((\sqrt[3]{(x - 2)} + 1) - 1)^3 + 2 = (\sqrt[3]{(x - 2)})^3 + 2 = x - 2 + 2 = x$

D. $f(g(x)) = \sqrt[3]{(((x - 1)^3 + 2) - 2)} + 1 = \sqrt[3]{((x - 1)^3)} + 1 = x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q1/5

$f(x) = (x - 1)^3 + 2$ and $g(x) = \sqrt[3]{(x - 2)} + 1$. Which computation of $g(f(x))$ is correct?

A. $g(f(x)) = \sqrt[3]{(((x - 1)^3 + 2) - 2)} + 1 = \sqrt[3]{((x - 1)^3)} + 1 = x - 1 + 1 = x$

B. $g(f(x)) = g(x) \cdot f(x) = (\sqrt[3]{(x - 2)} + 1)((x - 1)^3 + 2) = x$

C. $g(f(x)) = ((\sqrt[3]{(x - 2)} + 1) - 1)^3 + 2 = (\sqrt[3]{(x - 2)})^3 + 2 = x$

D. $g(f(x)) = \sqrt[3]{(((x - 1)^3 + 2) - 2)} + 2 = \sqrt[3]{((x - 1)^3)} + 2 = x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q2/5

True or false?

To verify inverses, both $f(g(x)) = x$ and $g(f(x)) = x$ must hold on the domains.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q3/5

For $f(x) = \sqrt{(x + 4)}$ and $g(x) = x^2 - 4$ $(x \geq 0)$, $g(f(x)) =$

A. x for all x

B. $|x|$

C. x for $x \geq -4$

D. $x^2 + 4$

Why is $g(x) = x^2 - 4$, without a domain restriction, not the inverse of $f(x) = \sqrt{(x + 4)}$?

A. Because $f(g(2)) \neq 2$, so the composition fails even for positive inputs.

B. $f(g(x)) = \sqrt{(x^2)} = |x|$, which is not x when $x < 0$; g is the inverse only on $x \geq 0$ (the range of f).

C. Because $g(x) = x^2 - 4$ is not a function, since it gives two outputs for each input.

D. Because $g(f(x)) = (\sqrt{(x + 4)})^2 - 4$ is undefined for every x, so the composition test cannot be done.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q5/5

Write exact logarithmic solutions of exponential equations

MAIN	Algebra II main question	**HSF-LE.A.4**
BACKWARD 1	Isolate the exponential expression	**HSA-REI.B.3**
BACKWARD 2	Powers of 2, 3, and 10	**8.EE.A.1**
BACKWARD 3	Logarithms as the inverse of exponentials	**HSF-BF.B.4.a**
FORWARD 1	Solve equations with different bases or variables on both sides	**HSF-LE.A.4**
FORWARD 2	Use the inverse relationship to solve logarithmic equations	**HSF-BF.B.5**

SET 59 • HSF-LE.A.4

MAIN

Algebra II main question • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $3 \cdot 2^{5t} = 48$ exactly.

$t =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q1/5

Which is the exact solution of $2 \cdot 10^{3t} = 9$?

A. $t = \log(9)/6$

B. $t = (\log 9 - 2)/3$

C. $t = 3 \log 4.5$

D. $t = \log(4.5)/3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q2/5

Which is the exact solution of $5e^{2t} = 30$?

A. $t = (\ln 6) \cdot 2$

B. $t = (\ln 6)/2$

C. $t = \ln 3$

D. $t = \ln 30/10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q3/5

Which is the exact solution of $100(1.05)^t = 300$?

A. $t = 3/1.05$

B. $t = \log 3 - \log 1.05$

C. $t = \log 3/\log 1.05$

D. $t = \log 300/\log 1.05$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q4/5

Which is the exact solution of $4 \cdot 3^{t/2} = 20$?

A. $t = 2 \log_3 5$

B. $t = \log_3 10$

C. $t = 2 \log_3 20$

D. $t = \log_3 5/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q5/5

SET 59 • HSF-LE.A.4

BACKWARD 1

Isolate the exponential expression • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve 3y = 48 for y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q1/5

Solve 5y = 30 for y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q2/5

True or false?

From $2 \cdot 10^{3t} = 9$, we get $10^{3t} = 4.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q3/5

From $4 \cdot 3^{t/2} = 20$, $3^{t/2}$ equals

A. 80

B. 5

C. 1/5

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q4/5

Solve 100y = 300 for y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q5/5

SET 59 • HSF-LE.A.4

BACKWARD 2

Powers of 2, 3, and 10 • Grade 8 • Expressions & Equations • 8.EE.A.1

Write 16 as a power of 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q1/5

Write 1000 as a power of 10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q2/5

True or false?

$3^4 = 12$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q3/5

$2^{5t} = 2^4$ gives

A. $t = 4$

B. $5t = 4$

C. $5t = 16$

D. $t = 20$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q4/5

Write 1/8 as a power of 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q5/5

SET 59 • HSF-LE.A.4

BACKWARD 3

Logarithms as the inverse of exponentials • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

Rewrite $10^x = 4.5$ as a logarithm.

A. $x = 10^{4.5}$

B. $x = 4.5/10$

C. $x = \log 10/4.5$

D. $x = \log 4.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q1/5

Rewrite $e^u = 6$ as a logarithm.

A. $u = \log 6$

B. $u = 6/e$

C. $u = \ln 6$

D. $u = e^6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q2/5

True or false?

$\log_2 8 = 4$ because $2^4 = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q3/5

$3^y = 5$ means y =

A. $\log_3 5$

B. 5/3

C. $\log_5 3$

D. 3/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q4/5

Evaluate $\log_2 32$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q5/5

SET 59 • HSF-LE.A.4

FORWARD 1

Solve equations with different bases or variables on both sides • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $2^x = 3^{x-1}$. Give x to the nearest thousandth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q1/5

Solve $5^{2x} = 7$. Give x to the nearest thousandth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q2/5

True or false?

$\log(ab) = \log a + \log b$ for positive a and b.

A. True

B. False

The exact solution of $e^{x+1} = 10$ is

A. 10/e

B. ln 10/1

C. ln 9

D. ln 10 − 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q4/5

Solve $3 \cdot 4^x = 2 \cdot 5^x$. Give x to the nearest thousandth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q5/5

SET 59 • HSF-LE.A.4

FORWARD 2

Use the inverse relationship to solve logarithmic equations • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5

Solve $\log_2(x + 3) = 5$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q1/5

Solve $\log x + \log(x - 3) = 1$. Give every value found after combining the logarithms and solving, and the actual solutions (enter none if there are none).

Values found: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q2/5

True or false?

$\log_b(b^x) = x$ for every real x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q3/5

$\ln(e^7)$ equals

A. $\ln 7$

B. 1

C. 7

D. e^7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q4/5

Solve $\ln(2x) = 3$ exactly.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q5/5

Evaluate logarithmic solutions of exponential models with technology

MAIN	Algebra II main question	**HSF-LE.A.4**
BACKWARD 1	Write exact logarithmic solutions	**HSF-LE.A.4**
BACKWARD 2	Evaluate logarithms with a calculator	**HSF-IF.A.2**
BACKWARD 3	Choose an appropriate level of accuracy	**HSN-Q.A.3**
FORWARD 1	Analyze how parameters change the solution	**HSF-LE.A.4**
FORWARD 2	Rewrite models to show doubling time or half-life	**HSF-IF.C.8.b**

SET 60 • HSF-LE.A.4

MAIN

Algebra II main question • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$2,500 earns 3.5% interest, credited once at the end of each year. Solve $2500(1.035)^t = 4000$ for t (to the nearest hundredth of a year), and give the number of yearly credits after which the balance first reaches at least $4,000. A calculator may be used.

t (years): _____

Number of yearly credits: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q1/5

Carbon-14 has a half-life of 5730 years. A sample has 30 of its original 80 units of carbon-14. Solve $80(0.5)^{t/5730} = 30$ to estimate its age to the nearest year. A calculator may be used.

Age (years): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q2/5

Coffee cools by $T(t) = 20 + 70(0.95)^t$ °C after t minutes. When is it 50°C? Give minutes to the nearest hundredth. A calculator may be used.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q3/5

A city of 1.2 million grows by $P(t) = 1.2e^{0.015t}$ million people. When does it reach 1.5 million? Give years to the nearest hundredth. A calculator may be used.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q4/5

A dose leaves $200(0.8)^t$ mg of a drug after t hours. At which whole hour is the amount first below 50 mg? A calculator may be used.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q5/5

SET 60 • HSF-LE.A.4

BACKWARD 1

Write exact logarithmic solutions • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Which is the exact solution of $1.035^t = 1.6$?

A. $t = \log(1.6 - 1.035)$

B. $t = \log 1.6/\log 1.035$

C. $t = \log 1.035/\log 1.6$

D. $t = 1.6/1.035$

Which is the exact solution of $e^{0.015t} = 1.25$?

A. t = ln 1.25/0.015

B. t = 1.25/0.015

C. t = ln(1.25/0.015)

D. t = 0.015 ln 1.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q2/5

True or false?

$0.8^t = 0.25$ gives $t = \log 0.25 / \log 0.8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q3/5

$(0.5)^{t/5730} = 0.375$ gives t =

A. $0.375 \cdot 5730$

B. $5730/\log 0.375$

C. $\log 0.375/5730$

D. $5730 \log 0.375/\log 0.5$

Which is the exact solution of $0.95^t = 3/7$?

A. $t = \log(3/7) - \log 0.95$

B. $t = (3/7)/0.95$

C. $t = \log 0.95/\log(3/7)$

D. $t = \log(3/7)/\log 0.95$

SET 60 • HSF-LE.A.4

BACKWARD 2

Evaluate logarithms with a calculator • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Evaluate log 1.6 to four decimal places. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q1/5

Evaluate ln 1.25 to four decimal places. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q2/5

True or false?

log 0.25 is positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q3/5

log 1.6/log 1.035 is about

A. 0.20

B. 46.0

C. 1.55

D. 13.66

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q4/5

Evaluate log 0.25/log 0.8 to two decimal places. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q5/5

SET 60 • HSF-LE.A.4

BACKWARD 3

Choose an appropriate level of accuracy • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3

Interest is added once a year. A model says a balance reaches its goal at t = 13.66 years. After how many whole years is the goal first reached?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q1/5

Round 8108.16 years to the nearest hundred years.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q2/5

True or false?

A time found from a model should be reported with more decimal places than the data justify.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q3/5

A drug amount first drops below 50 mg at t = 6.21 h. If it is checked every hour, it is first below 50 mg at

A. 6.21 h

B. 6 h

C. 7 h

D. 5 h

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q4/5

A population model gives t = 14.88 years. Why might this be reported as "about 15 years"?

A. Because rounding a model's answer up always gives the exact answer for the population.

B. Because 14.88 is not a real number of years, so it must be replaced by a whole number.

C. The model and its data are approximate, so reporting more precision than they justify would be misleading.

D. Because answers about years must always be whole numbers, whatever the model shows.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q5/5

SET 60 • HSF-LE.A.4

FORWARD 1

Analyze how parameters change the solution • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$2,500 is invested at 5% interest, credited once at the end of each year. Using the model $2500(1.05)^t$, find the time t when the model first equals $4,000 (to the nearest hundredth of a year), and the number of yearly credits after which the account first holds at least $4,000. A calculator may be used.

Model time t (years): _____

Number of yearly credits: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q1/5

Money earns 6% interest per year. Using the model $(1.06)^t = 2$, find the doubling time to the nearest hundredth of a year, the Rule of 72 estimate 72/6, and the number of yearly credits after which the money is first at least doubled when interest is credited once a year. A calculator may be used.

Model doubling time (years): _____

Rule of 72 estimate: _____

Number of yearly credits: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F1-Q2**

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q2/5

True or false?

Doubling the initial amount in $a(b)^t = d$ halves the time t needed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q3/5

In $200(0.8)^t = 50$, replacing 200 by 400 changes t by

A. doubling it

B. halving it

C. adding $\log 2/\log(1/0.8) \approx 3.11$ hours

D. nothing

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q4/5

At 3.5% yearly interest, why does $2,500 take as long to grow to $4,000 as $5,000 takes to grow to $8,000?

A. Because $5,000 is twice $2,500, and twice the money grows twice as fast, which exactly cancels the doubled target.

B. Both need the growth factor 1.6, and t = log 1.6/log 1.035 depends only on that factor, not on the starting amount.

C. Because both amounts earn the same interest in dollars each year, and both need to gain the same number of dollars.

D. They do not take the same time: the larger amount earns more interest each year, so it reaches its target sooner.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F1-Q5**

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q5/5

SET 60 • HSF-LE.A.4

FORWARD 2

Rewrite models to show doubling time or half-life • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b

Rewrite $2500(1.035)^{t}$ in the form $2500 \cdot 2^{t/D}$ and give D, in years to the nearest hundredth. A calculator may be used.

$D \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q1/5

Rewrite $200(0.8)^{t}$ in the form $200(0.5)^{t/H}$ and give H, in hours to the nearest hundredth. A calculator may be used.

H ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q2/5

True or false?

$80(0.5)^{t/5730}$ shows the half-life 5730 directly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q3/5

$1.2e^{0.015t}$ has doubling time about

A. 46.2 years

B. 66.7 years

C. 15 years

D. 0.015 years

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q4/5

A substance decays by $A = A_0(0.5)^{t/h}$, where h is its half-life. Why does this form make it easy to find how much remains after 3 half-lives?

A. After 3 half-lives $t/h = 3$, so $A = A_0(0.5)^3 = A_0/8$.

B. Because the substance is gone after 3 half-lives.

C. Because 3 half-lives leave 3/8 of the amount.

D. Because $A = A_0 - 3h$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q5/5

SET 61 • HSS-ID.A.4 • STATISTICS, INFERENCE, AND DECISIONS

Estimate population percentages with a normal distribution

MAIN	Algebra II main question	**HSS-ID.A.4**
BACKWARD 1	Interpret mean and standard deviation	**HSS-ID.A.2**
BACKWARD 2	Describe the shape of a distribution	**HSS-ID.A.1**
BACKWARD 3	Percents of a whole	**6.RP.A.3.c**
FORWARD 1	Find values from percentiles of a normal model	**HSS-ID.A.4**
FORWARD 2	Use a normal model for a margin of error	**HSS-IC.B.4**

SET 61 • HSS-ID.A.4

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Adult women's heights are approximately normal with mean 64 in and standard deviation 2.5 in. Estimate the percent between 59 and 69 inches. Use the 68-95-99.7 rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q1

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q1/5

Adult women's heights are approximately normal with mean 64 in and standard deviation 2.5 in. Estimate the percent of women taller than 66.5 inches. Use the 68-95-99.7 rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q2

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q2/5

Test scores are approximately normal with mean 500 and standard deviation 100. Estimate the percent of scores between 400 and 700. Use the 68-95-99.7 rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q3/5

Battery life is approximately normal with mean 40 h and standard deviation 4 h. Use the table to estimate the percent of batteries lasting less than 46 hours.

z	−1.5	−1	−0.5	0	0.5	1	1.5	2
Area to the left	0.0668	0.1587	0.3085	0.5000	0.6915	0.8413	0.9332	0.9772

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q4

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q4/5

Cereal box weights are approximately normal with mean 16.0 oz and standard deviation 0.2 oz. Use the table to estimate the percent between 15.8 and 16.3 oz.

z	−1.5	−1	−0.5	0	0.5	1	1.5	2
Area to the left	0.0668	0.1587	0.3085	0.5000	0.6915	0.8413	0.9332	0.9772

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q5

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q5/5

SET 61 • HSS-ID.A.4

BACKWARD 1

Interpret mean and standard deviation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

A data set has mean 64 and standard deviation 2.5. What value is one standard deviation above the mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q1/5

Which value is two standard deviations below a mean of 500 with standard deviation 100?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q2/5

True or false?

A larger standard deviation means the data are more spread out.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q3/5

46 is how many standard deviations above a mean of 40 when the standard deviation is 4?

A. 0.67

B. 4

C. 6

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q4/5

Compute $(15.8 - 16.0)/0.2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • HSS-ID.A.4

BACKWARD 2

Describe the shape of a distribution • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Which describes a histogram that is highest in the middle and falls off evenly on both sides?

A. skewed right

B. uniform

C. bimodal

D. symmetric and mound-shaped

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q1/5

In a symmetric distribution, how do the mean and the median compare?

A. The median is much larger than the mean.

B. The mean is much larger than the median.

C. They are about equal, both at the center.

D. They are at opposite ends of the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q2/5

True or false?

A normal curve is symmetric about its mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q3/5

A bell-shaped histogram is called

A. skewed right

B. uniform

C. mound-shaped and symmetric

D. bimodal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q4/5

A continuous distribution is symmetric about its median. What percent of the distribution lies below the median? Give a percent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q5/5

SET 61 • HSS-ID.A.4

BACKWARD 3

Percents of a whole • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Find 95% of 2,000.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q1/5

What is 34% + 13.5%?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q2/5

True or false?

Half of 68% is 36%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q3/5

100% − 95% split equally into two tails gives each tail

A. 5%

B. 47.5%

C. 0.5%

D. 2.5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q4/5

Write 0.7745 as a percent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q5/5

SET 61 • HSS-ID.A.4

FORWARD 1

Find values from percentiles of a normal model • High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Women's heights are approximately normal with mean 64 inches and standard deviation 2.5 inches. Above what height are the tallest 2.5% of women? Give the height in inches. Use the 68-95-99.7 rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q1

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q1/5

Battery life is approximately normal with mean 40 hours and standard deviation 4 hours. Use the table to find the battery life at the 84th percentile, in hours.

z	−1.5	−1	−0.5	0	0.5	1	1.5	2
Area to the left	0.0668	0.1587	0.3085	0.5000	0.6915	0.8413	0.9332	0.9772

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q2

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q2/5

True or false?

In a normal model, the median equals the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q3/5

The 16th percentile of scores with mean 500 and SD 100 is

A. 484

B. 600

C. 340

D. 400

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q4

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q4/5

A cereal box below 15.7 oz is rejected. Use the table to estimate the percent rejected (mean 16.0, SD 0.2).

z	−1.5	−1	−0.5	0	0.5	1	1.5	2
Area to the left	0.0668	0.1587	0.3085	0.5000	0.6915	0.8413	0.9332	0.9772

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q5

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q5/5

SET 61 • HSS-ID.A.4

FORWARD 2

Use a normal model for a margin of error • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

Simulated sample proportions are approximately normal with mean 0.58 and standard deviation 0.049. Using the 68-95-99.7 rule, within what interval do about 95% of them fall?

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q1/5

Simulated sample proportions from random samples of 100 have standard deviation 0.049. What margin of error does this give for the population proportion? Give a decimal. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q2/5

True or false?

When a simulated statistic is approximately normal and the margin of error is defined as the half-width of a 95% range, the margin of error is about two standard deviations of the simulated statistic.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q3/5

Simulated sample means are approximately normal with standard deviation 0.2 hours. Using a margin of error of 2 standard deviations (the 95% convention), the margin of error is

A. 0.1 hours

B. 0.4 hours

C. 2 hours

D. 0.2 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q4/5

Why can the 68-95-99.7 rule be used to find a 95% range for simulated sample proportions from large random samples?

A. The population values must be normal, even though the sample proportions themselves are not normal.

B. For large enough samples their distribution is approximately normal, so about 95% fall within 2 SD of the mean.

C. Sample proportions are exactly normal for every sample size, so the rule needs no conditions at all.

D. The 68-95-99.7 rule applies to every histogram, whatever its shape, so no check is needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F2-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q5/5

SET 62 • HSS-ID.A.4 • STATISTICS, INFERENCE, AND DECISIONS

Recognize data sets for which a normal model is not appropriate

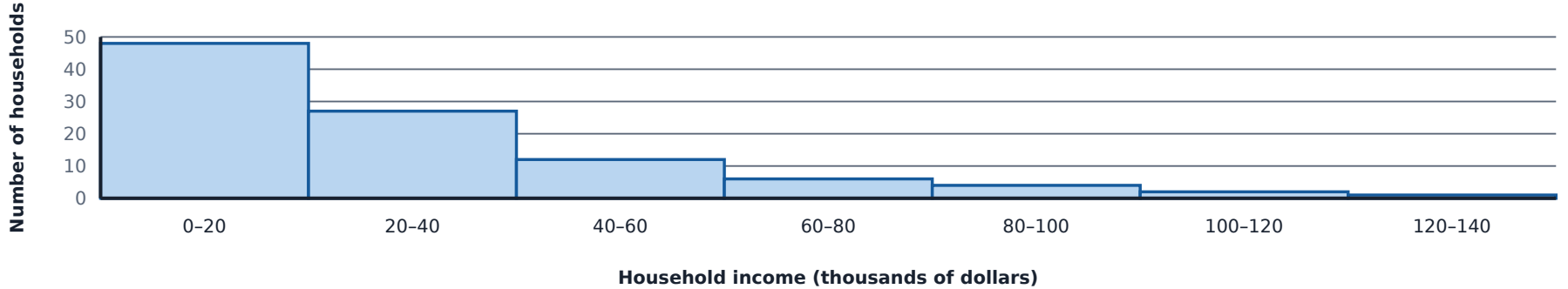
MAIN	Algebra II main question	**HSS-ID.A.4**
BACKWARD 1	Describe skewed, symmetric, bimodal, and uniform shapes	**HSS-ID.A.3**
BACKWARD 2	Measures of center and spread for a data set	**6.SP.A.2**
BACKWARD 3	Groups combined in one data set	**HSS-ID.A.1**
FORWARD 1	Check a normal model with the 68-95-99.7 rule	**HSS-ID.A.4**
FORWARD 2	Decide whether a probability model fits observed data	**HSS-IC.A.2**

SET 62 • HSS-ID.A.4

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

The histogram shows household incomes in a town. Is a normal model appropriate?



A. No: the distribution is strongly skewed right, not symmetric and mound-shaped.

B. No: it has two separate peaks, so it is bimodal rather than normal.

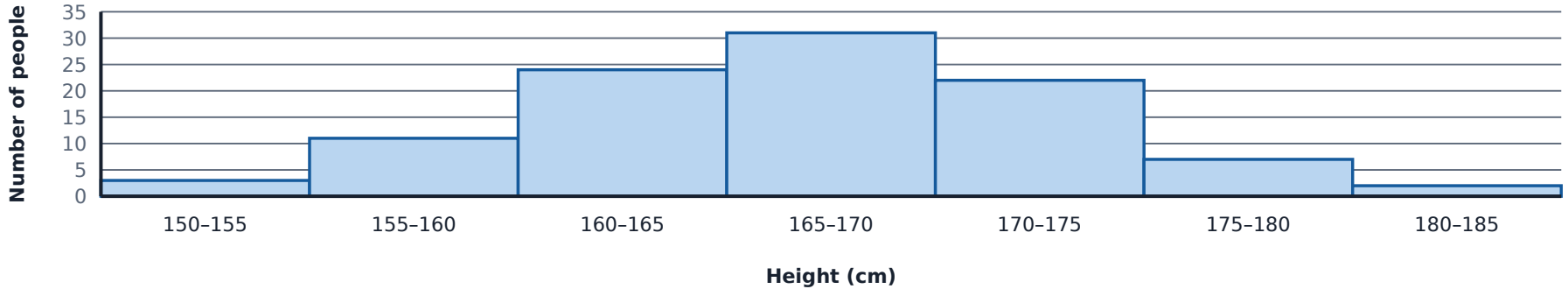
C. Yes: incomes are measured in dollars, and measurements are always normal.

D. Yes: it has a single peak, and any single-peaked distribution is normal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-M-Q1**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q1/5

The histogram shows heights of adults in a club. Is a normal model appropriate?



A. No: it has two separate peaks, one for each group in the club.

B. No: it is strongly skewed right with a long tail to the right.

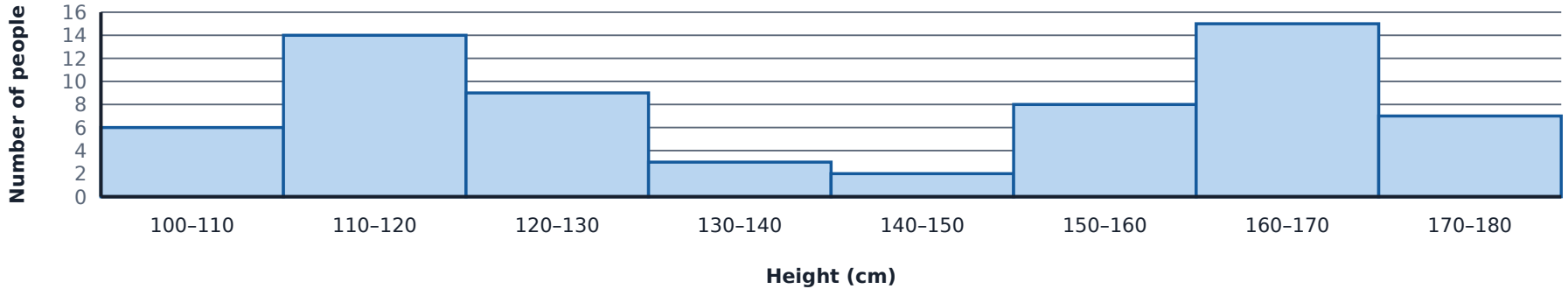
C. Yes: it is roughly symmetric and mound-shaped with a single peak.

D. No: the bars are not exactly equal on both sides of the peak.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-M-Q2**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q2/5

The histogram shows heights of all people at a school event (children and adults). Is a normal model appropriate?



A. No: it is uniform, with all of the bars the same height.

B. No: it has two peaks, from two different groups combined.

C. Yes: heights of people are always approximately normal.

D. Yes: it is roughly symmetric about its middle, so it is normal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-M-Q3**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q3/5

Waiting times at a clinic have mean 4 minutes, standard deviation 5 minutes, and minimum 0. Is a normal model appropriate?

A. No: a normal model requires the standard deviation to be smaller than 1 for any data set.

B. Yes: a normal model can describe any data set once its mean and standard deviation are known.

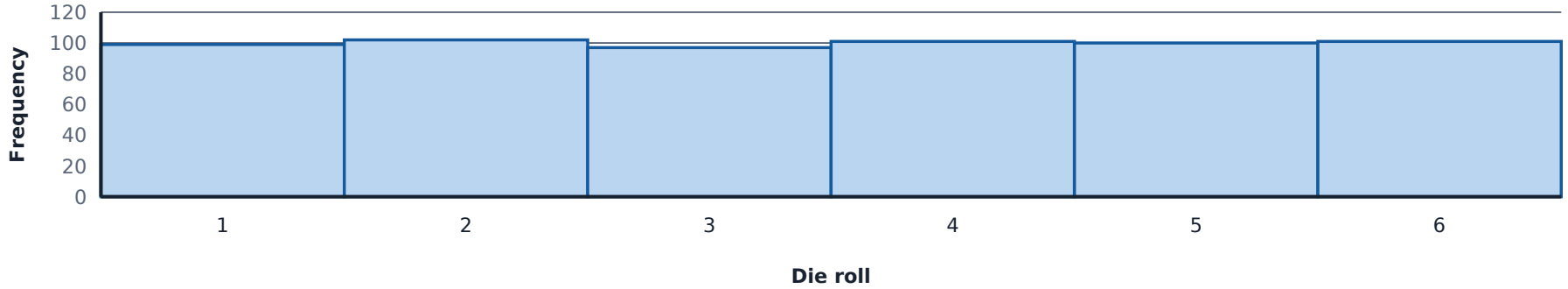
C. Yes: the mean and the standard deviation are both positive, so a normal model fits the times.

D. No: a normal model would put about 16% of times below $4 - 5 = -1$ minute, which is impossible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-M-Q4**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q4/5

The histogram shows the results of 600 rolls of a fair die. Is a normal model appropriate?



A. Yes: it is symmetric about 3.5, so it is normal.

B. No: it is strongly skewed left with a long tail.

C. No: it is roughly uniform (flat), not mound-shaped.

D. Yes: results of dice are always approximately normal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-M-Q5**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q5/5

SET 62 • HSS-ID.A.4

BACKWARD 1

Describe skewed, symmetric, bimodal, and uniform shapes • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

A histogram has a long tail to the right. Which describes its shape?

A. skewed right

B. symmetric

C. uniform

D. skewed left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q1/5

A histogram has two separate peaks. Which describes its shape?

A. symmetric with one peak

B. uniform

C. skewed left

D. bimodal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-B1-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q2/5

True or false?

In a right-skewed distribution, the mean is usually less than the median.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q3/5

A histogram with bars all about the same height is

A. skewed left

B. uniform

C. bimodal

D. normal

Which measure of center is more resistant to a long tail?

A. the maximum

B. the median

C. the mean

D. both equally

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q5/5

SET 62 • HSS-ID.A.4

BACKWARD 2

Measures of center and spread for a data set • Grade 6 • Statistics & Probability • 6.SP.A.2

What does the standard deviation of a data set describe?

A. how spread out the values are around the mean

B. the middle value when the data are in order

C. the number of values in the data set

D. the most common value in the data set

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q1/5

True or false?

A data set has all values at least 0 and mean 4. Some value in the data set can be more than 4 below the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q2/5

True or false?

A distribution can be described by its center, spread, and shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q3/5

Which is a measure of center?

A. median

B. range

C. standard deviation

D. IQR

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q4/5

Give one value two standard deviations below a mean of 4 with SD 5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • HSS-ID.A.4

BACKWARD 3

Groups combined in one data set • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Two groups with very different centers are combined into one data set. What will the histogram of the combined data most likely show?

A. a flat shape

B. a long right tail only

C. two peaks

D. one peak exactly in the middle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q1/5

Which data set is most likely to have a histogram with two peaks?

A. heights of the parents only

B. heights of 5-year-olds and their parents together

C. heights of the 5-year-olds only

D. ages of students in one 10th-grade class

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q2/5

True or false?

A histogram of one group can look different from a histogram of two groups combined.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q3/5

A bimodal histogram suggests

A. an error in counting

B. a small standard deviation

C. two groups in the data

D. a normal model

Heights of all people at a school event (young children and adults) form a two-peaked histogram. Which change is most likely to give mound-shaped distributions?

A. Add more children to the combined data set.

B. Round every height in the combined data to the nearest centimeter.

C. Remove the tallest person from the combined data.

D. Make separate histograms for the children and for the adults.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q5/5

SET 62 • HSS-ID.A.4

FORWARD 1

Check a normal model with the 68-95-99.7 rule • High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

In 1,000 measurements, 70% are within 1 SD of the mean and 96% are within 2 SDs. Is a normal model reasonable?

A. No: the percents must be exactly 68% and 95%.

B. Yes: any data set with 1,000 values is normal.

C. Yes: 70% and 96% are close to the normal 68% and 95%.

D. No: 96% is more than 95%, so the data cannot be normal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q1

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q1/5

A data set has 85% of its values within 1 SD of the mean and 91% within 2 SDs. Is a normal model reasonable?

A. Yes: 91% is close to 95%, so 85% does not matter.

B. Yes: both percents are above 50%.

C. No: a normal model needs 100% within 2 SDs.

D. No: 85% is far above 68% and 91% is below 95%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F1-Q2**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q2/5

True or false?

If a data set has an extreme outlier, the normal model may still fit the rest but the mean and SD are distorted.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q3/5

About what percent of normal data lies more than 3 SDs from the mean?

A. 5%

B. 3%

C. 0.3%

D. 32%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F1-Q4**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q4/5

A data set of 200 values has 25 values more than 2 SDs from the mean. What percent of the values is that, and about what percent does the 68-95-99.7 rule predict for a normal model?

Observed percent: _____

Normal model percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q5

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q5/5

SET 62 • HSS-ID.A.4

FORWARD 2

Decide whether a probability model fits observed data • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

A spinner is claimed to land on each of 4 colors equally often. In 400 spins the counts are 98, 105, 97, 100. Are the results consistent with the claim?

A. Yes: each count is close to the expected 100.

B. No: the counts are not all exactly 100.

C. No: 105 proves the spinner favors one color.

D. Yes: any 4 counts adding to 400 are consistent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q1/5

A model says a coin is fair. In 50 flips there were 34 heads. In 200 simulated runs of 50 fair flips, 3 runs had 34 or more heads. Use 5% as the cutoff for unusual. What do you conclude?

A. 1.5% is not unusual, so the data show that the fair-coin model fits.

B. 1.5% < 5%: unusual for a fair coin, so evidence against the model.

C. No simulated run had exactly 34 heads, so a fair coin has probability 0 of giving 34.

D. 1.5% < 5%: this proves the coin is unfair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F2-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q2/5

True or false?

A model is rejected only if the observed result would be unusual under the model.

A. True

B. False

Under a model, a simulated result at least as extreme as the observed one occurs in 34 of 200 runs. Use 5% as the cutoff for unusual. The model is

A. proved true

B. rejected

C. consistent with the data

D. impossible

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q4/5

How does judging a normal model for a data set differ from judging a probability model with a simulation?

A. Both check whether the data form a bell curve; the simulation just repeats the same comparison many more times.

B. Both compare the sample mean with the population mean and ignore the shape of the data and the simulation.

C. The first uses many simulated runs of the model; the second only compares the shape of the histogram with a bell curve.

D. The first compares the shape of the data with a normal curve; the second asks whether an observed result is unusual among simulated results.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F2-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q5/5

SET 63 • HSS-IC.A.1 • STATISTICS, INFERENCE, AND DECISIONS

Distinguish population, sample, parameter, and statistic

MAIN	Algebra II main question	**HSS-IC.A.1**
BACKWARD 1	Samples and populations	**7.SP.A.1**
BACKWARD 2	Draw inferences from random samples	**7.SP.A.2**
BACKWARD 3	Compute sample proportions and means	**6.SP.B.5.c**
FORWARD 1	Describe sampling variability	**HSS-IC.A.1**
FORWARD 2	Quantify variability with a margin of error	**HSS-IC.B.4**

SET 63 • HSS-IC.A.1

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

A city surveys 400 randomly chosen residents, and 62% of them support a new park. Which identifies the population, sample, parameter, and statistic correctly?

A. Population: all residents; sample: the 62% who support it; parameter: 400; statistic: 62%.

B. Population: all residents; sample: the 400 surveyed; parameter: percent of all residents in support; statistic: 62%.

C. Population: the 400 surveyed; sample: all residents; parameter: 62%; statistic: percent of all residents in support.

D. Population: all residents; sample: the 400 surveyed; parameter: 62%; statistic: percent of all residents in support.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-M-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q1/5

A factory measures 50 randomly selected bolts from a day's production; their mean length is 5.02 cm. Which identifies the parameter and the statistic?

A. Parameter: 5.02 cm, the mean of the 50 bolts; statistic: the mean length of all bolts made that day.

B. Parameter: the number of bolts measured, 50; statistic: the mean length of those bolts, 5.02 cm.

C. Parameter: the mean length of all bolts made that day; statistic: 5.02 cm, the mean of the 50 bolts.

D. Parameter: the 50 bolts that were measured; statistic: all of the bolts that were made that day.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q2/5

A news website posts a poll; 80% of the 2,000 visitors who answered want a later school start. Can this result be generalized to all students?

A. No: it is a voluntary-response sample, so people with strong opinions are over-represented.

B. Yes: 80% is so far above 50% that any bias could not change the overall conclusion.

C. No: a poll can never be used to estimate a population percent, even with random sampling.

D. Yes: 2,000 responses form a large sample, so the result must be accurate for all students.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q3/5

Why does a random sample allow conclusions about the whole population?

A. A random sample always includes every member of the population, so nothing is left out of it.

B. Chance decides who is chosen, so the sample tends to represent the population without systematic bias.

C. A random sample is always larger than a voluntary sample, so its result is always more exact.

D. A random sample always has exactly the same statistic as the population parameter it estimates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-M-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q4/5

A school uses the first 30 students arriving in the morning to estimate the students' average hours of sleep. Which statement is correct?

A. It is a random sample, because the times the students arrive vary by chance.

B. It is not random, but 30 students is enough to remove any bias from the estimate.

C. It is a random sample, because no one chose the students by name or by any other rule.

D. It is not a random sample: early arrivers may sleep differently, so the estimate may be biased.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q5/5

SET 63 • HSS-IC.A.1

BACKWARD 1

Samples and populations • Grade 7 • Statistics & Probability • 7.SP.A.1

In statistics, what is a population?

A. the number of people in a city

B. the part of the group that is measured

C. the entire group a study wants to describe

D. a number computed from a sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q1/5

In statistics, what is a sample?

A. the part of the population that is studied

B. the entire group of interest

C. a number describing the whole population

D. a list of every possible outcome

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q2/5

True or false?

A random sample gives every member of the population a chance to be chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q3/5

Surveying every student in a school is

A. a statistic

B. an experiment

C. a sample

D. a census

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q4/5

Why do studies often use a sample instead of the whole population?

A. Samples have no chance variation at all.

B. Samples always give the exact population value.

C. A population cannot be described by a number.

D. Studying everyone may be too costly, slow, or impossible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q5/5

SET 63 • HSS-IC.A.1

BACKWARD 2

Draw inferences from random samples • Grade 7 • Statistics & Probability • 7.SP.A.2

In a random sample of 50 students, 20 walk to school. Estimate how many of 600 students walk.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q1/5

Two random samples of 500 voters from the same city give 40% and 44% support for a plan. What is the best explanation for the difference?

A. The population percent changed between the two samples.

B. Random samples vary by chance, so a 4-point difference is not surprising.

C. One of the two samples must have been chosen incorrectly.

D. Random samples from one population always give the same percent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q2/5

True or false?

Different random samples from the same population usually give slightly different results.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q3/5

From a random sample, 30% of 200 voters prefer a candidate. The best estimate for all voters is

A. 30%

B. 200

C. cannot estimate

D. 60%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q4/5

Why is a larger random sample usually better for estimating a population value?

A. Its statistic varies less from sample to sample, so the estimate is more precise.

B. It always gives exactly the population value, with no chance variation.

C. It makes the population value larger, so the estimate is closer to it.

D. It removes all bias, even when the sample is not chosen at random.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q5/5

SET 63 • HSS-IC.A.1

BACKWARD 3

Compute sample proportions and means • Grade 6 • Statistics & Probability • 6.SP.B.5.c

248 of 400 people say yes. What proportion is that?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q1/5

Find the mean of 5.0, 5.1, 4.9, 5.08.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q2/5

True or false?

A proportion can be greater than 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q3/5

1,600 of 2,000 is

A. 16%

B. 1.25%

C. 80%

D. 20%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q4/5

Write 62% as a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q5/5

SET 63 • HSS-IC.A.1

FORWARD 1

Describe sampling variability • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

Two random samples of 400 residents of the same city give 62% and 59% support for a park. Which statement is correct?

A. One sample must be biased, since random samples agree exactly.

B. Each sample estimates a different population percent.

C. The samples differ by chance; both estimate the same population percent.

D. The population percent is exactly their average, 60.5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q1/5

Would random samples of 40 or random samples of 400 from the same population vary more from sample to sample?

A. Samples of 40: small samples are always biased.

B. Samples of 400: more people means more variation.

C. Samples of 40: smaller samples are affected more by chance.

D. They vary the same amount, since both are random.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q2/5

True or false?

A parameter changes from sample to sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q3/5

Which is a statistic?

A. the population size

B. the mean height of 30 sampled students

C. the mean height of all students

D. the population proportion

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q4/5

Why can a large voluntary-response sample be worse than a small random sample?

A. Voluntary samples are random, but a large one has too many answers to analyze well.

B. Large samples always have a larger margin of error than small random samples.

C. A large size does not remove bias; volunteers are systematically unrepresentative.

D. Small samples have no chance variation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F1-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q5/5

SET 63 • HSS-IC.A.1

FORWARD 2

Quantify variability with a margin of error • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

A city surveys a random sample of 400 residents, and 62% support a new park. Simulated sample proportions (samples of 400 with p = 0.62) have standard deviation 0.024. What is the margin of error? Give a decimal. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q1/5

In a random sample of 400 residents, 62% support a new park, and the margin of error is 4.8 percentage points. Give the interval estimate for the percent of all residents who support the park.

Lower end (%): _____

Upper end (%): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q2/5

True or false?

A margin of error describes chance variation, not bias.

A. True

B. False

If the sample size is multiplied by 4, the margin of error is about

A. unchanged

B. halved

C. doubled

D. divided by 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q4/5

A random sample of 400 residents gives 62% support for a park, with the 95% interval 57.2% to 66.8%. What does the interval say?

A. Exactly 62% of all of the residents of the city support the park.

B. The percent of all residents who support the park is plausibly between 57.2% and 66.8%.

C. 95% of all of the residents of the city support the park.

D. Between 57.2% and 66.8% of the 400 surveyed residents support the park.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F2-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q5/5

SET 64 • HSS-IC.A.2 • STATISTICS, INFERENCE, AND DECISIONS

Decide whether a model is consistent with observed results using a simulation

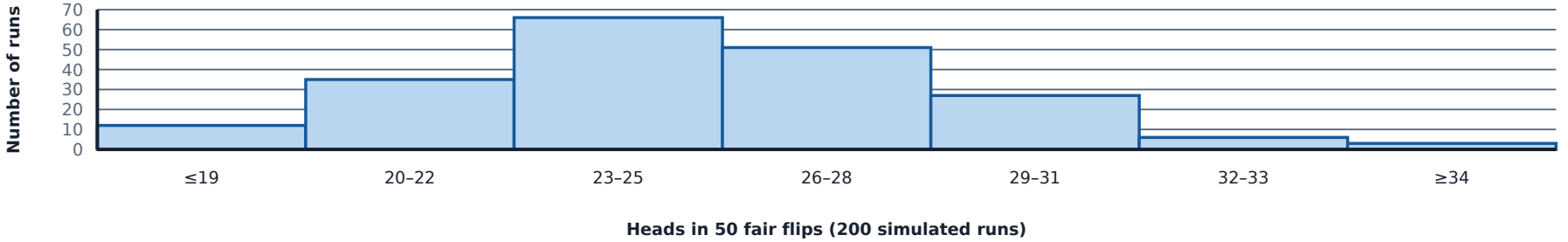
MAIN	Algebra II main question	**HSS-IC.A.2**
BACKWARD 1	Probability as long-run relative frequency	**7.SP.C.6**
BACKWARD 2	Use simulations of compound events	**7.SP.C.8.c**
BACKWARD 3	Relative frequency of a group of outcomes	**6.RP.A.3.c**
FORWARD 1	Design a simulation to test a model	**HSS-IC.A.2**
FORWARD 2	Compare two treatments with a re-randomization simulation	**HSS-IC.B.5**

SET 64 • HSS-IC.A.2

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

A coin is claimed to be fair. It landed heads 34 times in 50 flips. The histogram shows 200 simulated runs of 50 flips of a fair coin. Count runs with 34 or more heads. Use 5% as the cutoff for unusual. Which conclusion is justified?



A. 3 of 200 runs (1.5%) had 34 or more heads; 1.5% < 5%, so the data are evidence against a fair coin.

B. 3 of 200 runs (1.5%) had 34 or more heads, so the coin is proved to be unfair.

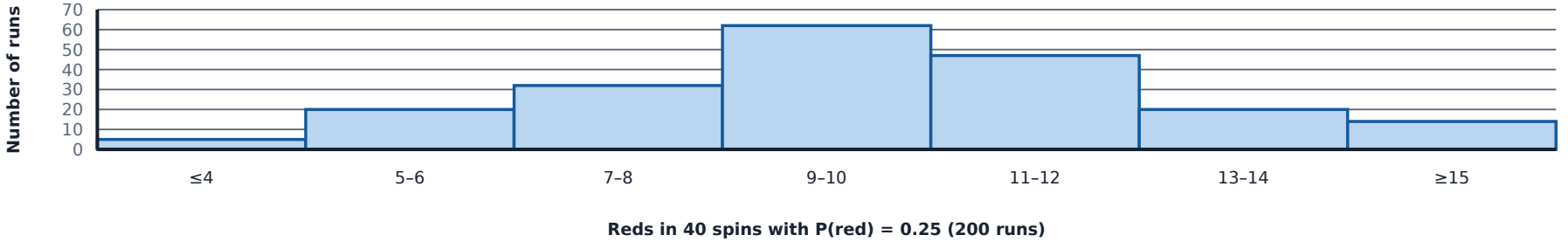
C. A fair coin can never land heads 34 times in 50 flips, so the claim is false.

D. 36 of 200 runs (18%) had 29 or more heads, so 34 heads is not unusual for a fair coin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-M-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q1/5

A spinner is claimed to land on red 25% of the time. It landed on red 13 times in 40 spins. The histogram shows 200 simulated runs of 40 spins under the claim. Count runs with 13 or more reds. Use 5% as the cutoff for unusual. Which conclusion is justified?



A. The expected count is 10, and 13 is not 10, so the claim is false.

B. 34 of 200 runs (17%) had 13 or more reds, so the claim is proved true.

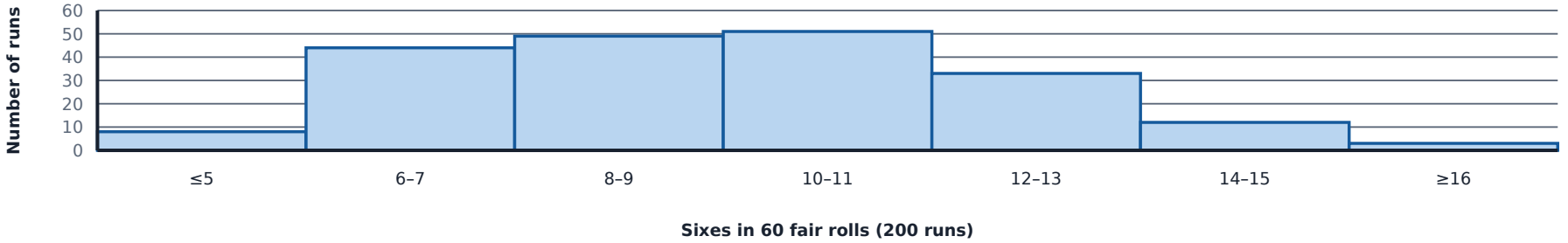
C. Only runs with exactly 13 reds count, which is rare, so the claim is rejected.

D. 34 of 200 runs (17%) had 13 or more reds; 17% > 5%, so the result is consistent with the claim.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-M-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q2/5

A die is claimed to be fair. It showed 16 sixes in 60 rolls. The histogram shows 200 simulated runs of 60 rolls of a fair die. Count runs with 16 or more sixes. Use 5% as the cutoff for unusual. Which conclusion is justified?



A. 15 of 200 runs (7.5%) had 14 or more sixes, so 16 sixes is not unusual.

B. 3 of 200 runs (1.5%) had 16 or more sixes; 1.5% < 5%, so the data are evidence against a fair die.

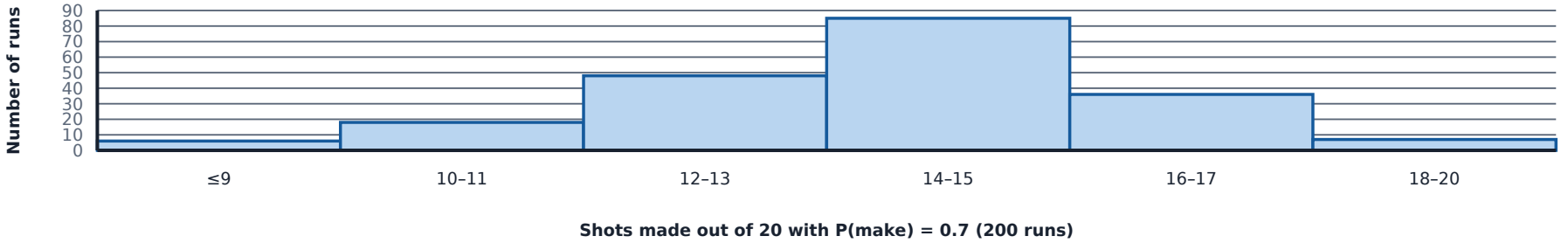
C. 3 of 200 runs (1.5%) had 16 or more sixes, so the die is proved to be unfair.

D. A fair die can never show 16 sixes in 60 rolls, so the claim is false.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-M-Q3**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q3/5

A player claims to make 70% of her free throws. She made 11 of 20. The histogram shows 200 simulated runs of 20 shots under the claim. Count runs with 11 or fewer makes. Use 5% as the cutoff for unusual. Which conclusion is justified?



A. She made 11, not the expected 14, so the claim is false.

B. 24 of 200 runs (12%) had 11 or fewer makes, so the 70% claim is proved true.

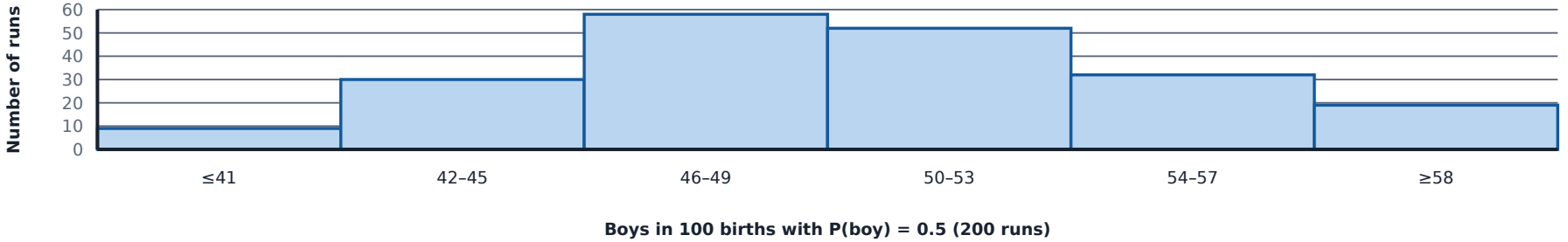
C. 24 of 200 runs (12%) had 11 or fewer makes; 12% > 5%, so the result is consistent with the claim.

D. 6 of 200 runs (3%) had 9 or fewer makes; 3% < 5%, so 11 makes is unusual.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-M-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q4/5

A model says boys and girls are equally likely. A hospital had 58 boys in 100 births. The histogram shows 200 simulated runs of 100 births under the model. Count runs with 58 or more boys. Use 5% as the cutoff for unusual. Which conclusion is justified?



A. 19 of 200 runs (9.5%) had 58 or more boys, so the model is proved true.

B. 19 runs had 58 or more boys, and 19 is more than 5, so the result is unusual.

C. There were 58 boys, not the expected 50, so the model is false.

D. 19 of 200 runs (9.5%) had 58 or more boys; 9.5% > 5%, so the result is consistent with the model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-M-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q5/5

SET 64 • HSS-IC.A.2

BACKWARD 1

Probability as long-run relative frequency • Grade 7 • Statistics & Probability • 7.SP.C.6

A fair coin is flipped 50 times. About how many heads do you expect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q1/5

A spinner has P(red) = 0.25. About how many reds in 40 spins?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q2/5

True or false?

Getting exactly the expected number every time is likely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q3/5

A fair die is rolled 60 times. Expected sixes:

A. 10

B. 16

C. 6

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q4/5

A fair coin is flipped 50 times and lands heads 26 times. Why is this not surprising?

A. 26 is close to the expected 25, and results vary by chance.

B. Every number of heads from 0 to 50 is equally likely.

C. A fair coin always gives exactly 26 heads in 50 flips.

D. 26 is more than half, which proves the coin is fair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q5/5

SET 64 • HSS-IC.A.2

BACKWARD 2

Use simulations of compound events • Grade 7 • Statistics & Probability • 7.SP.C.8.c

Which design simulates 50 flips of a fair coin with random digits 0-9?

A. Use 50 random digits; only the digit 5 = heads.

B. Use 50 random digits; even = heads, odd = tails.

C. Use 50 random digits; 0 = heads, 1-9 = tails.

D. Use 1 random digit and multiply it by 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q1/5

Which design simulates one spin of a spinner with P(red) = 0.25 using two-digit random numbers 00-99?

A. 00-25 = red; 26-99 = not red

B. 25 = red; all other numbers = not red

C. 00-49 = red; 50-99 = not red

D. 00-24 = red; 25-99 = not red

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q2/5

True or false?

Each run of a simulation should copy the real process under the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q3/5

A simulation has 200 runs. Each run represents

A. the final answer

B. one repetition of the whole experiment

C. one histogram bar

D. one flip

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q4/5

Why does a simulation use many runs instead of just one?

A. Because a single run always gives the wrong answer.

B. Because more runs change the model's probabilities.

C. To see the pattern of results the model produces by chance.

D. To make the model match the observed data exactly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q5/5

SET 64 • HSS-IC.A.2

BACKWARD 3

Relative frequency of a group of outcomes • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c